

Relations and Functions

Question1

If the domain of the function $f(x) = \sin^{-1} \left(\frac{5-x}{3+2x} \right) + \frac{1}{\log_e(10-x)}$ is $(-\infty, \alpha] \cup [\beta, \gamma) - \{\delta\}$, then $6(\alpha + \beta + \gamma + \delta)$ is equal to

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Options:

A.

66

B.

68

C.

70

D.

67

Answer: C

Solution:

$$f(x) = \sin^{-1} \left[\frac{5-x}{3+2x} \right] + \frac{1}{\log_e(10-x)}$$
$$-1 \leq \frac{5-x}{3+2x} \leq 1$$

$$\frac{2-3x}{3+2x} \leq 0$$

$$(3x-2)(2x+3) \geq 0$$

$$(3x-2)(2x+3) \geq 0 \Rightarrow x \in \left(-\infty, -\frac{3}{2}\right] \cup \left[\frac{2}{3}, \infty\right)$$

$$\frac{8+x}{3+2x} \geq 0$$

$$x \in (-\infty, -8] \cup \left[\frac{-3}{2}, \infty\right)$$

$$x \in (-\infty, -8] \cup \left[\frac{2}{3}, 10\right) - \{9\} = 6(\alpha + \beta + \gamma + \delta) = 70$$

Question2

Let f and g be functions satisfying $f(x+y) = f(x)f(y)$, $f(1) = 7$ and $g(x+y) = g(xy)$, $g(1) = 1$, for all $x, y \in \mathbf{N}$. If $\sum_{x=1}^n \left(\frac{f(x)}{g(x)}\right) = 19607$, then n is equal to :

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Options:

A.

6

B.

7

C.

4

D.

5

Answer: D

Solution:

$$f(x+y) = f(x) \cdot f(y) \Rightarrow f(x) = a^x$$
$$(\because f(1) = 7 \Rightarrow a^1 = 7)$$

$$\text{So } f(x) = 7^x$$

Now

$$g(x+y) = g(xy) \text{ (put } y = 1) \Rightarrow g(x+1) = g(x)$$

$$\text{So } g(1) = g(2) = g(3) = \dots\dots\dots = g(n) = 1$$

$$\text{Given } \sum_{x=1}^n \frac{f(x)}{g(x)} = 19607$$

$$\sum_{x=1}^n \frac{7^x}{1} = 19607 \Rightarrow 7 \left(\frac{7^n - 1}{7 - 1} \right) = 19607$$

$$7^n - 1 = \frac{6}{7} \times 19607$$

$$7^n = 16807 \Rightarrow n = 5$$

Question3

Let the domain of the function

$$f(x) = \log_3 \log_5 (7 - \log_2 (x^2 - 10x + 85)) + \sin^{-1} \left(\left| \frac{3x-7}{17-x} \right| \right) \text{ be } (\alpha, \beta]$$

. Then $\alpha + \beta$ is equal to :

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Options:

A.

12

B.

8

C.

10

D.

Answer: D

Solution:

$$\text{let } x^2 - 10x + 85 = \lambda$$

$$\therefore \text{Domain for first term } \lambda > 0 \dots\dots\dots (1)$$

$$\& 7 - \log_2 \lambda > 0 \Rightarrow \lambda < 2^7 \dots\dots\dots (2)$$

$$\& \log_5 (7 - \log_2 \lambda) > 0 \Rightarrow \lambda < 2^6 \dots\dots\dots (3)$$

$$\therefore \text{from (1), (2) \& (3)}$$

$$0 < \lambda < 2^6$$

$$0 < x^2 - 10x + 85 < 64 \Rightarrow x \in (3, 7) \dots\dots\dots (A)$$

$$\& \text{domain for second term } -1 \leq \frac{3x-7}{x-17} \leq 1 \Rightarrow x \in [-5, 6] \dots\dots\dots$$

From (A) \& (B), domain of function will be (3, 6]

$$\Rightarrow \alpha = 3, \beta = 6 \Rightarrow \alpha + \beta = 9$$

Question4

Let $f(x) = [x]^2 - [x+3] - 3, x \in \mathbf{R}$, where $[.]$ is the greatest integer funtion. Then

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Options:

A.

$$f(x) = 0 \text{ for finitely many values of } x$$

B.

$$f(x) < 0 \text{ only for } x \in [-1, 3)$$

C.

$$\int_0^2 f(x) dx = -6$$

D.

$$f(x) > 0 \text{ only for } x \in [4, \infty)$$

Answer: B

Solution:

$$(3) \Rightarrow f(x) = 0 \Rightarrow [x] = 3 \text{ or } [x] = -2 \\ \Rightarrow x \in [3, 4) \cup [-2, -1) \Rightarrow f(x) = 0 \text{ has infinite solutions}$$

$$f(x) = [x]^2 - [x + 3] - 3 \\ = [x^2] - [x] - 6 = ([x] - 3)([x] + 2)$$

$$(2) \Rightarrow \int_0^2 f(x) dx = \int_0^2 ([x]^2 - [x] - 6) dx$$

$$\int_0^2 f(x) dx = \int_0^2 ([x]^2 - [x] - 6) dx = \int_0^2 ([x]^2 - [x]) dx - 12 = -12$$

$$(4) \Rightarrow f(x) < 0 \Rightarrow ([x] - 3)([x] + 2) < 0 \Rightarrow -2 < [x] < 3 \Rightarrow -1 \leq [x] < 3$$

$$(1) \Rightarrow f(x) > 0 \Rightarrow [x] < -2 \text{ or } [x] > 3 \Rightarrow x < -2 \text{ or } x \geq 4 \Rightarrow x \in (-\infty, -2) \cup [4, \infty)$$

Question 5

Let f be a function such that $3f(x) + 2f\left(\frac{m}{19x}\right) = 5x, x \neq 0$, where

$m = \sum_{i=1}^9 (i)^2$. Then $f(5) - f(2)$ is equal to

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Options:

A.

36

B.

9

C.

-9

D.

18

Answer: D

Solution:

$$\text{Given } m = \sum_{i=1}^9 (i)^2 = 1^2 + 2^2 + \dots + 9^2$$

$$\text{We know, } 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\text{so, } m = \frac{9 \times 10 \times 19}{6} = 3 \times 5 \times 19 = 285$$

Given equation

$$3f(x) + 2f\left(\frac{m}{19x}\right) = 5x, x \neq 0$$

$$\text{put } m = 285, 3f(x) + 2f\left(\frac{285}{19x}\right) = 5x$$

$$3f(x) + 2f\left(\frac{15}{x}\right) = 5x - (1)$$

$$\text{put } x = \frac{15}{x}$$

$$3f\left(\frac{15}{x}\right) + 2f\left(\frac{15}{15/x}\right) = 5 \times \frac{15}{x}$$

$$3f\left(\frac{15}{x}\right) + 2f(x) = \frac{75}{x} - (2)$$

$$\text{from (1) } f\left(\frac{15}{x}\right) = \frac{5x - 3f(x)}{2} \text{ substitute in equation (2)}$$

$$3 \left[\frac{5x - 3f(x)}{2} \right] + 2f(x) = \frac{75}{x}$$

$$15x - 9f(x) + 4f(x) = \frac{150}{x}$$

$$5f(x) = 15x - \frac{150}{x}$$

$$f(x) = 3x - \frac{30}{x}$$

$$\text{to find } f(5) - f(2)$$

$$f(5) - f(2) = \left(3(5) - \frac{30}{5} \right) - \left(3 \times 2 - \frac{30}{2} \right)$$

$$= (15 - 6) - (6 - 15)$$

$$= 9 - (-9) = 18$$

Question6

If $g(x) = 3x^2 + 2x - 3$, $f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:

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Options:

A.

$$\frac{7}{2}$$

B.

$$-\frac{25}{6}$$

C.

$$\frac{25}{6}$$

D.

$$-\frac{7}{2}$$

Answer: A

Solution:

Given

$$g(x) = 3x^2 + 2x - 3, f(0) = -3, \text{ and } 4g(f(x)) = 3x^2 - 32x + 72$$

Since $g(x)$ is a quadratic function (degree 2) and the composite function $4g(f(x))$ is also a quadratic expression $(3x^2 - 32x + 72)$, $f(x)$ must be a linear function.

- Let $f(x) = ax + b$.
- Using $f(0) = -3 : a(0) + b = -3 \implies b = -3$.
- So, $f(x) = ax - 3$.

First, find $g(f(x))$ by substituting $f(x)$ into $g(x)$:

$$g(f(x)) = 3(f(x))^2 + 2f(x) - 3$$

Substitute this into the given equation $4g(f(x)) = 3x^2 - 32x + 72$:

$$4[3(ax - 3)^2 + 2(ax - 3) - 3] = 3x^2 - 32x + 72$$

Solve for the Coefficient ' a '

Expand the left side of the equation:

- $4[3(a^2x^2 - 6ax + 9) + 2ax - 6 - 3] = 3x^2 - 32x + 72$
- $4[3a^2x^2 - 18ax + 27 + 2ax - 9] = 3x^2 - 32x + 72$
- $4[3a^2x^2 - 16ax + 18] = 3x^2 - 32x + 72$
- $12a^2x^2 - 64ax + 72 = 3x^2 - 32x + 72$

Compare the coefficients:

- For $x^2 : 12a^2 = 3 \implies a^2 = \frac{1}{4} \implies a = \pm \frac{1}{2}$
- For $x : -64a = -32 \implies a = \frac{32}{64} = \frac{1}{2}$ Thus, $f(x) = \frac{1}{2}x - 3$. Calculate $f(g(2))$

Find $g(2)$

$$g(2) = 3(2)^2 + 2(2) - 3 = 12 + 4 - 3 = 13$$

Step B: Find $f(g(2))$ which is $f(13)$

$$f(13) = \frac{1}{2}(13) - 3 = 6.5 - 3 = 3.5 = \frac{7}{2}$$

Question7

The sum of all the elements in the range of

$$f(x) = \text{Sgn}(\sin x) + \text{Sgn}(\cos x) + \text{Sgn}(\tan x) + \text{Sgn}(\cot x),$$

$x \neq \frac{n\pi}{2}, n \in \mathbb{Z}$, **where**

$$\text{Sgn}(t) = \begin{cases} 1, & \text{if } t > 0 \\ -1, & \text{if } t < 0 \end{cases}$$

is :

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Options:

A.

4

B.

0

C.

2

D.

-2

Answer: C

Solution:

$$f(x) = \text{Sgn}(\sin x) + \text{Sgn}(\cos x) + \text{Sgn}(\tan x) + \text{Sgn}(\cot x)$$

Since the trig functions change signs depending on which quadrant x is in, we analyze the value of $f(x)$ quadrant by quadrant.

Given that $x \neq \frac{n\pi}{2}$, we don't have to worry about any of these terms being zero. Each Sgn term will simply be either 1 (if positive) or -1 (if negative).

Quadrant	Interval	$\sin x$	$\cos x$	$\tan x$	$\cot x$	Sum of Sgn values	$f(x)$
I	$0 < x < \frac{\pi}{2}$	+	+	+	+	$1 + 1 + 1 + 1$	4
II	$\frac{\pi}{2} < x < \pi$	+	-	-	-	$1 - 1 - 1 - 1$	-2
III	$\pi < x < \frac{3\pi}{2}$	-	-	+	+	$-1 - 1 + 1 + 1$	0
IV	$\frac{3\pi}{2} < x < 2\pi$	-	+	-	-	$-1 + 1 - 1 - 1$	-2

Range of $f(x)$:

From table, Range = $\{-2, 0, 4\}$

Sum of all elements in this range = $4 + (-2) + 0 = 2$

Question8

Given below are two statements :

Statement I : The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{1+|x|}$ is one-one.

Statement II : The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x^2+4x-30}{x^2-8x+18}$ is many-one.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Statement I is true but **Statement II** is false

B.

Both **Statement I** and **Statement II** are false

C.

Both **Statement I** and **Statement II** are true

D.

Statement I is false but **Statement II** is true

Answer: C

Solution:

Statement I: $f(x) = \frac{x}{1+|x|}$ is one-one

To check if a function is one-one (injective), we can check its monotonicity (whether it's always increasing or always decreasing).

1. For $x \geq 0$: $f(x) = \frac{x}{1+x}$. The derivative is $f'(x) = \frac{1}{(1+x)^2}$, which is always positive.

2. For $x < 0$: $f(x) = \frac{x}{1-x}$. The derivative is $f'(x) = \frac{1}{(1-x)^2}$, which is also always positive.

Since the derivative is positive for all x in the domain, the function is strictly increasing. A strictly monotonic function is always one-one.

Statement I is True.

Statement II: $f(x) = \frac{x^2+4x-30}{x^2-8x+18}$ is many-one

Discriminant of Denominator: For $x^2 - 8x + 18$, $D = (-8)^2 - 4(1)(18) = 64 - 72 = -8$. Since $D < 0$, the denominator is always positive and the function is continuous for all $x \in \mathbb{R}$.

Behavior at Infinity: As $x \rightarrow \infty$ and $x \rightarrow -\infty$, the value of $f(x)$ approaches 1 (the ratio of the leading coefficients).

Turning Points: Since the function starts near $y = 1$ at $-\infty$, moves through various values, and returns toward $y = 1$ at $+\infty$, it must have at least one local maximum or minimum. This means a horizontal line will intersect the graph at more than one point.

Also, let

$$y = \frac{x^2+4x-30}{x^2-8x+18}$$

Cross multiply:

$$y(x^2 - 8x + 18) = x^2 + 4x - 30$$

Rearrange:

$$(y - 1)x^2 + (-8y - 4)x + (18y + 30) = 0$$

This is a quadratic in x .

For many values of y , this quadratic has two distinct real solutions. Function is MANY-ONE

Question9

The number of functions $f : \{1, 2, 3, 4\} \rightarrow \{a, b, c\}$, which are not onto, is :

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Options:

A.

48

B.

45

C.

51

D.

35

Answer: B

Solution:

Total number of functions from $\{1, 2, 3, 4\}$ to $\{a, b, c\}$ is

$$3^4 = 81$$

Now we find the number of onto functions.

For a function to be onto, all the three elements a, b, c must appear as images of elements of $\{1, 2, 3, 4\}$.

Using inclusion-exclusion principle:

- Total functions: $3^4 = 81$

Let

- A = functions missing a
- B = functions missing b
- C = functions missing c

If one element is missing, then the function goes into only 2 elements.

So,

$$|A| = |B| = |C| = 2^4 = 16$$

Hence,

$$|A| + |B| + |C| = 3 \times 16 = 48$$

If two elements are missing, then all 4 elements of domain must go to only one element of codomain.

So,

$$|A \cap B| = |B \cap C| = |C \cap A| = 1^4 = 1$$

Thus,

$$|A \cap B| + |B \cap C| + |C \cap A| = 3$$

If all three are missing, that is impossible, so

$$|A \cap B \cap C| = 0$$

Therefore, number of functions which are not onto is

$$|A \cup B \cup C| = 48 - 3 = 45$$

So the correct answer is

$$\boxed{45}$$

Hence, **Option B** is correct.

Question10

Let $[\bullet]$ denote the greatest integer function. If the domain of the function

$f(x) = \cos^{-1} \left(\frac{4x+2[x]}{3} \right)$ is $[\alpha, \beta]$, then $12(\alpha + \beta)$ is equal to :

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Options:

A.

6

B.

8

C.

9

D.

4

Answer: A

Solution:

For the function

$$f(x) = \cos^{-1} \left(\frac{4x+2[x]}{3} \right),$$

the expression inside $\cos^{-1}$ must lie between -1 and 1 .

So, for domain:

$$-1 \leq \frac{4x+2[x]}{3} \leq 1$$

Multiplying by 3:

$$-3 \leq 4x + 2[x] \leq 3$$

Now let

$$[x] = n$$

where n is an integer and

$$n \leq x < n + 1.$$

Then the inequality becomes

$$-3 \leq 4x + 2n \leq 3.$$

Subtracting $2n$:

$$-3 - 2n \leq 4x \leq 3 - 2n$$

Dividing by 4:

$$\frac{-3-2n}{4} \leq x \leq \frac{3-2n}{4}$$

But also,

$$n \leq x < n + 1.$$

So we find possible integer values of n for which these intervals overlap.

Case 1: $n = -1$

Then

$$[x] = -1, \quad -1 \leq x < 0$$

and

$$\frac{-3-2(-1)}{4} \leq x \leq \frac{3-2(-1)}{4}$$

$$\frac{-1}{4} \leq x \leq \frac{5}{4}$$

Intersection with $[-1, 0)$ gives

$$-\frac{1}{4} \leq x < 0.$$

Case 2: $n = 0$

Then

$$0 \leq x < 1$$

and

$$\frac{-3}{4} \leq x \leq \frac{3}{4}$$

Intersection gives

$$0 \leq x \leq \frac{3}{4}.$$

Case 3: $n = -2$

Then

$$-2 \leq x < -1$$

and

$$\frac{-3-2(-2)}{4} \leq x \leq \frac{3-2(-2)}{4}$$

$$\frac{1}{4} \leq x \leq \frac{7}{4}$$

No intersection.

Case 4: $n = 1$

Then

$$1 \leq x < 2$$

and

$$\frac{-5}{4} \leq x \leq \frac{1}{4}$$

No intersection.

So the domain is

$$\left[-\frac{1}{4}, 0\right) \cup \left[0, \frac{3}{4}\right] = \left[-\frac{1}{4}, \frac{3}{4}\right].$$

Thus,

$$\alpha = -\frac{1}{4}, \quad \beta = \frac{3}{4}.$$

Therefore,

$$\alpha + \beta = \frac{1}{2}$$

and

$$12(\alpha + \beta) = 12 \cdot \frac{1}{2} = 6.$$

Hence the correct answer is

6

So, **Option A** is correct.

Question11

Let for some $\alpha \in \mathbb{R}$, $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $f(x + y) = f(x) + 2y^2 + y + \alpha xy$ for all $x, y \in \mathbb{R}$. If $f(0) = -1$ and $f(1) = 2$, then the value of $\sum_{n=1}^5 (\alpha + f(n))$ is :

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Options:

A.

110

B.

140

C.

150

D.

170

Answer: B

Solution:

Given

$$f(x+y) = f(x) + 2y^2 + y + \alpha xy \quad \text{for all } x, y \in \mathbb{R}$$

with

$$f(0) = -1, \quad f(1) = 2.$$

We have to find

$$\sum_{n=1}^5 (\alpha + f(n)).$$

Step 1: Use the given relation to find α

Put $x = 0$ in

$$f(x+y) = f(x) + 2y^2 + y + \alpha xy.$$

Then

$$f(y) = f(0) + 2y^2 + y + \alpha \cdot 0 \cdot y$$

so

$$f(y) = -1 + 2y^2 + y.$$

Thus, for every real y ,

$$f(y) = 2y^2 + y - 1.$$

Now use $f(1) = 2$:

$$f(1) = 2(1)^2 + 1 - 1 = 2.$$

This is already satisfied, so now we must use the original equation to determine α .

Substitute

$$f(x+y) = 2(x+y)^2 + (x+y) - 1$$

and

$$f(x) = 2x^2 + x - 1$$

into the given equation:

$$2(x+y)^2 + (x+y) - 1 = 2x^2 + x - 1 + 2y^2 + y + \alpha xy.$$

Expand the left side:

$$2x^2 + 4xy + 2y^2 + x + y - 1.$$

Right side is:

$$2x^2 + x - 1 + 2y^2 + y + \alpha xy.$$

Comparing both sides,

$$4xy = \alpha xy.$$

Hence,

$$\alpha = 4.$$

Step 2: Find $f(n)$ for $n = 1$ to 5

Since

$$f(n) = 2n^2 + n - 1,$$

we get:

$$f(1) = 2(1)^2 + 1 - 1 = 2$$

$$f(2) = 2(2)^2 + 2 - 1 = 8 + 2 - 1 = 9$$

$$f(3) = 2(3)^2 + 3 - 1 = 18 + 3 - 1 = 20$$

$$f(4) = 2(4)^2 + 4 - 1 = 32 + 4 - 1 = 35$$

$$f(5) = 2(5)^2 + 5 - 1 = 50 + 5 - 1 = 54$$

Step 3: Compute the required sum

We need

$$\sum_{n=1}^5 (\alpha + f(n)).$$

Since $\alpha = 4$,

$$\alpha + f(1) = 4 + 2 = 6$$

$$\alpha + f(2) = 4 + 9 = 13$$

$$\alpha + f(3) = 4 + 20 = 24$$

$$\alpha + f(4) = 4 + 35 = 39$$

$$\alpha + f(5) = 4 + 54 = 58$$

Therefore,

$$6 + 13 + 24 + 39 + 58 = 140.$$

So the required value is

$$\boxed{140}.$$

Hence, the correct option is **B**.

Question12

For the function $f : [1, \infty) \rightarrow [1, \infty)$ defined by $f(x) = (x - 1)^4 + 1$, among the two statements:

(I) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x)\}$ contains exactly two elements, and

(II) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x + 1)\}$ is an empty set,

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Options:

A.

only (I) is TRUE

B.

only (II) is TRUE

C.

both (I) and (II) are TRUE

D.

neither (I) nor (II) is TRUE

Answer: A

Solution:

For

$$f : [1, \infty) \rightarrow [1, \infty), \quad f(x) = (x - 1)^4 + 1,$$

we first find the inverse function.

Since for $x \geq 1$, the function $(x - 1)^4 + 1$ is strictly increasing, f is one-one and onto from $[1, \infty)$ to $[1, \infty)$.

Let

$$y = (x - 1)^4 + 1.$$

Then

$$y - 1 = (x - 1)^4.$$

Since $x \geq 1$, we take the positive fourth root:

$$x - 1 = (y - 1)^{1/4}.$$

Hence

$$f^{-1}(y) = 1 + (y - 1)^{1/4}.$$

So

$$f^{-1}(x) = 1 + (x - 1)^{1/4}.$$

Now check the two statements one by one.

Statement (I)

We need to solve

$$f(x) = f^{-1}(x).$$

That is,

$$(x - 1)^4 + 1 = 1 + (x - 1)^{1/4}.$$

So

$$(x - 1)^4 = (x - 1)^{1/4}.$$

Put

$$t = x - 1 \quad \text{where } t \geq 0.$$

Then the equation becomes

$$t^4 = t^{1/4}.$$

Now solve:

$$t^4 - t^{1/4} = 0.$$

This gives

$$t^{1/4} (t^{15/4} - 1) = 0.$$

Hence

$$t = 0 \quad \text{or} \quad t^{15/4} = 1.$$

Since $t \geq 0$, from $t^{15/4} = 1$ we get

$$t = 1.$$

Therefore

$$x - 1 = 0 \Rightarrow x = 1,$$

or

$$x - 1 = 1 \Rightarrow x = 2.$$

So the set

$$S = \{x \in [1, \infty) : f(x) = f^{-1}(x)\}$$

contains exactly two elements:

$$S = \{1, 2\}.$$

Thus, Statement (I) is TRUE.

Statement (II)

We need to solve

$$f(x) = f^{-1}(x + 1).$$

Now

$$f(x) = (x - 1)^4 + 1,$$

and

$$f^{-1}(x + 1) = 1 + ((x + 1) - 1)^{1/4} = 1 + x^{1/4}.$$

So the equation becomes

$$(x - 1)^4 + 1 = 1 + x^{1/4}.$$

Hence

$$(x - 1)^4 = x^{1/4}.$$

Again put

$$t = x^{1/4}, \quad t \geq 1 \text{ since } x \geq 1.$$

Then

$$x = t^4.$$

So the equation becomes

$$(t^4 - 1)^4 = t.$$

Instead of solving this directly, test simple values.

For $x = 1$:

$$f(1) = (1 - 1)^4 + 1 = 1,$$

and

$$f^{-1}(1 + 1) = f^{-1}(2) = 1 + (2 - 1)^{1/4} = 2.$$

So $x = 1$ is not a solution.

For $x = 2$:

$$f(2) = (2 - 1)^4 + 1 = 2,$$

and

$$f^{-1}(2 + 1) = f^{-1}(3) = 1 + (3 - 1)^{1/4} = 1 + 2^{1/4} \neq 2.$$

So $x = 2$ is not a solution.

Now define

$$g(x) = (x - 1)^4 - x^{1/4}, \quad x \geq 1.$$

We need to check whether $g(x) = 0$ has any solution.

Observe:

$$g(1) = 0 - 1 = -1 < 0,$$

and

$$g(2) = 1 - 2^{1/4} < 0,$$

while

$$g(3) = 16 - 3^{1/4} > 0.$$

Since g is continuous, there is at least one root in $(2, 3)$.

So the set

$$S = \{x \in [1, \infty) : f(x) = f^{-1}(x + 1)\}$$

is not empty.

Thus, Statement (II) is FALSE.

Therefore, the correct option is

Option A: only (I) is TRUE

Question13

Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x) = \sin^{-1} \left(\frac{x + [x]}{3} \right)$ is $[\alpha, \beta)$, then $\alpha^2 + \beta^2$ is equal to:

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Options:

A.

2

B.

5

C.

10

D.

13

Answer: B

Solution:

For the function

$$f(x) = \sin^{-1} \left(\frac{x + [x]}{3} \right),$$

the expression inside $\sin^{-1}$ must lie between -1 and 1 .

So, we need

$$-1 \leq \frac{x + [x]}{3} \leq 1$$

Multiplying throughout by 3,

$$-3 \leq x + [x] \leq 3$$

Now let

$$[x] = n$$

Then by definition of greatest integer function,

$$n \leq x < n + 1$$

and so

$$x + [x] = x + n$$

Hence the condition becomes

$$-3 \leq x + n \leq 3$$

But since $x \in [n, n + 1)$, let us check possible integer values of n .

Case 1: $n = -2$

Then

$$x \in [-2, -1)$$

and

$$x + [x] = x - 2$$

Since $x \in [-2, -1)$,

$$x - 2 \in [-4, -3)$$

This does not satisfy $-3 \leq x + [x]$ except at $x = -1$, but $-1 \notin [-2, -1)$.

So no value from this interval is allowed.

Case 2: $n = -1$

Then

$$x \in [-1, 0)$$

and

$$x + [x] = x - 1$$

So

$$x - 1 \in [-2, -1)$$

This lies completely within $[-3, 3]$.

Hence all $x \in [-1, 0)$ are allowed.

Case 3: $n = 0$

Then

$$x \in [0, 1)$$

and

$$x + [x] = x$$

So

$$x \in [0, 1)$$

This also lies within $[-3, 3]$.

Hence all $x \in [0, 1)$ are allowed.

Case 4: $n = 1$

Then

$$x \in [1, 2)$$

and

$$x + [x] = x + 1$$

So

$$x + 1 \in [2, 3)$$

This lies within $[-3, 3]$.

Hence all $x \in [1, 2)$ are allowed.

Case 5: $n = 2$

Then

$$x \in [2, 3)$$

and

$$x + [x] = x + 2$$

So

$$x + 2 \in [4, 5)$$

This exceeds 3, so no value is allowed.

Now let us also check the boundary points:

- At $x = -\frac{3}{2}$, actually $[x] = -2$, so this does not help.
- Better to solve more carefully near the ends.

We want the full set of x satisfying

$$-3 \leq x + [x] \leq 3$$

Let $x \in [n, n + 1)$, so $[x] = n$.

Then

$$-3 \leq x + n \leq 3$$

which gives

$$-3 - n \leq x \leq 3 - n$$

Together with $n \leq x < n + 1$, we need intersection of

$$[n, n + 1) \cap [-3 - n, 3 - n].$$

Now check relevant integers:

For $n = -2$

$$x \in [-2, -1) \cap [-1, 5] = \{-1\}$$

but -1 is not included in $[-2, -1)$, so no solution.

For $n = -1$

$$x \in [-1, 0) \cap [-2, 4] = [-1, 0)$$

For $n = 0$

$$x \in [0, 1) \cap [-3, 3] = [0, 1)$$

For $n = 1$

$$x \in [1, 2) \cap [-4, 2] = [1, 2)$$

For $n = 2$

$$x \in [2, 3) \cap [-5, 1] = \emptyset$$

Thus domain is

$$[-1, 0) \cup [0, 1) \cup [1, 2) = [-1, 2)$$

So,

$$\alpha = -1, \quad \beta = 2$$

Therefore,

$$\alpha^2 + \beta^2 = (-1)^2 + 2^2 = 1 + 4 = 5$$

Hence the correct answer is

5

So, **Option B** is correct.

Question14

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \frac{2x^2-3x+2}{3x^2+x+3}$. Then f is :

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Options:

A.

both one-one and onto

B.

one-one but not onto

C.

onto but not one-one

D.

neither one-one nor onto

Answer: D

Solution:

To determine the nature of the given function, let us carefully analyze it for both the one-one (injective) and onto (surjective) properties. The function is given by:

$$f(x) = \frac{2x^2-3x+2}{3x^2+x+3}$$

where the domain is the set of real numbers $\mathbf{R}$ and the codomain is also $\mathbf{R}$.

Checking if the function is one-one

A function $f : \mathbf{R} \rightarrow \mathbf{R}$ is said to be one-one if every element in the domain has a distinct image in the codomain. That is, if $f(x_1) = f(x_2)$, then we must have $x_1 = x_2$.

Let us check if we can find two different real values of x that give the same value for $f(x)$. Let us set $f(x) = 1$ and solve for x :

$$\frac{2x^2-3x+2}{3x^2+x+3} = 1$$

Cross-multiplying, we obtain:

$$2x^2 - 3x + 2 = 3x^2 + x + 3$$

Bringing all terms to one side, we get:

$$x^2 + 4x + 1 = 0$$

Using the quadratic formula to find the roots:

$$x = \frac{-4 \pm \sqrt{16 - 4(1)(1)}}{2} = \frac{-4 \pm \sqrt{12}}{2} = -2 \pm \sqrt{3}$$

Since we found two distinct real values, $x_1 = -2 + \sqrt{3}$ and $x_2 = -2 - \sqrt{3}$, such that $f(x_1) = f(x_2) = 1$, the function is not one-one.

Checking if the function is onto

A function is onto if its range is equal to its codomain. Here, the codomain is $\mathbf{R}$. To find the range, let us set $f(x) = y$:

$$y = \frac{2x^2-3x+2}{3x^2+x+3}$$

Cross-multiplying gives us:

$$y(3x^2 + x + 3) = 2x^2 - 3x + 2$$

$$3yx^2 + yx + 3y = 2x^2 - 3x + 2$$

Grouping the terms by the powers of x , we form a quadratic equation in x :

$$(3y - 2)x^2 + (y + 3)x + (3y - 2) = 0$$

Since x is a real number, the discriminant (D) of this quadratic equation must be greater than or equal to zero ($D \geq 0$).

$$D = (y + 3)^2 - 4(3y - 2)(3y - 2) \geq 0$$

$$(y + 3)^2 - 4(3y - 2)^2 \geq 0$$

This is in the form of $a^2 - b^2 \geq 0$, which can be factored as $(a - b)(a + b) \geq 0$. Here, $a = y + 3$ and $b = 2(3y - 2) = 6y - 4$:

$$[(y + 3) - (6y - 4)][(y + 3) + (6y - 4)] \geq 0$$

$$(-5y + 7)(7y - 1) \geq 0$$

To solve this inequality, we can multiply by -1 (which reverses the inequality sign):

$$(5y - 7)(7y - 1) \leq 0$$

The critical points are $y = \frac{1}{7}$ and $y = \frac{7}{5}$. Based on the inequality, y must lie between these two values:

$$\frac{1}{7} \leq y \leq \frac{7}{5}$$

This means the range of the function is the closed interval $\left[\frac{1}{7}, \frac{7}{5}\right]$.

Because the range is a small subset of the real numbers and not equal to the entire codomain $\mathbf{R}$, the function is not onto.

Conclusion:

Since $f(x)$ is neither one-one nor onto, the correct statement is given by Option D.

Option D

Question 15

If the domain of the function

$f(x) = \sqrt{\log_{(0.6)}\left(\left|\frac{2x-5}{x^2-4}\right|\right)}$ is $(-\infty, a] \cup \{b\} \cup [c, d) \cup (e, \infty)$, then the value of $a + b + c + d + e$ is _____.

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Answer: 4

Solution:

For domain $\log_{0.6} \left| \frac{2x-5}{x^2-4} \right| \geq 0$

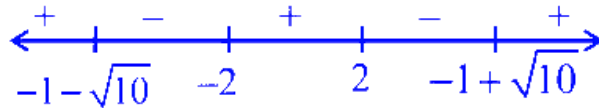
$$\frac{2x-5}{x^2-4} \leq 1 \quad \& \quad x \neq \frac{5}{2} \quad \dots\dots (1)$$

$$-1 \leq \frac{2x-5}{x^2-4} \leq 1$$

$$\frac{2x-5}{x^2-4} + 1 \geq 0$$

$$\frac{x^2+2x-9}{x^2-4} \geq 0$$

$$\frac{(x+1)^2-10}{(x-2)(x+2)} \geq 0$$



$$x \in \left(-\infty, -1 - \sqrt{10}\right] \cup (-2, 2) \cup \left[-1 + \sqrt{10}, \infty\right) \quad \dots\dots (2)$$

$$\frac{2x-5}{x^2-4} - 1 \leq 0$$

$$\frac{2x-5-x^2+4}{x^2-4} \leq 0$$

$$\frac{x^2-2x+1}{x^2-4} \geq 0$$

$$\frac{(x-1)^2}{(x-2)(x+2)} \geq 0$$

$$x \in (-\infty, -2) \cup (2, \infty) \cup \{1\} \quad \dots\dots (3)$$

$$(1) \cap (2) \cap (3)$$

$$x \in \left(-\infty, -1 - \sqrt{10}\right] \cup \{1\} \cup \left[-1 + \sqrt{10}, \frac{5}{2}\right) \cup \left(\frac{5}{2}, \infty\right)$$

$$a + b + c + d + e = -2 + 1 + 5 = 4$$

Question16

Let $A = \{1, 2, 3, 4, 5, 6\}$. The number of one-one functions $f : A \rightarrow A$ such that $f(1) \geq 3$, $f(3) \leq 4$ and $f(2) + f(3) = 5$, is ____ .

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Answer: 72

Solution:

We need the number of one-one functions $f : A \rightarrow A$, where

$$A = \{1, 2, 3, 4, 5, 6\}$$

and the conditions are:

1. $f(1) \geq 3$
2. $f(3) \leq 4$
3. $f(2) + f(3) = 5$

Since f is one-one and domain and codomain are the same finite set, such a function is a permutation of A .

Now let us count carefully.

From

$$f(2) + f(3) = 5$$

and since values are from $A = \{1, 2, 3, 4, 5, 6\}$, the possible pairs are:

$$(f(2), f(3)) = (1, 4), (2, 3), (3, 2), (4, 1)$$

Also $f(3) \leq 4$, which is already satisfied by all these cases.

Now we use the condition that f is one-one, so all function values must be distinct.

We also have:

$$f(1) \geq 3 \implies f(1) \in \{3, 4, 5, 6\}$$

But $f(1)$ must be different from both $f(2)$ and $f(3)$.

Case 1: $(f(2), f(3)) = (1, 4)$

Then $f(1) \in \{3, 4, 5, 6\}$, but it cannot be 4 because function is one-one.

So,

$$f(1) \in \{3, 5, 6\}$$

Number of choices for $f(1) = 3$.

After fixing $f(1), f(2), f(3)$, the remaining 3 elements of the domain can be mapped to the remaining 3 elements of the codomain in

$$3! = 6$$

ways.

So total in this case:

$$3 \times 6 = 18$$

Case 2: $(f(2), f(3)) = (2, 3)$

Then $f(1) \in \{3, 4, 5, 6\}$, but it cannot be 3.

So,

$$f(1) \in \{4, 5, 6\}$$

Number of choices for $f(1) = 3$.

Remaining mappings can be done in

$$3! = 6$$

ways.

Total in this case:

$$3 \times 6 = 18$$

Case 3: $(f(2), f(3)) = (3, 2)$

Then $f(1) \in \{3, 4, 5, 6\}$, but it cannot be 3.

So again number of choices for $f(1) = 3$.

Remaining mappings:

$$3! = 6$$

Total:

$$3 \times 6 = 18$$

Case 4: $(f(2), f(3)) = (4, 1)$

Then $f(1) \in \{3, 4, 5, 6\}$, but it cannot be 4.

So number of choices for $f(1) = 3$.

Remaining mappings:

$$3! = 6$$

Total:

$$3 \times 6 = 18$$

Total number of functions

$$18 + 18 + 18 + 18 = 72$$

Hence, the required number of one-one functions is

$$\boxed{72}$$

Question17

Let f be a polynomial function such that

$$\log_2(f(x)) = \left(\log_2\left(2 + \frac{2}{3} + \frac{2}{9} + \dots \infty\right)\right) \cdot \log_3\left(1 + \frac{f(x)}{f(1/x)}\right), x > 0$$

and $f(6) = 37$. Then $\sum_{n=1}^{10} f(n)$ is equal to ____ .

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Answer: 395

Solution:

Statement of Work :

1. $f(x)$ is a polynomial function

2. A logarithmic equation involving function

$$f : \log_2[f(x)] = \left[\log_2 \left(2 + \frac{2}{3} + \frac{2}{9} + \dots \infty \right) \right] \cdot \log_3 \left\{ 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)} \right\}; x > 0 \text{ and } f(6) = 37$$

3. To evaluate the sum : $\sum_{n=1}^{10} f(n)$

$$\begin{aligned} \log_2[f(x)] &= \left[\log_2 \left(2 + \frac{2}{3} + \frac{2}{9} + \dots \infty \right) \right] \cdot \left[\log_3 \left\{ 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)} \right\} \right] \\ &= \left[\log_2 2 \left\{ 1 + \frac{1}{3} + \left(\frac{1}{3} \right)^2 + \dots \infty \right\} \right] \cdot \left[\log_3 \left\{ 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)} \right\} \right] \\ &= \left[\log_2 2 \left(\frac{1}{1 - \frac{1}{3}} \right) \right] \cdot \left[\log_3 \left\{ 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)} \right\} \right] \\ &= [\log_2 3] \cdot \left[\log_3 \left\{ 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)} \right\} \right] \end{aligned}$$

$$\frac{\log_2[f(x)]}{\log_2 3} = \left[\log_3 \left\{ 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)} \right\} \right]$$

$$\Rightarrow \log_3[f(x)] = \log_3 \left\{ 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)} \right\}$$

$$\Rightarrow f(x) = 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}$$

$$\Rightarrow f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) \dots (i)$$

The function $f(x)$ is a polynomial function stated in SOW where $f(6) = 37 = (6)^2 + 1 \dots (ii)$

Equation (ii) on logical analysis reduces the polynomial $f(x)$ to the possible form below :

$$f(x) = x^2 + 1 \dots (iii) \leftarrow \text{Cubic powers of } x \text{ is not possible where } f(6) > 216$$

Now let us analyze the polynomial function $f(x)$ from preconditions set in SOW :

$$f(x) \cdot f\left(\frac{1}{x}\right) = (1 + x^2) \cdot \left(1 + \frac{1}{x^2}\right) = 1 + \frac{1}{x^2} + x^2 + 1 = (1 + x^2) + \left(1 + \frac{1}{x^2}\right) = f(x) + f\left(\frac{1}{x}\right) \dots (iv)$$

Equation (ii), (iii) and (iv) mathematically certify the polynomial $f(x) = 1 + x^2$:

$$f(x) = 1 + x^2$$

$$\begin{aligned}\Rightarrow s_{10} &= \sum_{n=1}^{10} f(n) = \sum_{n=1}^{10} (1 + n^2) = \sum_{n=1}^{10} (1) + \sum_{n=1}^{10} (n^2) \\ &= 10 + \frac{(10) \cdot (10+1) \cdot \{(2) \cdot (10) + 1\}}{6} = 10 + 385 = 395\end{aligned}$$

Question18

Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 4, 9, 16\}$. Then the number of many-one functions $f : A \rightarrow B$ such that $1 \in f(A)$ is equal to :

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Options:

A. 151

B. 139

C. 163

D. 127

Answer: A

Solution:

Step 1: Total Functions with $1 \in f(A)$

Any function $f : A \rightarrow B$ where $A = \{1, 2, 3, 4\}$ and $B = \{1, 4, 9, 16\}$ is defined by choosing one of the four elements of B for each element of A . Thus, the total number of functions is

$$4^4 = 256.$$

To count those functions where 1 appears at least once in the set $f(A)$, we can use the complementary counting method: subtract the functions that never use 1. If 1 is excluded, each element of A has only 3 choices (namely, $\{4, 9, 16\}$), so the number of such functions is

$$3^4 = 81.$$

Thus, the number of functions such that $1 \in f(A)$ is

$$256 - 81 = 175.$$

Step 2: Counting Many-One Functions

In this context, "many-one functions" are understood to be non-injective functions. Since an injective (one-to-one) function from A to B must be a permutation (because both sets have 4 elements), the number of one-to-one functions is

$$4! = 24.$$

It is important to note that every injective function $f : A \rightarrow B$ has $f(A) = B$ (a full permutation) which automatically means $1 \in f(A)$.

Thus, the number of many-one (non-injective) functions $f : A \rightarrow B$ with $1 \in f(A)$ is found by subtracting the one-to-one functions from the total functions that include 1:

$$175 - 24 = 151.$$

151

This detailed explanation shows that the number of many-one functions $f : A \rightarrow B$ such that $1 \in f(A)$ is indeed 151.

Question19

Let $f(x) = \log_e x$ and $g(x) = \frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1}$. Then the domain of $f \circ g$ is

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Options:

A. $(0, \infty)$

B. $[1, \infty)$

C. $\mathbb{R}$

D. $[0, \infty)$

Answer: C

Solution:

$$f(x) = \ln x$$

$$g(x) = \frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1}$$

$$D_g \in \mathbb{R}$$

$$D_f \in (0, \infty)$$

$$\text{For } D_{f \circ g} \Rightarrow g(x) > 0$$

$$\frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1} > 0$$

$$\Rightarrow x^4 - 2x^3 + 3x^2 - 2x + 2 > 0$$

Clearly $x < 0$ satisfies which are included in option (1) only.

Question 20

Let $f(x) = \frac{2^{x+2} + 16}{2^{2x+1} + 2^{x+4} + 32}$. Then the value of $8 \left(f\left(\frac{1}{15}\right) + f\left(\frac{2}{15}\right) + \dots + f\left(\frac{59}{15}\right) \right)$ is equal to

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Options:

A. 108

B. 92

C. 118

D. 102

Answer: C

Solution:

$$f(x) = \frac{42^x + 16}{2 \cdot 2^{2x} + 16 \cdot 2^x + 32}$$

$$f(x) = \frac{2(2^x + 4)}{2^{2x} + 8 \cdot 2^x + 16}$$

$$f(x) = \frac{2}{2^x + 4}$$

$$f(4 - x) = \frac{2^x}{2(2^x + 4)}$$

$$f(x) + f(4 - x) = \frac{1}{2}$$

$$\text{So, } f\left(\frac{1}{15}\right) + f\left(\frac{59}{15}\right) = \frac{1}{2}$$

$$\begin{aligned}\text{Similarly } &= f\left(\frac{29}{15}\right) + f\left(\frac{31}{15}\right) = \frac{1}{2} \\ f\left(\frac{30}{15}\right) &= f(2) = \frac{2}{2^2+4} = \frac{2}{8} = \frac{1}{4} \\ \Rightarrow &8\left(29 \times \frac{1}{2} + \frac{1}{4}\right)\end{aligned}$$

Ans. 118

Question21

The function $f : (-\infty, \infty) \rightarrow (-\infty, 1)$, defined by $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$ is :

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Options:

- A. One-one but not onto
- B. Onto but not one-one
- C. Both one-one and onto
- D. Neither one-one nor onto

Answer: A

Solution:

$$\begin{aligned}f(x) &= \frac{2^{2x} - 1}{2^{2x} + 1} \\ &= 1 - \frac{2}{2^{2x} + 1} \\ f'(x) &= \frac{2}{(2^{2x} + 1)^2} \cdot 2 \cdot 2^{2x} \cdot \ln 2 \text{ i.e always +ve}\end{aligned}$$

so $f(x)$ is $\uparrow$ function

$$\therefore f(-\infty) = -1$$

$$f(\infty) = 1$$

$$\therefore f(x) \in (-1, 1) \neq \text{co-domain}$$

so function is one-one but not onto

Question22

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = (2 + 3a)x^2 + \left(\frac{a+2}{a-1}\right)x + b, a \neq 1$. If

$f(x + y) = f(x) + f(y) + 1 - \frac{2}{7}xy$, then the value of $28 \sum_{i=1}^5 |f(i)|$ is

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Options:

A. 735

B. 675

C. 715

D. 545

Answer: B

Solution:

$$f(x) = (3a + 2)x^2 + \left(\frac{a+2}{a-1}\right)x + b$$

$$f\left(x + \frac{1}{2}\right) = f(x) + f(y) + 1 - \frac{2}{7}xy \quad \dots (1)$$

$$\text{In (1) Put } x = y = 0 \Rightarrow f(0) = 2f(0) + 1 \Rightarrow f(0) = -1$$

$$\text{So, } f(0) = 0 + 0 + b = -1 \Rightarrow b = -1$$

$$\text{In (1) Put } y = -x \Rightarrow f(0) = f(x) + f(-x) + 1 + \frac{2}{7}x^2$$

$$-1 = 2(3a + 2)x^2 + 2b + 1 + \frac{2}{7}x^2$$

$$-1 = \left(2(3a + 2) + \frac{2}{7}\right)x^2 + 1 - 2$$

$$\Rightarrow 6a + 4 + \frac{2}{7} = 0$$

$$a = -\frac{5}{7}$$

$$\text{So } f(x) = -\frac{1}{7}x^2 - \frac{3}{4}x - 1$$

$$\Rightarrow |f(x)| = \frac{1}{28}|4x^2 + 21x + 28|$$

Now, $28 \sum_{i=1}^5 |f(i)| = 28(|f(1)| + |f(2)| + \dots + |f(5)|)$

$$28 \cdot \frac{1}{28} \cdot 675 = 675$$

Question 23

If $f(x) = \frac{2^x}{2^x + \sqrt{2}}$, $x \in \mathbb{R}$, then $\sum_{k=1}^{81} f\left(\frac{k}{82}\right)$ is equal to

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Options:

A. 82

B. $81\sqrt{2}$

C. 41

D. $\frac{81}{2}$

Answer: D

Solution:

$$f(x) = \frac{2^x}{2^x + \sqrt{2}}$$

$$\begin{aligned} f(x) + f(1-x) &= \frac{2^x}{2^x + \sqrt{2}} + \frac{2^{1-x}}{2^{1-x} + \sqrt{2}} \\ &= \frac{2^x}{2^x + \sqrt{2}} + \frac{2}{2 + \sqrt{2}2^x} = \frac{2^x + \sqrt{2}}{2^x + \sqrt{2}} = 1 \end{aligned}$$

$$\begin{aligned} \text{Now, } \sum_{k=1}^{81} f\left(\frac{k}{82}\right) &= f\left(\frac{1}{82}\right) + f\left(\frac{2}{82}\right) + \dots + f\left(\frac{81}{82}\right) \\ &= f\left(\frac{1}{82}\right) + f\left(\frac{2}{82}\right) + \dots + f\left(1 - \frac{2}{82}\right) + f\left(1 - \frac{1}{82}\right) \\ &= \left[f\left(\frac{1}{82}\right) + f\left(1 - \frac{1}{82}\right)\right] + \left[f\left(\frac{2}{82}\right) + f\left(1 - \frac{2}{82}\right)\right] + \dots + 40 \text{ cases} + f\left(\frac{41}{82}\right) \end{aligned}$$

$$= (1 + 1 + \dots 40 \text{ times}) + \frac{2^{1/2}}{2^{1/2} + 2^{1/2}}$$

$$40 + \frac{1}{2} = \frac{81}{2}$$

Question24

Let $f : [0, 3] \rightarrow A$ be defined by $f(x) = 2x^3 - 15x^2 + 36x + 7$ and $g : [0, \infty) \rightarrow B$ be defined by $g(x) = \frac{x^{2025}}{x^{2025} + 1}$, If both the functions are onto and $S = \{x \in Z; x \in A \text{ or } x \in B\}$, then $n(S)$ is equal to :

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Options:

A.

29

B.

31

C.

30

D.

36

Answer: C

Solution:

as $f(x)$ is onto hence A is range of $f(x)$

$$\begin{aligned} \text{now } f'(x) &= 6x^2 - 30x + 36 \\ &= 6(x - 2)(x - 3) \end{aligned}$$

$$f(2) = 16 - 60 + 72 + 7 = 35$$

$$f(3) = 54 - 135 + 108 + 7 = 34$$

$$f(0) = 7$$

hence range $\in [7, 35] = A$

also for range of $g(x)$

$$g(x) = 1 - \frac{1}{x^{2025} + 1} \in [0, 1) = B$$

$$s = \{0, 7, 8, \dots, 35\} \text{ hence } n(s) = 30$$

Question 25

If the domain of the function $\log_5(18x - x^2 - 77)$ is (α, β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$ is (γ, δ) , then $\alpha^2 + \beta^2 + \gamma^2$ is equal to:

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

186

B.

179

C.

195

D.

174

Answer: A

Solution:

$$f_1(x) = \log_5(18x - x^2 - 77)$$

$$\therefore 18x - x^2 - 77 > 0$$

$$x^2 - 18x + 77 < 0$$

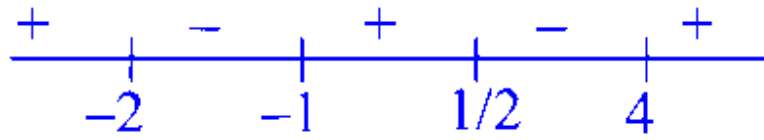
$$x \in (7, 11) \alpha = 7, \beta = 11$$

$$f_2(x) = \log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$$

$$\therefore x - 1 > 0, x - 1 \neq 1, \frac{2x^2+3x-2}{x^2-3x-4} > 0$$

$$x > 1, x \neq 2, \frac{(2x-1)(x+2)}{(x-4)(x+1)} > 0$$

$$x > 1, x \neq 2,$$



$$\therefore x \in (4, \infty)$$

$$\therefore \gamma = 4$$

$$\begin{aligned}\therefore \alpha^2 + \beta^2 + \gamma^2 &= 49 + 121 + 16 \\ &= 186\end{aligned}$$

Question 26

If the domain of the function $f(x) = \frac{1}{\sqrt{10+3x-x^2}} + \frac{1}{\sqrt{x+|x|}}$ is (a, b) , then $(1+a)^2 + b^2$ is equal to :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. 29

B. 30

C. 25

D. 26

Answer: D

Solution:

$$x + |x| = \begin{cases} 2x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$\Rightarrow \frac{1}{\sqrt{x+|x|}}, \text{ domain is } x > 0, \text{ as } 2x \neq 0$$

Similarly,

$$\frac{1}{\sqrt{3x+10-x^2}} \text{ is defined when } 3x+10-x^2 > 0$$

$$\Rightarrow x^2 - 3x - 10 < 0$$

$$(x-5)(x+2) < 0$$

$$\Rightarrow x \in (-2, 5)$$

$$\Rightarrow \text{Domain will be } (0, \infty) \cap (-2, 5) = (0, 5)$$

$$\Rightarrow (1+a)^2 + b^2 = 1 + 25 = 26$$

Question27

If the domain of the function is

$f(x) = \log_e \left(\frac{2x-3}{5+4x} \right) + \sin^{-1} \left(\frac{4+3x}{2-x} \right)$ is $[\alpha, \beta)$, then $\alpha^2 + 4\beta$ equal to

Options:

A. 4

B. 3

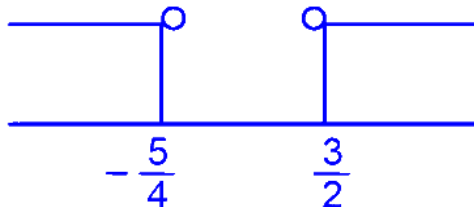
C. 7

D. 5

Answer: A

Solution:

$$\frac{2x-3}{4x+5} > 0$$



$$\therefore x \in \left(-\infty, -\frac{5}{4} \right) \cup \left(\frac{3}{2}, \infty \right) \dots\dots\dots (i)$$

$$-1 \leq \frac{3x+4}{2-x} \leq 1$$

$$\frac{3x+4}{2-x} \leq 1$$

$$\Rightarrow \frac{3x+4}{2-x} - 1 \leq 0$$

$$\Rightarrow \frac{3x+4-2+x}{x-2} \geq 0$$

$$\Rightarrow \frac{4x+2}{x-2} \geq 0$$

$$\Rightarrow x \in \left(-\infty, -\frac{1}{2} \right] \cup (2, \infty) \dots\dots\dots (ii)$$

$$\begin{aligned}\frac{3x+4}{2-x} &\geq -1 \\ \Rightarrow \frac{3x+4}{2-x} + 1 &\geq 0 \\ \Rightarrow \frac{3x+4+2-x}{2-x} &\geq 0 \\ \Rightarrow \frac{2x+6}{x-2} &\leq 0 \\ \therefore x &\in [-3, 2) \dots\dots\dots (iii)\end{aligned}$$

Taking intersection of (i), (ii) and (iii)

$$\begin{aligned}x &\in \left[-3, -\frac{5}{4}\right) \\ \alpha &= -3, \beta = -\frac{5}{4} \\ \alpha^2 + 4\beta &= 4\end{aligned}$$

Question28

If the domain of the function $f(x) = \log_7 (1 - \log_4 (x^2 - 9x + 18))$ is $(\alpha, \beta) \cup (\gamma, o)$, then $\alpha + \beta + \gamma + \hat{o}$ is equal to

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

- A. 17
- B. 15
- C. 16
- D. 18

Answer: D

Solution:

$$1 - \log_4 (x^2 - 9x + 18) > 0$$

$$\log_4 (x^2 - 9x + 18) < 1$$

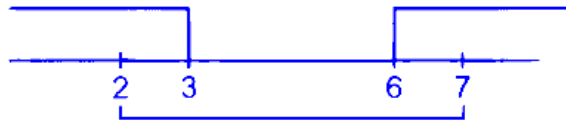
$$x^2 - 9x + 18 < 4$$

$$x^2 - 9x + 14 < 0$$

$$x \in (2, 7)$$

$$x^2 - 9x + 18 > 0$$

$$x \in (-\infty, 3) \cup (6, \infty)$$



$$x \in (2, 3) \cup (6, 7)$$

$$\alpha + \beta + \gamma + \delta = 18$$

Question 29

Let f be a function such that $f(x) + 3f\left(\frac{24}{x}\right) = 4x, x \neq 0$. Then $f(3) + f(8)$ is equal to

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. 13

B. 11

C. 10

D. 12

Answer: B

Solution:

$$f(x) + 3f\left(\frac{24}{x}\right) = 4x, x \neq 0 \quad \dots (1)$$

replace x by $\frac{24}{x}$

$$f\left(\frac{24}{x}\right) + 3f\left(\frac{24}{24}\right) = 4\left(\frac{24}{x}\right) = \frac{96}{x} \quad \dots (2)$$

$$3 \times (2) - (1)$$

$$\Rightarrow 8f(x) = \frac{96 \cdot 3}{x} - 4x \Rightarrow f(x) = \frac{36}{x} - \frac{x}{2}$$

$$f(3) + f(8) = \left(12 - \frac{3}{2}\right) + \left(\frac{36}{8} - 4\right)$$

$$= 8 + \frac{36}{8} - \frac{12}{8} = 11$$

Question30

Let $f, g : (1, \infty) \rightarrow \mathbb{R}$ be defined as $f(x) = \frac{2x+3}{5x+2}$ and $g(x) = \frac{2-3x}{1-x}$. If the range of the function fog: $[2, 4] \rightarrow \mathbb{R}$ is $[\alpha, \beta]$, then $\frac{1}{\beta-\alpha}$ is equal to

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. 56

B. 2

C. 29

D. 68

Answer: A

Solution:

$$g(2) = 4, g(4) = \frac{10}{3}$$

$$f \text{ is decreasing in } \left(\frac{10}{3}, 4\right)$$

$$\begin{aligned}\therefore \alpha &= f(4) = \frac{1}{2} \\ \beta &= f\left(\frac{10}{3}\right) = \frac{29}{56} \\ \frac{1}{\beta - \alpha} &= \frac{1}{\frac{29}{56} - \frac{1}{2}} = 56\end{aligned}$$

Question31

Let the domains of the functions

$f(x) = \log_4 \log_3 \log_7 (8 - \log_2 (x^2 + 4x + 5))$ and
 $g(x) = \sin^{-1} \left(\frac{7x+10}{x-2} \right)$ be (α, β) and $[\gamma, \delta]$, respectively. Then
 $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is equal to :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

- A. 15
- B. 13
- C. 16
- D. 14

Answer: A

Solution:

$$\begin{aligned}f(x) &= \log_4 (\log_3 (\log_7 (8 - \log_2 (x^2 + 4x + 5)))) \\ \log_3 (\log_1 (8 - \log_2 (x^2 + 4x + 5))) &> 0 \\ \log_7 (8 - \log_2 (x^2 + 4x + 5)) &> 1 \\ 8 - \log_2 (x^2 + 4x + 5) &> 7 \\ -\log_2 (x^2 + 4x + 5) &> -1\end{aligned}$$

$$\log_2 (x^2 + 4x + 5) < 1$$

$$x^2 + 4x + 5 < 2$$

$$x^2 + 4x + 3 < 0$$

$$\Rightarrow (x+3)(x+1) < 0 \quad \dots (1)$$

$$\log_7 (8 - \log_2 (x^2 + 4x + 5)) > 0$$

$$8 - \log_2 (x^2 + 4x + 5) > 1$$

$$\log_2 (x^2 + 4x + 5) < 9$$

$$x^2 + 4x + 5 < 2^9$$

$$x^2 + 4x + 5 < 512$$

$$\Rightarrow x^2 + 4x - 507 < 0$$

$$\Rightarrow x = -4 \pm \sqrt{16 + 2028}$$

$$x = \frac{-4 \pm \sqrt{2044}}{2} \quad \dots (2)$$

$$\Rightarrow \left(x - \left(\frac{-4 + \sqrt{2044}}{2} \right) \right) \left(x - \left(\frac{-4 - \sqrt{2044}}{2} \right) \right) < 0$$

$$x^2 + 4x + 5 > 0$$

$$D > 0$$

$$x \in R$$

$$\text{Also, } 8 - \log_2 (x^2 + 4x + 5) > 0$$

$$\log_2 (x^2 + 4x + 5) < 8$$

$$x^2 + 4x + 5 < 256$$

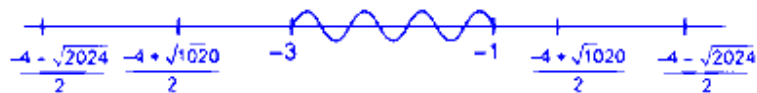
$$\Rightarrow x^2 + 4x - 251 < 0$$

$$\Rightarrow x = -4 \pm \sqrt{16 + 1004}$$

$$\Rightarrow x = \frac{-4 \pm \sqrt{1020}}{2}$$

$$\Rightarrow \left(x - \left(\frac{-4 + \sqrt{1020}}{2} \right) \right) \left(x - \left(\frac{-4 - \sqrt{1020}}{2} \right) \right) < 0$$

$\therefore$ Intersection of (1), (2) and (3)



$$\therefore x \in (-3, -1)$$

$$-1 \leq \frac{7x + 10}{x - 2} \leq 1$$

$$\Rightarrow x \in [-2, -1]$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (-3)^2 + (-1)^2 + (-2)^2 + (-1)^2$$

$$= 9 + 1 + 4 + 1$$

$$= 15$$

Question32

If the range of the function $f(x) = \frac{5-x}{x^2-3x+2}$, $x \neq 1, 2$, is $(-\infty, \alpha] \cup [\beta, \infty)$, then $\alpha^2 + \beta^2$ is equal to :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

188

B.

192

C.

190

D.

194

Answer: D

Solution:

$$y = \frac{5-x}{x^2-3x+2}$$

$$yx^2 - 3xy + 2y + x - 5 = 0$$

$$yz^2 + (-3y+1)x + (2y-5) = 0$$

Case I : If $y = 0$ (Accepted)

$$\Rightarrow x = 5$$

Case II : If $y \neq 0$

$$D \geq 0$$

$$(-3y + 1)^2 - 4(y)(2y - 5) \geq 0$$

$$9y^2 + 1 - 6y - 8y^2 + 20y \geq 0$$

$$y^2 + 14y + 1 \geq 0$$

$$(y + 7)^2 - 48 \geq 0$$

$$|y + 7| \geq 4\sqrt{3}$$

$$\Rightarrow y + 7 \geq 4\sqrt{3} \text{ or } y + 7 \leq -4\sqrt{3}$$

$$\Rightarrow y \geq 4\sqrt{3} - 7 \text{ or } y \leq -4\sqrt{3} - 7$$

From Case I and Case II

$$y \in (-\infty, -4\sqrt{3} - 7] \cup [4\sqrt{3} - 7, \infty)$$

$$\text{So } \alpha = -4\sqrt{3} - 7$$

$$\beta = 4\sqrt{3} - 7$$

$$\begin{aligned} \Rightarrow a^2 + b^2 &= (-4\sqrt{3} - 7)^2 + (4\sqrt{3} - 7)^2 \\ &= 2(48 + 49) \\ &= 194 \end{aligned}$$

Question33

Let the domain of the function $f(x) = \cos^{-1} \left(\frac{4x+5}{3x-7} \right)$ be $[\alpha, \beta]$ and the domain of $g(x) = \log_2 (2 - 6 \log_{27}(2x + 5))$ be (γ, δ) .

Then $|7(\alpha + \beta) + 4(\gamma + \delta)|$ is equal to _____.

JEE Main 2025 (Online) 8th April Evening Shift

Answer: 96

Solution:

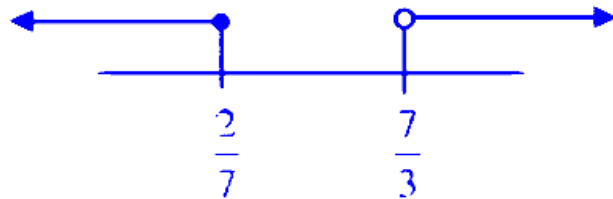
$$f(x) = \cos^{-1} \left(\frac{4x + 5}{3x - 7} \right)$$

$$\Rightarrow -1 \leq \left(\frac{4x+5}{3x-7} \right) \leq 1$$

$$\left(\frac{4x+5}{3x-7} \right) \geq -1$$

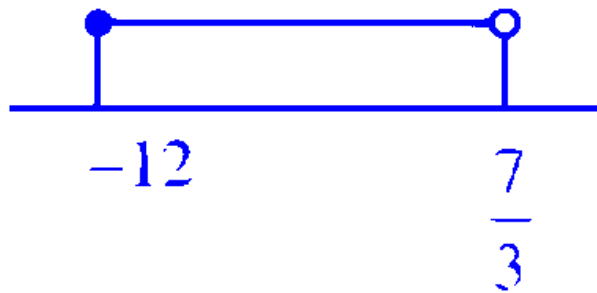
$$\frac{4x+5+3x-7}{3x-7} \geq 0$$

$$\Rightarrow \frac{7x-2}{3x-7} \geq 0$$



$$x \in \left(-\infty, \frac{2}{7} \right] \cup \left(\frac{7}{3}, \infty \right)$$

$$\& \frac{4x+5}{3x-7} \leq 1 \Rightarrow \frac{x+12}{3x-7} \leq 0$$



$\therefore$ Domain of $f(x)$ is

$$\left[-12, \frac{2}{7} \right] \cup \left(\frac{7}{3}, \infty \right)$$

$$g(x) = \log_2 (2 - 6 \log_{27} (2x + 5))$$

Domain

$$2 - 6 \log_{27} (2x + 5) > 0$$

$$\Rightarrow 6 \log_{27} (2x + 5) < 2$$

$$\Rightarrow \log_{27} (2x + 5) < \frac{1}{3}$$

$$\Rightarrow 2x + 5 < 3$$

$$\Rightarrow x < -1$$

$$\& 2x + 5 > 0 \Rightarrow x > -\frac{5}{2}$$

$$\text{Domain is } x \in \left(-\frac{5}{2}, -1 \right)$$

$$\gamma = -\frac{5}{2}, \delta = -1$$

$$|7(\alpha + \beta) + 4(\gamma + \delta)| = \left| 7\left(-12 + \frac{2}{7} + 4\left(-\frac{5}{2} - 1\right)\right) \right|$$

$$|-82 - 14| = 96$$

Question34

The function $f : \mathbb{N} - \{1\} \rightarrow \mathbb{N}$; defined by $f(n)$ = the highest prime factor of n , is :

[27-Jan-2024 Shift 1]

Options:

A.

both one-one and onto

B.

one-one only

C.

onto only

D.

neither one-one nor onto

Answer: D

Solution:

$$f: \mathbb{N} - \{1\} \rightarrow \mathbb{N}$$

$f(n) =$ The highest prime factor of n .

$$f(2) = 2$$

$$f(4) = 2$$

$\Rightarrow$ many one

4 is not image of any element

$\Rightarrow$ into

Hence many one and into

Neither one-one nor onto.

Question 35

Let $f: \mathbb{R} - \left\{ \frac{-1}{2} \right\} \rightarrow \mathbb{R}$ and $g: \mathbb{R} - \left\{ \frac{-5}{2} \right\} \rightarrow \mathbb{R}$ be defined as $f(x) = \frac{2x+3}{2x+1}$ and $g(x) = \frac{|x|+1}{2x+5}$. Then the domain of the function $f \circ g$ is :

[27-Jan-2024 Shift 2]

Options:

A.

$$\mathbb{R} - \left\{ -\frac{5}{2} \right\}$$

B.

$$\mathbb{R}$$

C.

$$\mathbb{R} - \left\{ -\frac{7}{4} \right\}$$

D.

$$\mathbb{R} - \left\{ -\frac{5}{2}, -\frac{7}{4} \right\}$$

Answer: A

Solution:

$$f(x) = \frac{2x+3}{2x+1}; x \neq -\frac{1}{2}$$

$$g(x) = \frac{|x|+1}{2x+5}; x \neq -\frac{5}{2}$$

Domain of $f(g(x))$

$$f(g(x)) = \frac{2g(x)+3}{2g(x)+1}$$

$$x \neq -\frac{5}{2} \text{ and } \frac{|x|+1}{2x+5} \neq -\frac{1}{2}$$

$$x \in \mathbb{R} - \left\{ -\frac{5}{2} \right\} \text{ and } x \in \mathbb{R}$$

$$\therefore \text{Domain will be } \mathbb{R} - \left\{ -\frac{5}{2} \right\}$$

Question36

Consider the function $f : [1/2, 1] \rightarrow \mathbb{R}$ defined by $f(x) = 4\sqrt{2x} - 3\sqrt{2x}^{-1}$.
Consider the statements

(I) The curve $y = f(x)$ intersects the x-axis exactly at one point

(II) The curve $y = f(x)$ intersects the x-axis at $x = \cos \pi/12$

Then

[29-Jan-2024 Shift 1]

Options:

A.

Only (II) is correct

B.

Both (I) and (II) are incorrect

C.

Only (I) is correct

D.

Both (I) and (II) are correct

Answer: D

Solution:

$$f'(x) = 12\sqrt{2}x^2 - 3\sqrt{2} \geq 0 \text{ for } \left[\frac{1}{2}, 1 \right]$$

$$f\left(\frac{1}{2}\right) < 0$$

$f(1) > 0 \Rightarrow$ (A) is correct.

$$f(x) = \sqrt{2}(4x^3 - 3x) - 1 = 0$$

Let $\cos \alpha = x$,

$$\cos 3\alpha = \cos \frac{\pi}{4} \Rightarrow \alpha = \frac{\pi}{12}$$

$$x = \cos \frac{\pi}{12}$$

(4) is correct.

Question 37

$$\text{If } f(x) = \begin{cases} 2+2x & -1 \leq x < 0 \\ 1 - \frac{x}{3} & 0 \leq x \leq 3 \end{cases}$$

$$g(x) = \begin{cases} -x & -3 \leq x \leq 0 \\ x & 0 < x \leq 1 \end{cases} \dots$$

then range of $(f \circ g(x))$ is

[29-Jan-2024 Shift 1]

Options:

A.

$(0, 1]$

B.

$[0, 3)$

C.

$[0, 1]$

D.

$[0, 1)$

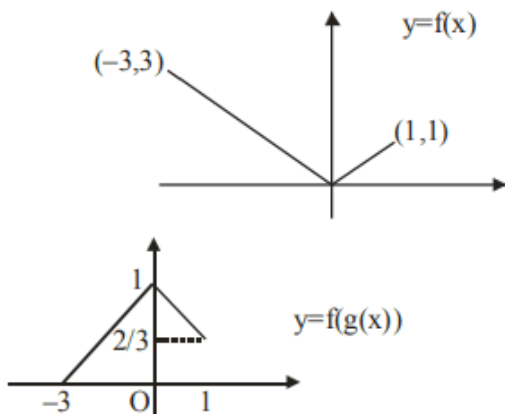
Answer: C

Solution:

$$f(g(x)) = \begin{cases} 2 + 2g(x), & -1 \leq g(x) < 0 \dots\dots (1) \\ 1 - \frac{g(x)}{3}, & 0 \leq g(x) \leq 3 \dots\dots (2) \end{cases}$$

By (1) $x \in \phi$

And by (2) $x \in [-3, 0]$ and $x \in [0, 1]$



Range of $f(g(x))$ is $[0, 1]$

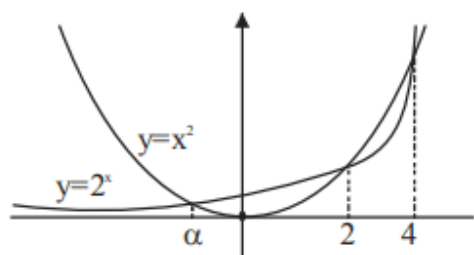
Question38

Let $f(x) = 2^x - x^2$, $x \in \mathbb{R}$. If m and n are respectively the number of points at which the curves $y = f(x)$ and $y = f'(x)$ intersects the x -axis, then the value of $m + n$ is

[29-Jan-2024 Shift 1]

Answer: 5

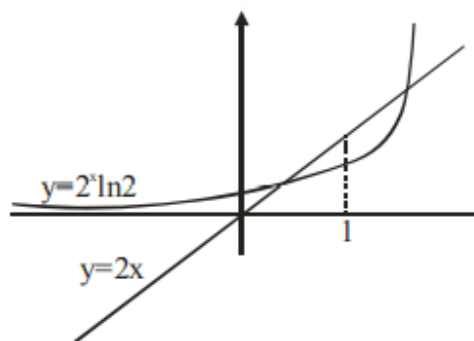
Solution:



$$\therefore m = 3$$

$$f'(x) = 2^x \ln 2 - 2x = 0$$

$$2^x \ln 2 = 2x$$



$$\therefore n = 2$$

$$\Rightarrow m + n = 5$$

Question39

If the domain of the function $f(x) = \cos^{-1}(2 - |x|/4) + (\log_e(3 - x))^{-1}$ is $[-\alpha, \beta) - \{\gamma\}$, then $\alpha + \beta + \gamma$ is equal to :

[30-Jan-2024 Shift 1]

Options:

A.

12

B.

9

C.

11

D.

8

Answer: C

Solution:

$$-1 \leq \left| \frac{2-|x|}{4} \right| \leq 1$$

$$\Rightarrow \left| \frac{2-|x|}{4} \right| \leq 1$$

$$-4 \leq 2-|x| \leq 4$$

$$-6 \leq -|x| \leq 2$$

$$-2 \leq |x| \leq 6$$

$$|x| \leq 6$$

$$\Rightarrow x \in [-6, 6] \dots\dots(1)$$

$$\text{Now, } 3-x \neq 1$$

$$\text{And } x \neq 2 \dots\dots(2)$$

$$\text{and } 3-x > 0$$

$$x < 3 \dots\dots(3)$$

$$\text{From (1), (2) and (3)}$$

$$\Rightarrow x \in [-6, 3) - \{2\}$$

$$\alpha = 6$$

$$\beta = 3$$

$$\gamma = 2$$

$$\alpha + \beta + \gamma = 11$$

Question40

Let $A = \{1, 2, 3, \dots, 7\}$ and let $P(A)$ denote the power set of A . If the number of functions $f : A \rightarrow P(A)$ such that $a \in f(a), \forall a \in A$ is m , $n \in \mathbb{N}$ and m is least, then $m + n$ is equal to _____

[30-Jan-2024 Shift 1]

Answer: 44

Solution:

$$f : A \rightarrow P(A)$$

$$a \in f(a)$$

That means 'a' will connect with subset which contain element 'a'.

Total options for 1 will be 2^6 . (Because 2^6 subsets contains 1)

Similarly, for every other element

$$\text{Hence, total is } 2^6 \times 2^6 \times 2^6 \times 2^6 \times 2^6 \times 2^6 \times 2^6 = 2^{42}$$

$$\text{Ans. } 2 + 42 = 44$$

Question 41

If the domain of the function $f(x) = \log_e \left(\frac{2x+3}{4x^2+x-3} \right) + \cos^{-1} \left(\frac{2x-1}{x+2} \right)$ is $(\alpha, \beta]$, then the value of $5\beta - 4\alpha$ is equal to

[30-Jan-2024 Shift 2]

Options:

A.

10

B.

12

C.

11

D.

9

Answer: B

Solution:

$$\frac{2x+3}{4x^2+x-3} > 0 \text{ and } -1 \leq \frac{2x-1}{x+2} \leq 1$$

$$\frac{2x+3}{(4x-3)(x+1)} > 0 \quad \frac{3x+1}{x+2} \geq 0 \text{ \& } \frac{x-3}{x+2} \leq 0$$

$$\begin{array}{ccccccc} & - & & + & & - & & + \\ & \bullet & & \bullet & & \bullet & & \bullet \\ -3/2 & & -1 & & 3/4 & & & \end{array}$$

$$(-\infty, -2) \cup \left[\frac{-1}{3}, \infty \right) \dots\dots (1)$$

$$(-2, 3] \dots\dots (2)$$

$$\left[\frac{-1}{3}, 3 \right] \dots\dots (3) \quad (1) \cap (2) \cap (3)$$

$$\left(\frac{3}{4}, 3 \right]$$

$$\alpha = \frac{3}{4}\beta = 3$$

$$5\beta - 4\alpha = 15 - 3 = 12$$

Question42

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined $f(x) = \frac{x}{(1+x^4)^{1/4}}$ and $g(x) = \int_0^{\sqrt{2\sqrt{5}}} x^2 g(x) dx$
 $f(f(f(f(x))))$ then 18

[30-Jan-2024 Shift 2]

Options:

A.

33

B.

36

C.

42

D.

39

Answer: D

Solution:

$$f(x) = \frac{x}{(1+x^4)^{1/4}}$$

$$f \circ f(x) = \frac{f(x)}{(1+f(x)^4)^{1/4}} = \frac{\frac{x}{(1+x^4)^{1/4}}}{\left(1 + \frac{x^4}{1+x^4}\right)^{1/4}} = \frac{x}{(1+2x^4)^{1/4}}$$

$$f(f(f(f(x)))) = \frac{x}{(1+4x^4)^{1/4}}$$

$$18 \int_0^{\sqrt[4]{2\sqrt{5}}} \frac{x^3}{(1+4x^4)^{1/4}} dx$$

$$\text{Let } 1+4x^4 = t^4$$

$$16x^3 dx = 4t^3 dt$$

$$\frac{18}{4} \int_1^3 \frac{t^3 dt}{t}$$

$$= \frac{9}{2} \left(\frac{t^3}{3} \right)_1^3$$

$$= \frac{3}{2} [26] = 39$$

Question43

If $f(x) = \frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$ and $(f \circ f)(x) = g(x)$, where $g : \mathbb{R} - \left\{ \frac{2}{3} \right\} \rightarrow \mathbb{R} - \left\{ \frac{2}{3} \right\}$, then $(g \circ g \circ g)(4)$ is equal to

[31-Jan-2024 Shift 1]

Options:

A.

$$-\frac{19}{20}$$

B.

$$19/20$$

C.

$$-4$$

D.

$$4$$

Answer: D

Solution:

$$f(x) = \frac{4x+3}{6x-4}$$

$$g(x) = \frac{4\left(\frac{4x+3}{6x-4}\right) + 3}{6\left(\frac{4x+3}{6x-4}\right) - 4} = \frac{34x}{34} = x$$

$$g(x) = x \therefore g(g(g(4))) = 4$$

Question44

If the function $f : (-\infty, -1] \rightarrow (a, b]$ defined by $f(x) = e^{x^3 - 3x + 1}$ is one-one and onto, then the distance of the point

$P(2b + 4, a + 2)$ from the line $x + e^{-3}y = 4$ is :

[31-Jan-2024 Shift 2]

Options:

A.

$$2\sqrt{1 + e^6}$$

B.

$$4\sqrt{1 + e^6}$$

C.

$$3\sqrt{1 + e^6}$$

D.

$$\sqrt{1 + e^6}$$

Answer: A

Solution:

$$f(x) = e^{x^3 - 3x + 1}$$

$$f'(x) = e^{x^3 - 3x + 1} \cdot (3x^2 - 3)$$

$$= e^{x^3 - 3x + 1} \cdot 3(x - 1)(x + 1)$$

$$\text{For } f'(x) \geq 0$$

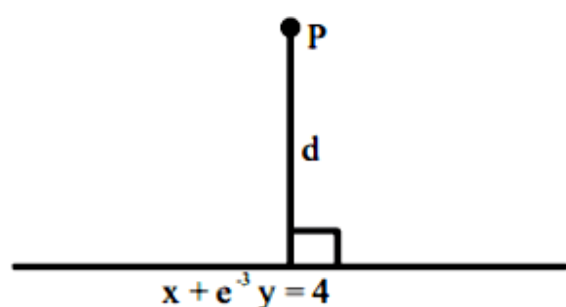
$\therefore f(x)$ is increasing function

$$\therefore a = e^{-\infty} = 0 = f(-\infty)$$

$$b = e^{-1+3+1} = e^3 = f(-1)$$

$$P(2b + 4, a + 2)$$

$$\therefore P(2e^3 + 4, 2)$$



$$d = \frac{(2e^3 + 4) + 2e^{-3} - 4}{\sqrt{1 + e^{-6}}} = 2\sqrt{1 + e^6}$$

Question45

Let $f : R \rightarrow R$ and $g : R \rightarrow R$ be defined as

$$f(x) = \begin{cases} \log_e x & , \quad x > 0 \\ e^{-x} & , \quad x \leq 0 \end{cases}.$$

and

$$g(x) = \begin{cases} x & , \quad x \geq 0 \\ e^x & , \quad x < 0 \end{cases}.$$

Then, $g \circ f : R \rightarrow R$ is :

[1-Feb-2024 Shift 1]

Options:

A.

one-one but not onto

B.

neither one-one nor onto

C.

onto but not one-one

D.

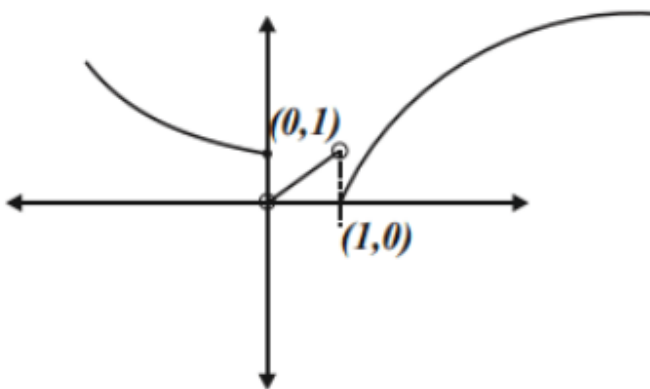
both one-one and onto

Answer: B

Solution:

$$g(f(x)) = \begin{cases} f(x) & f(x) \geq 0 \\ e^{f(x)} & f(x) < 0 \end{cases}.$$

$$g(f(x)) = \begin{cases} e^{-x} & (-\infty, 0] \\ e^{\ln x} & (0, 1) \\ \ln x & [1, \infty) \end{cases}.$$



Graph of $g(f(x))$

$g(f(x)) \Rightarrow$ Many one into

Question46

If the domain of the function $f(x) = \frac{\sqrt{x^2 - 25}}{(4 - x^2)} + \log_{10}(x^2 + 2x - 15)$ is $(-\infty, \alpha) \cup [\beta, \infty)$, then $\alpha^2 + \beta^3$ is equal to :

[1-Feb-2024 Shift 2]

Options:

A.

140

B.

175

C.

150

D.

125

Answer: C

Solution:

$$f(x) = \frac{\sqrt{x^2 - 25}}{4 - x^2} + \log_{10}(x^2 + 2x - 15)$$

$$\text{Domain : } x^2 - 25 \geq 0 \Rightarrow x \in (-\infty, -5] \cup [5, \infty)$$

$$4 - x^2 \neq 0 \Rightarrow x \neq \{-2, 2\}$$

$$x^2 + 2x - 15 > 0 \Rightarrow (x + 5)(x - 3) > 0$$

$$\Rightarrow x \in (-\infty, -5) \cup (3, \infty)$$

$$\therefore x \in (-\infty, -5) \cup [5, \infty)$$

$$\alpha = -5; \beta = 5$$

$$\therefore \alpha^2 + \beta^3 = 150$$

Question47

Let $A = \{1, 2, 3, 4, \dots, 10\}$ and $B = \{0, 1, 2, 3, 4\}$. The number of elements in the relation $R = \{(a, b) \in A \times A : 2(a - b)^2 + 3(a - b) \in B\}$ is _____.

[6-Apr-2023 shift 1]

Answer: 18

Solution:

$$A = \{1, 2, 3, \dots, 10\}$$

$$B = \{0, 1, 2, 3, 4\}$$

$$R = \{(a, b) \in A \times A : 2(a - b)^2 + 3(a - b) \in B\}$$

$$\text{Now } 2(a - b)^2 + 3(a - b) = (a - b)(2(a - b) + 3)$$

$$\Rightarrow a = b \text{ or } a - b = -2$$

$$\text{When } a = b \Rightarrow 10 \text{ order pairs}$$

$$\text{When } a - b = -2 \Rightarrow 8 \text{ order pairs}$$

$$\text{Total} = 18$$

Question48

Let $A = \{0, 3, 4, 6, 7, 8, 9, 10\}$ and R be the relation defined on A such that $R = \{(x, y) \in A \times A : x - y \text{ is odd positive integer or } x - y = 2\}$. The minimum number of elements that must be added to the relation R , so that it is a symmetric relation, is equal to _____.

[8-Apr-2023 shift 1]

Answer: 19

Solution:

$$A = \{0, 3, 4, 6, 7, 8, 9, 10\} \quad 3, 7, 9 \rightarrow \text{odd}$$

$$R = \{x - y = \text{odd} + \text{ve or } x - y = 2\} \quad 0, 4, 6, 8, 10 \rightarrow \text{even}$$

$${}^3C_1 \cdot {}^5C_1 = 15 + (6, 4), (8, 6), (10, 8), (9, 7)$$

Min^m ordered pairs to be added must be
:15 + 4 = 19

Question49

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$. Then the relation
 $R = \{(x, y) \in A \times A : x + y = 7\}$ is
[8-Apr-2023 shift 2]

Options:

- A. Symmetric but neither reflexive nor transitive
- B. Transitive but neither symmetric nor reflexive
- C. An equivalence relation
- D. Reflexive but neither symmetric nor transitive

Answer: A

Solution:

$$R = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

Question50

Let $A = \{2, 3, 4\}$ and $B = \{8, 9, 12\}$. Then the number of elements in
the relation $R = \{((a_1, b_1), (a_2, b_2)) \in (A \times B, A \times B) : a_1 \text{ divides } b_2 \text{ and } a_2 \text{ divides } b_1\}$ is
[10-Apr-2023 shift 2]

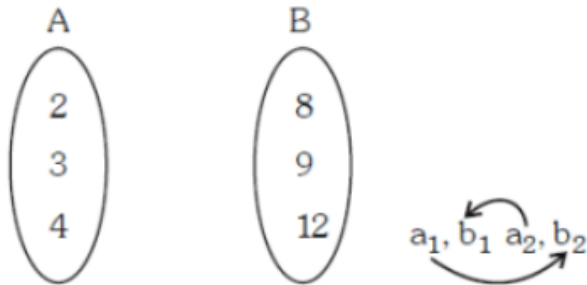
Options:

- A. 18
- B. 24
- C. 12

D. 36

Answer: D

Solution:



a_1 divides b_2

Each element has 2 choices

$$\Rightarrow 3 \times 2 = 6$$

a_2 divides b_1

Each element has 2 choices

$$\Rightarrow 3 \times 2 = 6$$

$$\text{Total} = 6 \times 6 = 36$$

Question 51

Let $A = \{1, 3, 4, 6, 9\}$ and $B = \{2, 4, 5, 8, 10\}$. Let R be a relation defined on $A \times B$ such that $R = \{((a_1, b_1), (a_2, b_2, \dots)): a_1 \leq b_2 \text{ and } b_1 \leq a_2\}$. Then the number of elements in the set R is

[11-Apr-2023 shift 2]

Options:

A. 52

B. 160

C. 26

D. 180

Answer: B

Solution:

Let $a_1 = 1 \Rightarrow 5$ choices of b_2

$a_1 = 3 \Rightarrow 4$ choices of b_2

$a_1 = 4 \Rightarrow 4$ choices of b_2

$a_1 = 6 \Rightarrow 2$ choices of b_2

$a_1 = 9 \Rightarrow 1$ choices of b_2

For (a_1, b_2) 16 ways .

Similarly, $b_1 = 2 \Rightarrow 4$ choices of a_2

$b_1 = 4 \Rightarrow 3$ choices of a_2

$b_1 = 5 \Rightarrow 2$ choices of a_2

$b_1 = 8 \Rightarrow 1$ choices of a_2

Required elements in $R = 160$

Question52

The number of the relations, on the set $\{1, 2, 3\}$ containing $(1, 2)$ and $(2, 3)$, which are reflexive and transitive but not symmetric, is

[12-Apr-2023 shift 1]

Answer: 3

Solution:

$A = \{1, 2, 3\}$

For Reflexive $(1, 1)(2, 2), (3, 3) \in R$

For transitive : $(1, 2)$ and $(2, 3) \in R \Rightarrow (1, 3) \in R$

Not symmetric : $(2, 1)$ and $(3, 2) \notin R$

$R_1 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$

$R_2 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)(2, 1)\}$

$R_3 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)(2, 1)\}$

Question53

Let $A = \{-4, -3, -2, 0, 1, 3, 4\}$ and $R = \{(a, b) \in A \times A : b = |a| \text{ or } b^2 = a + 1\}$ be a relation on A . Then the minimum number of elements, that must be added to the relation R so that it becomes reflexive and

symmetric, is _____
[13-Apr-2023 shift 2]

Answer: 7

Solution:

$R = [(-4, 4), (-3, 3), (3, -2), (0, 1), (0, 0), (1, 1), (4, 4), (3, 3)]$
For reflexive, add $\Rightarrow (-2, -2), (-4, -4), (-3, -3)$
For symmetric, add $\Rightarrow (4, -4), (3, -3), (-2, 3), (1, 0)$

Question54

Let $A = \{1, 2, 3, 4\}$ and R be a relation on the set $A \times A$ defined by $R = \{(a, b, (c, d) : 2a + 3b = 4c + 5d\}$. Then the number of elements in R is _____
[15-Apr-2023 shift 1]

Answer: 6

Solution:

$A = \{1, 2, 3, 4\}$
 $R = \{(a, b), (c, d)\}$
 $2a + 3b = 4c + 5d = \alpha$ let
 $2a = \{2, 4, 6, 8\}$ $4c = \{4, 8, 12, 16\}$
 $3b = \{3, 6, 9, 12\}$ $5d = \{5, 10, 15, 20\}$

$$2a + 3b = \left\{ \begin{array}{cccc} 5 & 8 & 11 & 14 \\ 7 & 10 & 13 & 16 \\ 9 & 12 & 15 & 18 \\ 11 & 14 & 17 & 20 \end{array} \right\} \quad 4c + 5d = \left\{ \begin{array}{cccc} 9 & 14 & 19 & 24 \\ 13 & 18 & & \\ 17 & 22 & & \\ 21 & 26 & & \end{array} \right\}$$

Possible value of $\alpha = 9, 13, 14, 14, 17, 18$
Pairs of $\{(a, b), (c, d)\} = 6$

Question55

Let $5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3, x > 0$. Then $18 \int_1^2 f(x) dx$ is equal to :
[6-Apr-2023 shift 1]

Options:

A. $10\log_e 2 - 6$

B. $10\log_e 2 + 6$

C. $5\log_e 2 - 3$

D. $5\log_e 2 + 3$

Answer: A

Solution:

$$5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3 \dots (1)$$

$$x \rightarrow \frac{1}{x}$$

$$5f\left(\frac{1}{x}\right) + 4f(x) = x + 3 \dots (2)$$

$$(1) \times 5 - (2) \times 4$$

$$\Rightarrow f(x) = \frac{5}{9x} - \frac{4}{9}x + \frac{1}{3}$$

$$\begin{aligned} \Rightarrow 18 \int_1^2 f(x) dx &= 18 \left(\frac{5}{9} \ln 2 - \frac{4}{9} \times \frac{3}{2} + \frac{1}{3} \right) \\ &= 10 \ln 2 - 6 \end{aligned}$$

Question56

Let $A = \{x \in \mathbb{R} : [x+3] + [x+4] \leq 3\}$,

$B = \left\{ x \in \mathbb{R} : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^x} \right)^{x-3} < 3^{-3x} \right\}$, where $[t]$ denotes greatest integer function. Then,
[6-Apr-2023 shift 1]

Options:

A. $A \subset B, A \neq B$

B. $A \cap B = \varnothing$

C. $A = B$

D. $B \subset C, A \neq B$

Answer: C

Solution:

$$A = \{x \in \mathbb{R} : [x+3] + [x+4] \leq 3\}$$

$$2[x] + 7 \leq 3$$

$$2[x] \leq -4$$

$$[x] \leq -2 \Rightarrow x < -1 \dots (A)$$

$$B = \left\{ x \in \mathbb{R} : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^x} \right)^{x-3} < 3^{-3x} \right\}$$

$$3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^x} \right)^{x-3} < 3^{-3x}$$

$$3^{2x-3} \left(\frac{10}{10} \right)^{x-3}$$

$$\Rightarrow \left(\frac{1}{9} \right)^{x-3} < 3^{-5x}$$

$$\Rightarrow 3^{6-2x} < 3^{3-5x}$$

$$\Rightarrow 6-2x < 3-5x$$

$$\Rightarrow 3 < -3x$$

$$\Rightarrow \left(\frac{1}{10} \right)^x < -1 \dots (B)$$

$$A = B$$

Question 57

Let the sets A and B denote the domain and range respectively of the function $f(x) = \frac{1}{\sqrt{[x]-x}}$, where $[x]$ denotes the smallest integer greater than or equal to x . Then among the statements :

(S1) : $A \cap B = (1, \infty) - \mathbb{N}$ and

(S2) : $A \cup B = (1, \infty)$

[6-Apr-2023 shift 2]

Options:

- A. only (S1) is true
- B. neither (S1) nor (S2) is true
- C. only (S2) is true
- D. both (S1) and (S2) are true

Answer: A

Solution:

$$f(x) = \frac{1}{\sqrt{[x] - x}}$$

If $x \in I$ $[x] = [x]$ (greatest integer function)

If $x \notin I$ $[x] = [x] + 1$

$$\Rightarrow f(x) = \begin{cases} \frac{1}{\sqrt{[x] - x}} & x \in I \\ \frac{1}{\sqrt{[x] + 1 - x}} & x \notin I \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} \frac{1}{\sqrt{-\{x\}}} & x \in I \quad (\text{does not exist}) \\ \frac{1}{\sqrt{1 - \{x\}}} & x \notin I \end{cases}$$

$\Rightarrow$ domain of $f(x) = \mathbb{R} - I$

Now, $f(x) = \frac{1}{\sqrt{1 - \{x\}}}$, $x \notin I$

$\Rightarrow x < \{x\} < 1$

$\Rightarrow 0 < 1 - \{x\} < 1$

$\Rightarrow \frac{1}{\sqrt{1 - \{x\}}} > 1$

$\Rightarrow \text{Range}(1, \infty)$

$\Rightarrow A = \mathbb{R} - I$

$B = (1, \infty)$

So, $A \cap B = (1, \infty) - \mathbb{N}$

$A \cup B \neq (1, \infty)$

$\Rightarrow S1$ is only correct.

Question 58

Let $f, g : \mathbb{N} - \{1\} \rightarrow \mathbb{N}$ be functions defined by $f(a) = \alpha$, where α is the maximum of the powers of those primes p such that p^α divides a , and

$g(a) = a + 1$, for all $a \in \mathbb{N} - \{1\}$. Then, the function $f + g$ is
[27-Jul-2022-Shift-1]

Options:

- A. one-one but not onto
- B. onto but not one-one
- C. both one-one and onto
- D. neither one-one nor onto

Answer: D

Solution:

$f, g : \mathbb{N} - \{1\} \rightarrow \mathbb{N}$ defined as

$f(a) = \alpha$, where α is the maximum power of those primes p such that p^α divides a .

$g(a) = a + 1$

Now,

$$f(2) = 1, \quad g(2) = 3 \Rightarrow (f + g)(2) = 4$$

$$f(3) = 1, \quad g(3) = 4 \Rightarrow (f + g)(3) = 5$$

$$f(4) = 2, \quad g(4) = 5 \Rightarrow (f + g)(4) = 7$$

$$f(5) = 1, \quad g(5) = 6 \Rightarrow (f + g)(5) = 7$$

$$\therefore (f + g)(5) = (f + g)(4)$$

$\therefore f + g$ is not one-one

Now, $\because f_{\min} = 1, g_{\min} = 3$

So, there does not exist any $x \in \mathbb{N} - \{1\}$ such that $(f + g)(x) = 1, 2, 3$

$\therefore f + g$ is not onto

Question59

If domain of the function $\log_e \left(\frac{6x^2 + 5x + 1}{2x - 1} \right) + \cos^{-1} \left(\frac{2x^2 - 3x + 4}{3x - 5} \right)$ is

$(\alpha, \beta) \cup (\gamma, \delta]$, then, $18(\alpha^2 + \beta^2 + \gamma^2 + \delta^2)$ is equal to
[8-Apr-2023 shift 2]

Answer: 20

Solution:

$$\frac{6x^2 + 5x + 1}{2x - 1} > 0$$

$$\frac{(3x + 1)(2x + 1)}{2x - 1} > 0$$

$$x \in \left[\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right] \cup \left(\frac{5}{3}, \infty \right) \dots (B)$$

$$x < \frac{5}{3} \dots (C)$$

$$A \cap B \cap C \equiv \left(\frac{-1}{2}, \frac{-1}{3} \right) \cup \left(\frac{1}{2}, \frac{1}{\sqrt{2}} \right]$$

$$\text{So } 18(\alpha^2 + \beta^2 + \gamma^2 + \delta^2) = 18 \left(\frac{1}{4} + \frac{1}{9} + \frac{1}{4} + \frac{1}{2} \right)$$

$$= 18 + 2 = 20$$

Question60

Let $R = \{a, b, c, d, e\}$ and $S = \{1, 2, 3, 4\}$. Total number of onto functions $f : R \rightarrow S$ such that $f(a) \neq 1$, is equal to _____.

[8-Apr-2023 shift 2]

Answer: 180

Solution:

Total onto function

$$\frac{{}^4P_5}{{}^4P_3} \times {}^4P_4 = 240$$

Now when $f(a) = 1$

$$\left[4 + \frac{{}^4P_4}{{}^4P_2} \times {}^4P_3 = 24 + 36 = 60. \right]$$

$$\text{so required } f^n = 240 - 60 = 180$$

Question61

If the domain of the function $f(x) = \sec^{-1}\left(\frac{2x}{5x+3}\right)$ is $[\alpha, \beta) \cup (\gamma, \delta]$, then $|3\alpha + 10(\beta + \gamma) + 21\delta|$ is equal to _____.
 [10-Apr-2023 shift 2]

Answer: 24

Solution:

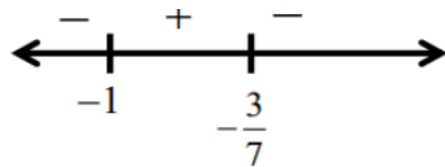
$$f(x) = \sec^{-1} \frac{2x}{5x+3}$$

$$\left| \frac{2x}{5x+3} \right|$$

$$\left| \frac{2x}{5x+3} \right| \geq 1 \Rightarrow |2x| \geq |5x+3|$$

$$(2x)^2 - (5x+3)^2 \geq 0$$

$$(7x+3)(-3x-3) \geq 0$$



$$\therefore \text{domain } \left[-1, \frac{-3}{5}\right) \cup \left(\frac{-3}{5}, \frac{-3}{7}\right]$$

$$\alpha = -1, \beta = \frac{-3}{5}, \gamma = \frac{-3}{5}, \delta = \frac{-3}{7}$$

$$3\alpha + 10(\beta + \gamma) + 21\delta = -3$$

$$-3 + 10\left(\frac{-6}{5}\right) + \left(\frac{-3}{7}\right)21 = -24$$

Question62

The domain of the function $f(x) = \frac{1}{\sqrt{[x]^2 - 3[x] - 10}}$ is (where $[x]$ denotes the greatest integer less than or equal to x)
 [11-Apr-2023 shift 2]

Options:

A. $(-\infty, -3] \cup [6, \infty)$

B. $(-\infty, -2) \cup (5, \infty)$

C. $(-\infty, -3] \cup (5, \infty)$

D. $(-\infty, -2) \cup [6, \infty)$

Answer: D

Solution:

$$F(x) = \frac{1}{\sqrt{[x]^2 - 3[x] - 10}}$$

$$[x]^2 - 3[x] - 10 > 0$$

$$([x] + 2)([x] - 5) > 0$$



$$[x] < -2 \text{ or } [x] > 5$$

$$[x] \leq -3 \text{ or } [x] \geq 6$$

$$x < -2 \text{ or } x \geq 6$$

$$x \in (-\infty, -2) \cup [6, \infty)$$

Question63

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3, 4, 5, 6\}$. Then the number of functions $f : A \rightarrow B$ satisfying $f(1) + f(2) = f(4) - 1$ is equal to

 .
[11-Apr-2023 shift 2]

Answer: 360

Solution:

$$f(1) + f(2) + 1 = f(4) \leq 6$$

$$f(1) + f(2) \leq 5$$

Case (i) $f(1) = 1 \Rightarrow f(2) = 1, 2, 3, 4 \Rightarrow 4$ mappings

Case (ii) $f(1) = 2 \Rightarrow f(2) = 1, 2, 3 \Rightarrow 3$ mappings

Case (iii) $f(1) = 3 \Rightarrow f(2) = 1, 2 \Rightarrow 2$ mappings

Case (iv) $f(1) = 4 \Rightarrow f(2) = 1 \Rightarrow 1$ mapping

$f(5)$ & $f(6)$ both have 6 mappings each
 Number of functions $= (4 + 3 + 2 + 1) \times 6 \times 6 = 360$

Question 64

Let D be the domain of the function $f(x) = \sin^{-1} \left(\log_{3x} \left(\frac{6 + 2\log_3 x}{-5x} \right) \right)$. If the range of the function $g : D \rightarrow \mathbb{R}$ defined by $g(x) = x - [x]$, ($[x]$ is the greatest integer function), is (α, β) , then $\alpha^2 + \frac{5}{\beta}$ is equal to

[12-Apr-2023 shift 1]

Options:

- A. 46
- B. 135
- C. 136
- D. 45

Answer: B

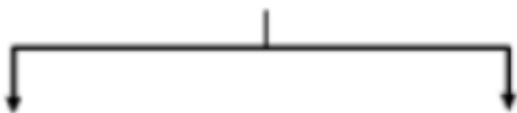
Solution:

$$\frac{6 + 2\log_3 x}{-5x} > 0 \text{ \& } x > 0 \text{ \& } x \neq \frac{1}{3}$$

$$\text{this gives } x \in \left(0, \frac{1}{27} \right) \dots (1)$$

$$-1 \leq \log_{3x} \left(\frac{6 + 2\log_3 x}{-5x} \right) \leq 1$$

$$3x \leq \frac{6 + 2\log_3 x}{-5x} \leq \frac{1}{3x}$$



$$15x^2 + 6 + 2\log_3 x \geq 0 \quad 6 + 2\log_3 x + \frac{5}{3} \geq 0$$

$$x \in \left(0, \frac{1}{27} \right) \dots (2) \quad x \geq 3^{-\frac{23}{6}} \dots (3)$$

form (1), (2) & (3)

$$x \in \left[3^{-\frac{23}{6}}, \frac{1}{27} \right)$$

$\therefore \alpha$ is small positive quantity

$$\&\beta = \frac{1}{27}$$

$\therefore \alpha^2 + \frac{5}{\beta}$ is just greater than 135

Question65

For $x \in \mathbb{R}$, two real valued functions $f(x)$ and $g(x)$ are such that, $g(x) = \sqrt{x} + 1$ and $f \circ g(x) = x + 3 - \sqrt{x}$. Then $f(0)$ is equal to
[13-Apr-2023 shift 1]

Options:

A. 5

B. 0

C. -3

D. 1

Answer: A

Solution:

$$\begin{aligned} g(x) &= \sqrt{x} + 1 \\ f \circ g(x) &= x + 3 - \sqrt{x} \\ &= (\sqrt{x} + 1)^2 - 3(\sqrt{x} + 1) + 5 \\ &= g^2(x) - 3g(x) + 5 \\ \Rightarrow f(x) &= x^2 - 3x + 5 \\ \therefore f(0) &= 5 \end{aligned}$$

But, if we consider the domain of the composite function $f \circ g(x)$ then in that case $f(0)$ will be not defined as $g(x)$ cannot be equal to zero.

Question66

For the differentiable function $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}$, let

$$3f(x) + 2f\left(\frac{1}{x}\right) = \frac{1}{x} - 10, \text{ then } \left| f(3) + f'\left(\frac{1}{4}\right) \right| \text{ is equal to}$$

[13-Apr-2023 shift 1]

Options:

A. 13

B. $\frac{29}{5}$

C. $\frac{33}{5}$

D. 7

Answer: A

Solution:

$$\left[3f(x) + 2f\left(\frac{1}{x}\right) = \frac{1}{x} - 10 \right] \times 3$$

$$\left[2f(x) + 3f\left(\frac{1}{x}\right) = x - 10 \right] \times 2$$

$$5f(x) = \frac{3}{x} - 2x - 10$$

$$f(x) = \frac{1}{5} \left(\frac{3}{x} - 2x - 10 \right)$$

$$f'(x) = \frac{1}{5} \left(-\frac{3}{x^2} - 2 \right)$$

$$\begin{aligned} \left| f(3) + f'\left(\frac{1}{4}\right) \right| &= \left| \frac{1}{5}(1 - 6 - 10) + \frac{1}{5}(-48 - 2) \right| \\ &= |-3 - 10| = 13 \end{aligned}$$

Question67

The range of $f(x) = 4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is

[13-Apr-2023 shift 2]

Options:

A. $[0, \pi)$

B. $[0, \pi]$

C. $[0, 2\pi)$

D. $[0, 2\pi]$

Answer: C

Solution:

$$f(x) = 4\sin^{-1}\left(\frac{x^2}{1+x^2}\right)$$

$$0 \leq \frac{x^2}{1+x^2} < 1$$

$$\Rightarrow 0 \leq \sin^{-1}\left(\frac{x^2}{1+x^2}\right) < \frac{\pi}{2}$$

$$\Rightarrow 0 \leq 4\sin^{-1}\left(\frac{x^2}{1+x^2}\right) < 2\pi$$

Range : $[0, 2\pi)$

Question68

If the domain of the function

$f(x) = \log_e(4x^2 + 11x + 6) + \sin^{-1}(4x + 3) + \cos^{-1}\left(\frac{10x+6}{3}\right)$ is $(\alpha, \beta]$, then

$36 | \alpha + \beta$ is equal to

[15-Apr-2023 shift 1]

Options:

A. 72

B. 63

C. 45

D. 54

Answer: C

Solution:

$$f(x) = \ln(4x^2 + 11x + 6) + \sin^{-1}(4x + 3) + \cos^{-1}\left(\frac{10x+6}{3}\right)$$

$$(i) \quad 4x^2 + 11x + 6 > 0$$

$$4x^2 + 8x + 3x + 6 > 0$$

$$(4x + 3)(x + 2) > 0$$

$$x \in (-\infty, -2) \cup \left(-\frac{3}{4}, \infty\right)$$

$$(ii) 4x + 3 \in [-1, 1]$$

$$x \in [-1, -1/2]$$

$$(iii) \frac{10x+6}{3} \in [-1, 1]$$

$$x \in \left[-\frac{9}{10}, -\frac{3}{10}\right]$$

$$x \in \left(-\frac{3}{4}, -\frac{1}{2}\right] \quad \alpha = -\frac{3}{4}, \beta = -\frac{1}{2}$$

$$\alpha + \beta = -\frac{5}{4}$$

$$36 \mid \alpha + \beta \mid = 45$$

Question69

The relation $R = \{(a, b) : \gcd(a, b) = 1, 2a \neq b, a, b \in \mathbb{Z}\}$ is:
[24-Jan-2023 Shift 1]

Options:

- A. transitive but not reflexive
- B. symmetric but not transitive
- C. reflexive but not symmetric
- D. neither symmetric nor transitive

Answer: D

Solution:

Reflexive : $(a, a) \quad \gcd(a, a) = a$

Which is not true for every $a \in \mathbb{Z}$.

Symmetric:

Take $a = 2, b = 1 \quad \gcd(2, 1) = 1$

Also $2a = 4 \neq b$

Now when $a = 1, b = 2 \quad \gcd(1, 2) = 1$

Also now $2a = 2 = b$

Hence $a = 2b$

$\Rightarrow R$ is not Symmetric

Transitive:

Let $a = 14, b = 19, c = 21$

$\gcd(a, b) = 1$

$\gcd(b, c) = 1$

$\gcd(a, c) = 7$

Hence not transitive

R is neither symmetric nor transitive.

Question70

Let R be a relation defined on $\mathbb{N}$ as $a R b$ is $2a + 3b$ is a multiple of 5, $a, b \in \mathbb{N}$. Then R is
[29-Jan-2023 Shift 2]

Options:

- A. not reflexive
- B. transitive but not symmetric
- C. symmetric but not transitive
- D. an equivalence relation

Answer: D

Solution:

$a R a \Rightarrow 5a$ is multiple of 5

So reflexive

$a R b \Rightarrow 2a + 3b = 5\alpha$,

Now $b R a$

$$\begin{aligned} 2b + 3a &= 2b + \left(\frac{5\alpha - 3b}{2} \right) \cdot 3 \\ &= \frac{15}{2}\alpha - \frac{5}{2}b = \frac{5}{2}(3\alpha - b) \\ &= \frac{5}{2}(2a + 2b - 2\alpha) \\ &= 5(a + b - \alpha) \end{aligned}$$

Hence symmetric

$a R b \Rightarrow 2a + 3b = 5\alpha$.

$b R c \Rightarrow 2b + 3c = 5\beta$

Now $2a + 5b + 3c = 5(\alpha + \beta)$

$\Rightarrow 2a + 5b + 3c = 5(\alpha + \beta)$

$\Rightarrow 2a + 3c = 5(\alpha + \beta - b)$

$\Rightarrow a R c$

Hence relation is equivalence relation.

Question71

The minimum number of elements that must be added to the relation $R = \{(a, b), (b, c)\}$ on the set $\{a, b, c\}$ so that it becomes symmetric and transitive is:

[30-Jan-2023 Shift 1]

Options:

- A. 4
- B. 7
- C. 5
- D. 3

Answer: B

Solution:

For Symmetric $(a, b), (b, c) \in R$

$\Rightarrow (b, a), (c, b) \in R$

For Transitive $(a, b), (b, c) \in R$

$\Rightarrow (a, c) \in R$

Now

1. Symmetric

$\therefore (a, c) \in R \Rightarrow (c, a) \in R$

2. Transitive

$\therefore (a, b), (b, a) \in R$

$\Rightarrow (a, a) \in R \& (b, c), (c, b) \in R$

$\Rightarrow (b, b) \& (c, c) \in R$

$\therefore$ Elements to be added

$$\left\{ \begin{array}{l} (b, a) \quad (c, b) \quad (a, c) \quad (c, a) \\ , (a, a) \quad (b, b) \quad (c, c) \end{array} \right\}$$

Number of elements to be added = 7

Question72

Let R be a relation on $N \times N$ defined by $(a, b)R(c, d)$ if and only if $ad(b - c) = bc(a - d)$. Then R is

[31-Jan-2023 Shift 1]

Options:

- A. symmetric but neither reflexive nor transitive

- B. transitive but neither reflexive nor symmetric
- C. reflexive and symmetric but not transitive
- D. symmetric and transitive but not reflexive

Answer: A

Solution:

$$(a, b)R(c, d) \Rightarrow ad(b - c) = bc(a - d)$$

Symmetric:

$$(c, d)R(a, b) \Rightarrow cb(d - a) = da(c - b) \Rightarrow$$

Symmetric

Reflexive:

$$(a, b)R(a, b) \Rightarrow ab(b - a) \neq ba(a - b) \Rightarrow$$

Not reflexive

Transitive: $(2, 3)R(3, 2)$ and $(3, 2)R(5, 30)$ but

$((2, 3), (5, 30)) \notin R \Rightarrow$ Not transitive

Question 73

Among the relations

$$S = \left\{ (a, b) : a, b \in \mathbb{R} - \{0\}, 2 + \frac{a}{b} > 0 \right\}$$

**And $T = \{(a, b) : a, b \in \mathbb{R}, a^2 - b^2 \in \mathbb{Z}\}$,
[31-Jan-2023 Shift 2]**

Options:

- A. S is transitive but T is not
- B. T is symmetric but S is not
- C. Neither S nor T is transitive
- D. Both S and T are symmetric

Answer: B

Solution:

For relation $T = a^2 - b^2 = -I$

Then, (b, a) on relation R

$$\Rightarrow b^2 - a^2 = -1$$

$\therefore T$ is symmetric

$$S = \left\{ (a, b) : a, b \in \mathbb{R} - \{0\}, 2 + \frac{a}{b} > 0 \right\}$$

$$2 + \frac{a}{b} > 0 \Rightarrow \frac{a}{b} > -2, \Rightarrow \frac{b}{a} < \frac{-1}{2}$$

If $(b, a) \in S$ then

$$2 + \frac{b}{a} \text{ not necessarily positive}$$

$\therefore S$ is not symmetric

Question 74

Let R be a relation on $\mathbb{R}$, given by $R = \{(a, b) : 3a - 3b + \sqrt{7} \text{ is an irrational number}\}$. Then R is
[1-Feb-2023 Shift 1]

Options:

- A. Reflexive but neither symmetric nor transitive
- B. Reflexive and transitive but not symmetric
- C. Reflexive and symmetric but not transitive
- D. An equivalence relation

Answer: A

Solution:

Check for reflexivity:

As $3(a-a) + \sqrt{7} = \sqrt{7}$ which belongs to relation so relation is reflexive

Check for symmetric:

$$\text{Take } a = \frac{\sqrt{7}}{3}, b = 0$$

Now $(a, b) \in R$ but $(b, a) \notin R$

As $3(b-a) + \sqrt{7} = 0$ which is rational so relation is not symmetric.

Check for Transitivity:

$$\text{Take } (a, b) \text{ as } \left(\frac{\sqrt{7}}{3}, 1 \right)$$

$$\&(b, c) \text{ as } \left(1, \frac{2\sqrt{7}}{3} \right)$$

So now $(a, b) \in R$ & $(b, c) \in R$ but $(a, c) \notin R$ which means relation is not transitive

Question 75

Let $P(S)$ denote the power set of $S = \{1, 2, 3, \dots, 10\}$. Define the relations R_1 and R_2 on $P(S)$ as $A R_1 B$ if $(A \cap B^c) \cup (B \cap A^c) = \emptyset$ and $A R_2 B$ if $A \cup B^c = B \cup A^c, \forall A, B \in P(S)$. Then :

[1-Feb-2023 Shift 2]

Options:

- A. both R_1 and R_2 are equivalence relations
- B. only R_1 is an equivalence relation
- C. only R_2 is an equivalence relation
- D. both R_1 and R_2 are not equivalence relations

Answer: A

Solution:

$$S = \{1, 2, 3, \dots, 10\}$$

$P(S)$ = power set of S

$$A R_1 B \Rightarrow (A \cap \overrightarrow{B}) \cup (\overrightarrow{A} \cap B) = \emptyset$$

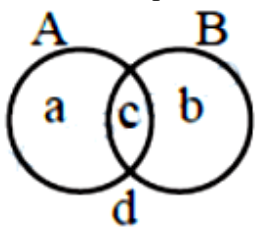
R_1 is reflexive, symmetric

For transitive

$$(A \cap \overrightarrow{B}) \cup (\overrightarrow{A} \cap B) = \emptyset; \{a\} = \emptyset = \{b\} \Rightarrow A = B$$

$$(B \cap \overrightarrow{C}) \cup (\overrightarrow{B} \cap C) = \emptyset \therefore B = C$$

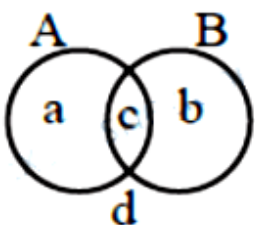
$\therefore A = C$ equivalence.



$$R_2 \equiv A \cup \overrightarrow{B} = \overrightarrow{A} \cup B$$

$R_2 \rightarrow$ Reflexive, symmetric

for transitive



$$\begin{aligned}
A \cup \overrightarrow{B} &= \overrightarrow{A} \cup B \Rightarrow \{a, c, d\} = \{b, c, d\} \\
\{a\} &= \{b\} \therefore A = B \\
B \cup \overrightarrow{C} &= \overrightarrow{B} \cup C \Rightarrow B = C \\
\therefore A &= C \quad \therefore A \cup \overrightarrow{C} = \overrightarrow{A} \cup C \therefore \text{Equivalence}
\end{aligned}$$

Question76

The equation $x^2 - 4x + [x] + 3 = x[x]$, where $[x]$ denotes the greatest integer function, has:
[24-Jan-2023 Shift 1]

Options:

- A. exactly two solutions in $(-\infty, \infty)$
- B. no solution
- C. a unique solution in $(-\infty, 1)$
- D. a unique solution in $(-\infty, \infty)$

Answer: D

Solution:

$$\begin{aligned}
x^2 - 4x + [x] + 3 &= x[x] \\
\Rightarrow x^2 - 4x + 3 &= x[x] - [x] \\
(x-1)(x-3) &= [x] \cdot (x-1) \\
\Rightarrow x = 1 \text{ or } x-3 &= [x] \\
\Rightarrow x - [x] &= 3 \\
\{x\} &= 3 \text{ (Not Possible)} \\
\text{Only one solution } x &= 1 \text{ in } (-\infty, \infty)
\end{aligned}$$

Question77

Let $f(x)$ be a function such that $f(x+y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{N}$. If $f(1) = 3$ and $\sum_{k=1}^n f(k) = 3279$, then the value of n is
[24-Jan-2023 Shift 2]

Options:

A. 6

B. 8

C. 7

D. 9

Answer: C

Solution:

$$f(x+y) = f(x) \cdot f(y) \quad \forall x, y \in \mathbb{N}, f(1) = 3$$

$$f(2) = f^2(1) = 3^2$$

$$f(3) = f(1)f(2) = 3^3$$

$$f(4) = 3^4$$

$$f(k) = 3^k$$

$$\sum_{k=1}^n f(k) = 3279$$

$$f(1) + f(2) + f(3) + \dots + f(k) = 3279$$

$$3 + 3^2 + 3^3 + \dots + 3^k = 3279$$

$$\frac{3(3^k - 1)}{3 - 1} = 3279$$

$$\frac{3^k - 1}{2} = 1093$$

$$3^k - 1 = 2186$$

$$3^k = 2187$$

$$k = 7$$

Question78

If $f(x) = \frac{2^{2x}}{2^{2x} + 2}$, $x \in \mathbb{R}$ then $f\left(\frac{1}{2023}\right) + f\left(\frac{2}{2023}\right) + \dots + f\left(\frac{2022}{2023}\right)$ equal to

[24-Jan-2023 Shift 2]

Options:

A. 2011

B. 1010

C. 2010

D. 1011

Answer: D

Solution:

$$f(x) = \frac{4^x}{4^x + 2}$$

$$f(x) + f(1-x) = \frac{4^x}{4^x + 2} + \frac{4^{1-x}}{4^{1-x} + 2}$$

$$= \frac{4^x}{4^x + 2} + \frac{4}{4 + 2(4^x)}$$

$$= \frac{4^x}{4^x + 2} + \frac{2}{2 + 4^x}$$

$$= 1$$

$$\Rightarrow f(x) + f(1-x) = 1$$

$$\text{Now } f\left(\frac{1}{2023}\right) + f\left(\frac{2}{2023}\right) + f\left(\frac{3}{2023}\right) + \dots +$$

$$\dots + f\left(1 - \frac{3}{2023}\right) + f\left(1 - \frac{2}{2023}\right) + f\left(1 - \frac{1}{2023}\right)$$

Now sum of terms equidistant from beginning and end is 1

$$\text{Sum} = 1 + 1 + 1 + \dots + 1 \text{ (1011 times)}$$

$$= 1011$$

Question 79

For some $a, b, c \in \mathbb{N}$, let $f(x) = ax - 3$ and $g(x) = x^b + c$, $x \in \mathbb{R}$. If

$(f \circ g)^{-1}(x) = \left(\frac{x-7}{2}\right)^{1/3}$ then $(f \circ g)(ac) + (g \circ f)(b)$ is equal to

[25-Jan-2023 Shift 1]

Answer: 2039

Solution:

$$\text{Let } f \circ g(x) = h(x)$$

$$\Rightarrow h^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$$

$$\Rightarrow h(x) = f \circ g(x) = 2x^3 + 7$$

$$f \circ g(x) = a(x^b + c) - 3$$

$$\begin{aligned} \Rightarrow a &= 2, b = 3, c = 5 \\ \Rightarrow \text{fog}(ac) &= \text{fog}(10) = 2007 \\ g(f(x)) &= (2x - 3)^3 + 5. \\ \Rightarrow \text{gof}(b) &= \text{gof}(3) = 32 \\ \Rightarrow \text{sum} &= 2039 \end{aligned}$$

Question80

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \log_{\sqrt{m}}\{\sqrt{2}(\sin x - \cos x) + m - 2\}$, for some m , such that the range of f is $[0, 2]$. Then the value of m is _____
[25-Jan-2023 Shift 2]

Options:

- A. 5
- B. 3
- C. 2
- D. 4

Answer: A

Solution:

Since,

$$-\sqrt{2} \leq \sin x - \cos x \leq \sqrt{2}$$

$$\therefore -2 \leq \sqrt{2}(\sin x - \cos x) \leq 2$$

(Assume $\sqrt{2}(\sin x - \cos x) = k$)

$$-2 \leq k \leq 2 \quad \dots (i)$$

$$f(x) = \log_{\sqrt{m}}(k + m - 2)$$

Given,

$$0 \leq f(x) \leq 2$$

$$0 \leq \log_{\sqrt{m}}(k + m - 2) \leq 2$$

$$1 \leq k + m - 2 \leq m$$

$$-m + 3 \leq k \leq 2 \dots \dots (ii)$$

From eq. (i) & (ii), we get $-m + 3 = -2$
 $\Rightarrow m = 5$

Question81

The number of functions $f : \{1, 2, 3, 4\} \rightarrow \{a \in \mathbb{Z} : |a| \leq 8\}$ satisfying $f(n) + \frac{1}{n}f(n+1) = 1, \forall n \in \{1, 2, 3\}$ is
[25-Jan-2023 Shift 2]

Options:

- A. 3
- B. 4
- C. 1
- D. 2

Answer: D

Solution:

$f : \{1, 2, 3, 4\} \rightarrow \{a \in \mathbb{Z} : |a| \leq 8\}$
 $f(n) + \frac{1}{n}f(n+1) = 1, \forall n \in \{1, 2, 3\}$
 $f(n+1)$ must be divisible by n
 $f(4) \Rightarrow -6, -3, 0, 3, 6$
 $f(3) \Rightarrow -8, -6, -4, -2, 0, 2, 4, 6, 8$
 $f(2) \Rightarrow -8, \dots, 8$
 $f(1) \Rightarrow -8, \dots, 8$
 $\frac{f(4)}{3}$ must be odd since $f(3)$ should be even therefore 2 solution possible.

Question82

Let $f(x) = 2x^n + \lambda, \lambda \in \mathbb{R}, n \in \mathbb{N}$, and $f(4) = 133, f(5) = 255$. Then the sum of all the positive integer divisors of $(f(3) - f(2))$ is
[25-Jan-2023 Shift 2]

Options:

- A. 61
- B. 60
- C. 58
- D. 59

Answer: B

Solution:

$$f(x) = 2x^n + \lambda$$

$$f(4) = 133$$

$$f(5) = 255$$

$$133 = 2 \times 4^n + \lambda \dots (1)$$

$$255 = 2 \times 5^n + \lambda \dots (2)$$

$$(2) - (1)$$

$$122 = 2(5^n - 4^n)$$

$$\Rightarrow 5^n - 4^n = 61$$

$$\therefore n = 3 \text{ \& } \lambda = 5$$

$$\text{Now, } f(3) - f(2) = 2(3^3 - 2^3) = 38$$

Number of Divisors is 1, 2, 19, 38; & their sum is 60

Question83

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \frac{x^2 + 2x + 1}{x^2 + 1}$. Then

[29-Jan-2023 Shift 1]

Options:

A. $f(x)$ is many-one in $(-\infty, -1)$

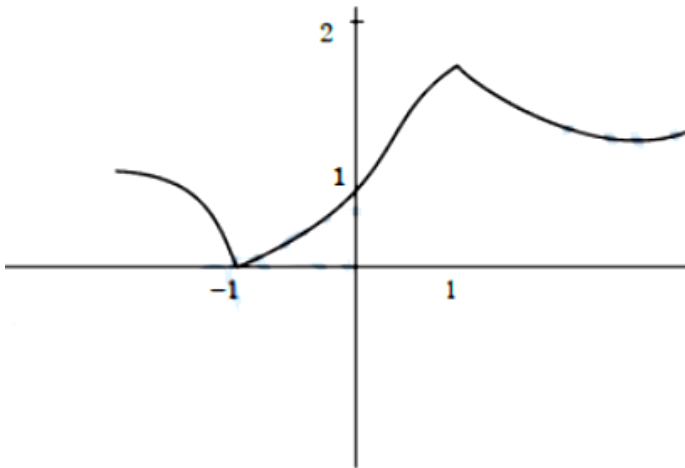
B. $f(x)$ is many-one in $(1, \infty)$

C. $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$

D. $f(x)$ is one-one in $(-\infty, \infty)$

Answer: C

Solution:



$$f(x) = \frac{(x+1)^2}{x^2+1} = 1 + \frac{2x}{x^2+1}$$

$$f(x) = 1 + \frac{2}{x + \frac{1}{x}}$$

Question84

The domain of $f(x) = \frac{\log_{(x+1)}(x-2)}{e^{2\log_e x} - (2x+3)}$, $x \in \mathbb{R}$ is

[29-Jan-2023 Shift 1]

Options:

- A. $\mathbb{R} - \{1-3\}$
- B. $(2, \infty) - \{3\}$
- C. $(-1, \infty) - \{3\}$
- D. $\mathbb{R} - \{3\}$

Answer: B

Solution:

$$\begin{aligned} x-2 &> 0 \Rightarrow x > 2 \\ x+1 &> 0 \Rightarrow x > -1 \\ x+1 &\neq 1 \Rightarrow x \neq 0 \text{ and } x > 0 \\ \text{Denominator} \\ x^2 - 2x - 3 &\neq 0 \end{aligned}$$

$$(x-3)(x+1) \neq 0$$

$$x \neq -1, 3$$

So Ans $(2, \infty) - \{3\}$

Question85

Consider a function $f : \mathbb{N} \rightarrow \mathbb{R}$, satisfying

$f(1) + 2f(2) + 3f(3) + \dots + xf(x) = x(x+1)f(x)$; $x \geq 2$ with $f(1) = 1$. Then

$\frac{1}{f(2022)} + \frac{1}{f(2028)}$ is equal to

[29-Jan-2023 Shift 2]

Options:

A. 8200

B. 8000

C. 8400

D. 8100

Answer: D

Solution:

Given for $x \geq 2$

$$f(1) + 2f(2) + \dots + xf(x) = x(x+1)f(x)$$

replace x by $x+1$

$$\Rightarrow x(x+1)f(x) + (x+1)f(x+1)$$

$$= (x+1)(x+2)f(x+1)$$

$$\Rightarrow \frac{x}{f(x+1)} + \frac{1}{f(x)} = \frac{(x+2)}{f(x)}$$

$$\Rightarrow xf(x) = (x+1)f(x+1) = \frac{1}{2}, x \geq 2$$

$$f(2) = \frac{1}{4}, f(3) = \frac{1}{6}$$

$$\text{Now } f(2022) = \frac{1}{4044}$$

$$f(2028) = \frac{1}{4056}$$

$$\text{So, } \frac{1}{f(2022)} + \frac{1}{f(2028)} = 4044 + 4056 = 8100$$

Question86

Suppose $f : \mathbb{R} \rightarrow (0, \infty)$ be a differentiable function such that $5f(x+y) = f(x) \cdot f(y)$, $\forall x, y \in \mathbb{R}$. If $f(3) = 320$, then $\sum_{n=0}^5 f(n)$ is equal to:
[30-Jan-2023 Shift 1]

Options:

A. 6875

B. 6575

C. 6825

D. 6528

Answer: C

Solution:

Option (3)

$$5f(x+y) = f(x) \cdot f(y)$$

$$5f(0) = f(0)^2 \Rightarrow f(0) = 5$$

$$5f(x+1) = f(x) \cdot f(1)$$

$$\Rightarrow \frac{f(x+1)}{f(x)} = \frac{f(1)}{5}$$

$$\Rightarrow \frac{f(1)}{f(0)} \cdot \frac{f(2)}{f(1)} \cdot \frac{f(3)}{f(2)} = \left(\frac{f(1)}{5} \right)^3$$

$$\Rightarrow \frac{320}{5} = \frac{(f(1))^3}{5^3} \Rightarrow f(1) = 20$$

$$\therefore 5f(x+1) = 20 \cdot f(x) \Rightarrow f(x+1) = 4f(x)$$

$$\sum_{n=0}^5 f(n) = 5 + 5.4 + 5.4^2 + 5.4^3 + 5.4^4 + 5.4^5$$

$$= \frac{5[4^6 - 1]}{3} = 6825$$

Question87

Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the number of one functions $f : S \rightarrow P(S)$, where $P(S)$ denote the power set of S , such that $f(n) \subset f(m)$ where $n < m$ is _____.

[30-Jan-2023 Shift 1]

Answer: 3240

Solution:

Let $S = \{1, 2, 3, 4, 5, 6\}$, then the number of one-one functions, $f: S \rightarrow P(S)$, where $P(S)$ denotes the power set of S , such that $f(n) \subset f(m)$ where $n < m$ is

$$n(S) = 6$$

$$P(S) = \left\{ \begin{array}{cccccc} \phi & \{1\} & \dots & \{6\} & \{1, 2\} & \dots \\ \{5, 6\} & \dots & \{1, 2, 3, 4, 5, 6\} & & & \end{array} \right\}$$

– 64 elements

case –1

$f(6) = S$ i.e. 1 option,

$f(5) =$ any 5 element subset A of S i.e. 6 options,

$f(4) =$ any 4 element subset B of A i.e. 5 options,

$f(3) =$ any 3 element subset C of B i.e. 4 options,

$f(2) =$ any 2 element subset D of C i.e. 3 options,

$f(1) =$ any 1 element subset E of D or empty subset i.e. 3 options,

Total functions = 1080

Case - 2

$f(6) =$ any 5 element subset A of S i.e. 6 options,

$f(5) =$ any 4 element subset B of A i.e. 5 options,

$f(4) =$ any 3 element subset C of B i.e. 4 options,

$f(3) =$ any 2 element subset D of C i.e. 3 options,

$f(2) =$ any 1 element subset E of D i.e. 2 options,

$f(1) =$ empty subset i.e. 1 option

Total functions = 720

Case –3

$f(6) = S$

$f(5) =$ any 4 element subset A of S i.e. 15 options,

$f(4) =$ any 3 element subset B of A i.e. 4 options,

$f(3) =$ any 2 element subset C of B i.e. 3 options,

$f(2) =$ any 1 element subset D of C i.e. 2 options,

$f(1) =$ empty subset i.e. 1 option

Total functions = 360

Case –4

$f(6) = S$

$f(5) =$ any 5 element subset A of S i.e. 6 options,

$f(4) =$ any 3 element subset B of A i.e. 10 options,

$f(3) =$ any 2 element subset C of B i.e. 3 options,

$f(2) =$ any 1 element subset D of C i.e. 2 options,

$f(1) =$ empty subset i.e. 1 option

Total functions = 360

Case –5

$f(6) = S$

$f(5) =$ any 5 element subset A of S i.e. 6 options,

$f(4) =$ any 4 element subset B of A i.e. 5 options,

$f(3) =$ any 2 element subset C of B i.e. 6 options,

$f(2) =$ any 1 element subset D of C i.e. 2 options,

$f(1) =$ empty subset i.e. 1 option

Total functions = 360

Case - 6

$$f(6) = S$$

$f(5)$ = any 5 element subset A of S i.e. 6 options,

$f(4)$ = any 4 element subset B of A i.e. 5 options,

$f(3)$ = any 3 element subset C of B i.e. 4 options,

$f(2)$ = any 1 element subset D of C i.e. 3 options,

$f(1)$ = empty subset i.e. 1 option

Total functions = 360

$\therefore$ Number of such functions = 3240

Question88

Let $f^1(x) = \frac{3x+2}{2x+3}$, $x \in \mathbb{R} - \left\{ \frac{-3}{2} \right\}$

For $n \geq 2$, define $f^n(x) = f^1 \circ f^{n-1}(x)$.

If $f^5(x) = \frac{ax+b}{bx+a}$, $\gcd(a, b) = 1$, then $a + b$ is equal to _____.

[30-Jan-2023 Shift 1]

Answer: 3125

Solution:

$$f^1(x) = \frac{3x+2}{2x+3}$$

$$\Rightarrow f^2(x) = \frac{13x+12}{12x+13}$$

$$\Rightarrow f^3(x) = \frac{63x+62}{62x+63}$$

$$\therefore f^5(x) = \frac{1563x+1562}{1562x+1563}$$

$$a + b = 3125$$

Question89

The range of the function $f(x) = \sqrt{3-x} + \sqrt{2+x}$ is

[30-Jan-2023 Shift 2]

Options:

A. $[\sqrt{5}, \sqrt{10}]$

B. $[2\sqrt{2}, \sqrt{11}]$

C. $[\sqrt{5}, \sqrt{13}]$

D. $[\sqrt{2}, \sqrt{7}]$

Answer: A

Solution:

$$y^2 = 3 - x + 2 + x + 2\sqrt{(3-x)(2+x)}$$
$$= 5 + 2\sqrt{6+x-x^2}$$

$$y^2 = 5 + 2\sqrt{\frac{25}{4} - \left(x - \frac{1}{2}\right)^2}$$

$$y_{\max} = \sqrt{5+5} = \sqrt{10}$$

$$y_{\min} = \sqrt{5}$$

Question90

Let $A = \{1, 2, 3, 5, 8, 9\}$. Then the number of possible functions $f : A \rightarrow A$ such that $f(m \cdot n) = f(m) \cdot f(n)$ for every $m, n \in A$ with $m \cdot n \in A$ is equal to _____.

[30-Jan-2023 Shift 2]

Answer: 432

Solution:

$$f(1) = 1; f(9) = f(3) \times f(3)$$

$$\text{i.e., } f(3) = 1 \text{ or } 3$$

$$\text{Total function} = 1 \times 6 \times 2 \times 6 \times 6 \times 1 = 432$$

Question91

If the domain of the function $f(x) = \frac{[x]}{1+x^2}$, where $[x]$ is greatest integer $\leq x$, is $[2, 6)$, then its range is
[31-Jan-2023 Shift 1]

Options:

A. $\left(\frac{5}{26}, \frac{2}{5} \right] - \left\{ \frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53} \right\}$

B. $\left(\frac{5}{26}, \frac{2}{5} \right]$

C. $\left(\frac{5}{37}, \frac{2}{5} \right] - \left\{ \frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53} \right\}$

D. $\left(\frac{5}{37}, \frac{2}{5} \right]$

Answer: D

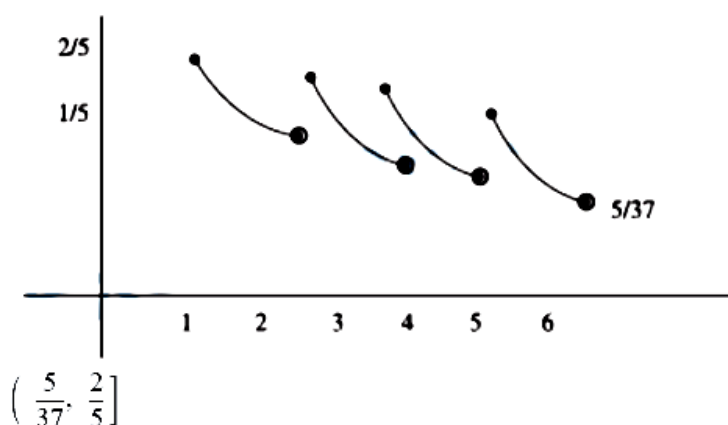
Solution:

$$f(x) = \frac{2}{1+x^2} \quad x \in [2, 3)$$

$$f(x) = \frac{3}{1+x^2} \quad x \in [3, 4)$$

$$f(x) = \frac{4}{1+x^2} \quad x \in [4, 5)$$

$$f(x) = \frac{5}{1+x^2} \quad x \in [5, 6)$$



Question92

The absolute minimum value, of the function

$f(x) = x^2 - x + 1 | + [x^2 - x + 1]$, where $[t]$ denotes the greatest integer function, in the interval $[-1, 2]$, is :

[31-Jan-2023 Shift 2]

Options:

A. $\frac{3}{4}$

B. $\frac{3}{2}$

C. $\frac{1}{4}$

D. $\frac{5}{4}$

Answer: A

Solution:

$$f(x) = |x^2 - x + 1| + [x^2 - x + 1]; x \in [-1, 2]$$

$$\text{Let } g(x) = x^2 - x + 1$$

$$= \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$\because |x^2 - x + 1| \text{ and } [x^2 - x + 2]$$

Both have minimum value at $x = 1/2$

$$\Rightarrow \text{Minimum } f(x) = \frac{3}{4} + 0$$

$$= \frac{3}{4}$$

Question93

Let $f : \mathbb{R} - \{2, 6\} \rightarrow \mathbb{R}$ be real valued function defined as

$$f(x) = \frac{x^2 + 2x + 1}{x^2 - 8x + 12}. \text{ Then range of } f \text{ is}$$

[31-Jan-2023 Shift 2]

Options:

A. $\left(-\infty, -\frac{21}{4}\right] \cup [0, \infty)$

B. $\left(-\infty, -\frac{21}{4}\right) \cup (0, \infty)$

C. $\left(-\infty, -\frac{21}{4}\right] \cup \left[\frac{21}{4}, \infty\right)$

D. $\left(-\infty, -\frac{21}{4}\right] \cup [1, \infty)$

Answer: A

Solution:

Let $y = \frac{x^2 + 2x + 1}{x^2 - 8x + 12}$

By cross multiplying

$$yx^2 - 8xy + 12y - x^2 - 2x - 1 = 0$$

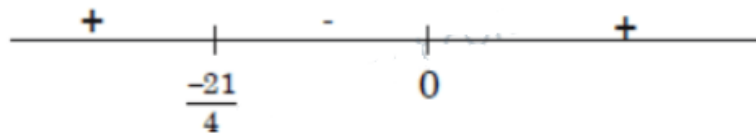
$$x^2(y - 1) - x(8y + 2) + (12y - 1) = 0$$

Case 1, $y \neq 1$

$$D \geq 0$$

$$\Rightarrow (8y + 2)^2 - 4(y - 1)(12y - 1) \geq 0$$

$$\Rightarrow y(4y + 21) \geq 0$$



$$y \in \left(-\infty, -\frac{21}{4}\right] \cup [0, \infty) - \{1\}$$

Case 2, $y = 1$

$$x^2 + 2x + 1 = x^2 - 8x + 12$$

$$10x = 11$$

$$x = \frac{11}{10} \quad \text{So, } y \text{ can be } 1$$

$$\text{Hence } y \in \left(-\infty, -\frac{21}{4}\right] \cup [0, \infty)$$

Question94

Let $f : \mathbb{R} - \{0, 1\} \rightarrow \mathbb{R}$ be a function such that $f(x) + f\left(\frac{1}{1-x}\right) = 1 + x$.

Then $f(2)$ is equal to :
[1-Feb-2023 Shift 2]

Options:

A. $\frac{9}{2}$

B. $\frac{9}{4}$

C. $\frac{7}{4}$

D. $\frac{7}{3}$

Answer: B

Solution:

$$f(x) + f\left(\frac{1}{1-x}\right) = 1 + x$$

$$x = 2 \Rightarrow f(2) + f(-1) = 3$$

$$x = -1 \Rightarrow f(-1) + f\left(\frac{1}{2}\right) = 0 \dots (2)$$

$$x = \frac{1}{2} \Rightarrow f\left(\frac{1}{2}\right) + f(2) = \frac{3}{2} \dots (3)$$

$$(1) + (3) - (2) \Rightarrow 2f(2) = \frac{9}{2}$$

$$\therefore f(2) = \frac{9}{4}$$

Question95

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x - 1$ and $g : \mathbb{R} - \{1, -1\} \rightarrow \mathbb{R}$ be defined as $g(x) = \frac{x^2}{x^2 - 1}$. Then the function fog is :

[26-Jun-2022-Shift-2]

Options:

- A. one-one but not onto
- B. onto but not one-one
- C. both one-one and onto
- D. neither one-one nor onto

Answer: D

Solution:

$f : R \rightarrow R$ defined as

$$f(x) = x - 1 \text{ and } g : R \rightarrow \{1, -1\} \rightarrow R, g(x) = \frac{x^2}{x^2 - 1}$$

$$\text{Now } f \circ g(x) = \frac{x^2}{x^2 - 1} - 1 = \frac{1}{x^2 - 1}$$

$$\therefore \text{Domain of } f \circ g(x) = R - \{-1, 1\}$$

$$\text{And range of } f \circ g(x) = (-\infty, -1] \cup (0, \infty)$$

$$\text{Now, } \frac{d}{dx}(f \circ g(x)) = \frac{-1}{x^2 - 1} \cdot 2x = \frac{2x}{1 - x^2}$$

$$\therefore \frac{d}{dx}(f \circ g(x)) > 0 \text{ for } \frac{2x}{(1-x)(1+x)} > 0$$

$$\Rightarrow \frac{x}{(x-1)(x+1)} < 0$$

$$\therefore x \in (-\infty, -1) \cup (0, 1)$$

$$\text{and } \frac{d}{dx}(f \circ g(x)) < 0 \text{ for } x \in (-1, 0) \cup (1, \infty)$$

$\therefore f \circ g(x)$ is neither one-one nor onto.

Question 96

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{2e^{2x}}{e^{2x} + e}$.

Then

$$f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + f\left(\frac{3}{100}\right) + \dots + f\left(\frac{99}{100}\right)$$

is equal to _____

[27-Jun-2022-Shift-1]

Answer: 99

Solution:

Given,

$$f(x) = \frac{2e^{2x}}{e^{2x} + e}$$

$$\therefore f(1-x) = \frac{2e^{2(1-x)}}{e^{2(1-x)} + e}$$

$$= \frac{2 \cdot \frac{e^2}{e^{2x}}}{\frac{e^2}{e^{2x}} + e}$$

$$= \frac{2e^2}{e^2 + e^{2x} \cdot e}$$

$$= \frac{2e^2}{e(e + e^{2x})}$$

$$= \frac{2e}{e + e^{2x}}$$

$$\therefore f(x) + f(1-x) = \frac{2e^{2x}}{e^{2x} + e} + \frac{2e}{e^{2x} + e}$$

$$= \frac{2(e^{2x} + e)}{e^{2x} + e}$$

$$= 2 \dots \dots (1)$$

Now,

$$f\left(\frac{1}{100}\right) + f\left(\frac{99}{100}\right)$$

$$= f\left(\frac{1}{100}\right) + f\left(1 - \frac{1}{100}\right)$$

$$= 2 \text{ [as } f(x) + f(1-x) = 2 \text{]}$$

$$f\left(\frac{2}{100}\right) + f\left(1 - \frac{2}{100}\right) = 2$$

⋮

$$f\left(\frac{49}{100}\right) + f\left(1 - \frac{49}{100}\right) = 2$$

$$\therefore \text{Total sum} = 49 \times 2$$

$$\text{Remaining term} = f\left(\frac{50}{100}\right) = f\left(\frac{1}{2}\right)$$

Put $x = \frac{1}{2}$ in equation (1), we get

$$f\left(\frac{1}{2}\right) + f\left(1 - \frac{1}{2}\right) = 2$$

$$\Rightarrow 2f\left(\frac{1}{2}\right) = 2$$

$$\Rightarrow f\left(\frac{1}{2}\right) = 1$$

$$\therefore \text{Sum} = 49 \times 2 + 1 = 99$$

Question 97

Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Define $f : S \rightarrow S$ as

$$f(n) = \begin{cases} 2n & , \text{ if } n = 1, 2, 3, 4, 5 \\ 2n - 11 & , \text{ if } n = 6, 7, 8, 9, 10, \end{cases}$$

Let $g : S \rightarrow S$ be a function such that $g(n) = \begin{cases} n + 1 & , \text{ if } n \text{ is odd} \\ n - 1 & , \text{ if } n \text{ is even} \end{cases}$

Then $g(10)g(1) + g(2) + g(3) + g(4) + g(5)$ is equal to
[27-Jun-2022-Shift-2]

Answer: 190

Solution:

$$\therefore f(n) = \begin{cases} 2n & n = 1, 2, 3, 4, 5 \\ 2n - 11 & n = 6, 7, 8, 9, 10 \end{cases}$$

$$\therefore f(1) = 2, f(2) = 4, \dots, f(5) = 10$$

$$\text{and } f(6) = 1, f(7) = 3, f(8) = 5, \dots, f(10) = 9$$

$$\text{Now, } f(g(n)) = \begin{cases} n + 1 & \text{if } n \text{ is odd} \\ n - 1 & \text{if } n \text{ is even} \end{cases}$$

$$\begin{aligned}
f(g(10)) &= 9 \Rightarrow g(10) = 10 \\
f(g(1)) &= 2 \Rightarrow g(1) = 1 \\
f(g(2)) &= 1 \Rightarrow g(2) = 6 \\
f(g(3)) &= 4 \Rightarrow g(3) = 2 \\
f(g(4)) &= 3 \Rightarrow g(4) = 7 \\
f(g(5)) &= 6 \Rightarrow g(5) = 3 \\
\therefore g(10) + g(1) + g(2) + g(3) + g(4) + g(5) &= 190
\end{aligned}$$

Question 98

Let a function $f : \mathbb{N} \rightarrow \mathbb{N}$ be defined by

$$f(n) = \begin{cases} 2n & n = 2, 4, 6, 8, \dots \\ n-1 & n = 3, 7, 11, 15, \dots \\ \frac{n+1}{2} & n = 1, 5, 9, 13, \dots \end{cases}$$

then, f is

[28-Jun-2022-Shift-1]

Options:

- A. one-one but not onto
- B. onto but not one-one
- C. neither one-one nor onto
- D. one-one and onto

Answer: D

Solution:

When $n = 1, 5, 9, 13$ then $\frac{n+1}{2}$ will give all odd numbers.

When $n = 3, 7, 11, 15, \dots$

$n-1$ will be even but not divisible by 4

When $n = 2, 4, 6, 8, \dots$

Then $2n$ will give all multiples of 4

So range will be $\mathbb{N}$.

And no two values of n give same y , so function is one-one and onto.

Question99

The probability that a randomly chosen one-one function from the set $\{a, b, c, d\}$ to the set $\{1, 2, 3, 4, 5\}$ satisfies $f(a) + 2f(b) - f(c) = f(d)$ is :
[28-Jun-2022-Shift-2]

Options:

A. $\frac{1}{24}$

B. $\frac{1}{40}$

C. $\frac{1}{30}$

D. $\frac{1}{20}$

Answer: D

Solution:

Number of one-one function from $\{a, b, c, d\}$ to set $\{1, 2, 3, 4, 5\}$ is ${}^5P_4 = 120n(s)$.

The required possible set of value $(f(a), f(b), f(c), f(d))$ such that $f(a) + 2f(b) - f(c) = f(d)$ are $(5, 3, 2, 1), (5, 1, 2, 3), (4, 1, 3, 5), (3, 1, 4, 5), (5, 4, 3, 2)$ and $(3, 4, 5, 2)$

$$\therefore n(E) = 6$$

$$\therefore \text{Required probability} = \frac{n(E)}{n(S)} = \frac{6}{120} = \frac{1}{20}$$

Question100

Let $S = \{1, 2, 3, 4\}$. Then the number of elements in the set $\{f : S \times S \rightarrow S : f \text{ is onto and } f(a, b) = f(b, a) \geq a \forall (a, b) \in S \times S\}$ is _____
[28-Jun-2022-Shift-2]

Answer: 37

Solution:

There are 16 ordered pairs in $S \times S$. We write all these ordered pairs in 4 sets as follows.

$$A = \{(1, 1)\}$$

$$B = \{(1, 4), (2, 4), (3, 4), (4, 4), (4, 3), (4, 2), (4, 1)\}$$

$$C = \{(1, 3), (2, 3), (3, 3), (3, 2), (3, 1)\}$$

$$D = \{(1, 2), (2, 2), (2, 1)\}$$

All elements of set B have image 4 and only element of A has image 1 .

All elements of set C have image 3 or 4 and all elements of set D have image 2 or 3 or 4 .

We will solve this question in two cases.

Case I: When no element of set C has image 3 .

Number of onto functions = 2 (when elements of set D have images 2 or 3)

Case II: When atleast one element of set C has image 3 .

Number of onto functions = $(2^3 - 1)(1 + 2 + 2) = 35$

Total number of functions = 37

Question101

The domain of the function $\cos^{-1} \left(\frac{2\sin^{-1} \left(\frac{1}{4x^2 - 1} \right)}{\pi} \right)$ is :

[29-Jun-2022-Shift-1]

Options:

A. $\mathbb{R} - \left\{ -\frac{1}{2}, \frac{1}{2} \right\}$

B. $(-\infty, -1] \cup [1, \infty) \cup \{0\}$

C. $\left(-\infty, \frac{-1}{2}\right) \cup \left(\frac{1}{2}, \infty\right) \cup \{0\}$

D. $\left(-\infty, \frac{-1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, \infty\right) \cup \{0\}$

Answer: D

Solution:

$$-1 \leq \frac{2\sin^{-1} \left(\frac{1}{4x^2 - 1} \right)}{\pi} \leq 1$$

$$\Rightarrow -\frac{\pi}{2} \leq \sin^{-1} \left(\frac{1}{4x^2 - 1} \right) \leq \frac{\pi}{2}$$

$$\Rightarrow -1 \leq \frac{1}{4x^2 - 1} \leq 1$$

$$\therefore \frac{1}{4x^2 - 1} + 1 \geq 0$$

$$\Rightarrow \frac{1 + 4x^2 - 1}{4x^2 - 1} \geq 0$$

$$\Rightarrow \frac{4x^2}{4x^2 - 1} \geq 0$$

$$\Rightarrow \frac{4x^2}{(2x+1)(2x-1)} \geq 0 \dots\dots (1)$$

$$\therefore x \in \left(-\alpha, -\frac{1}{2}\right) \cup \{0\} \cup \left(\frac{1}{2}, \alpha\right)$$

$$\text{And } \frac{1}{4x^2 - 1} - 1 \leq 0$$

$$\Rightarrow \frac{1 - 4x^2 + 1}{4x^2 - 1} \leq 0$$

$$\Rightarrow \frac{2 - 4x^2}{4x^2 - 1} \leq 0$$

$$\Rightarrow \frac{2x^2 - 1}{4x^2 - 1} \geq 0$$

$$\Rightarrow \frac{(\sqrt{2}x+1)(\sqrt{2}x-1)}{(2x+1)(2x-1)} \geq 0$$

$$x \in \left(-\alpha, -\frac{1}{\sqrt{2}}\right) \cup \left(-\frac{1}{2}, \frac{1}{2}\right) \cup \left(\frac{1}{\sqrt{2}}, \alpha\right)$$

From (3) and (4), we get

$$\therefore x \in \left[-\alpha, -\frac{1}{\sqrt{2}}\right) \cup \left[\frac{1}{\sqrt{2}}, \alpha\right) \cup \{0\}$$

Question102

Let $c, k \in \mathbb{R}$. If $f(x) = (c+1)x^2 + (1-c^2)x + 2k$ and $f(x+y) = f(x) + f(y) - xy$, for all $x, y \in \mathbb{R}$, then the value of $|2(f(1) + f(2) + f(3) + \dots + f(20))|$ is equal to _____
[29-Jun-2022-Shift-1]

Answer: 3395

Solution:

$f(x)$ is polynomial

Put $y = 1/x$ in given functional equation we get

$$\begin{aligned}
f\left(x + \frac{1}{x}\right) &= f(x) + f\left(\frac{1}{x}\right) - 1 \\
\Rightarrow (c+1)\left(x + \frac{1}{x}\right)^2 + (1-c^2)\left(x + \frac{1}{x}\right) + 2K \\
&= (c+1)x^2 + (1-c^2)x + 2K + (c+1)\frac{1}{x^2} + (1-c^2)\frac{1}{x} + 2K - 1 \\
\Rightarrow 2(c+1) &= 2K - 1 \dots (1) \\
\text{and put } x &= y = 0 \text{ we get} \\
f(0) &= 2 + f(0) - 0 \Rightarrow f(0) = 0 \Rightarrow k = 0 \\
\therefore k &= 0 \text{ and } 2c = -3 \Rightarrow c = -3/2 \\
f(x) &= -\frac{x^2}{2} - \frac{5x}{4} = \frac{1}{4}(5x + 2x^2) \\
\left| 2 \sum_{i=1}^{20} f(i) \right| &= \left| \frac{-2}{4} \left(\frac{5 \cdot 20 \cdot 21}{2} + \frac{2 \cdot 20 \cdot 21 \cdot 41}{6} \right) \right| \\
&= \left| \frac{-1}{2}(2730 + 5740) \right| \\
&= \left| -\frac{6790}{2} \right| = 3395.
\end{aligned}$$

Question103

Let $f(x)$ and $g(x)$ be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$ and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is _____
[29-Jun-2022-Shift-2]

Answer: 18

Solution:

$$\begin{aligned}
f(g(x)) &= 8x^2 - 2x. \\
g(f(x)) &= 4x^2 + 6x + 1. \\
\text{So, } g(x) &= 2x - 1 \\
&\&f(x) = 2x^2 + 3x + 1 \\
f(2) &= 8 + 6 + 1 = 15 \\
\text{Ans. } &18
\end{aligned}$$

Question104

The domain of the function

$$f(x) = \frac{\cos^{-1}\left(\frac{x^2-5x+6}{x^2-9}\right)}{\log_e(x^2-3x+2)} \text{ is :}$$

[24-Jun-2022-Shift-1]

Options:

A. $(-\infty, 1) \cup (2, \infty)$

B. $(2, \infty)$

C. $\left[-\frac{1}{2}, 1\right) \cup (2, \infty)$

D. $\left[-\frac{1}{2}, 1\right) \cup (2, \infty) - \left\{3, \frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}\right\}$

Answer: D

Solution:

$$-1 \leq \frac{x^2-5x+6}{x^2-9} \leq 1 \text{ and } x^2-3x+2 > 0, \neq 1$$

$$\frac{(x-3)(2x+1)}{x^2-9} \geq 0 \mid \frac{5(x-3)}{x^2-9} \geq 0$$

The solution to this inequality is

$$x \in \left[-\frac{1}{2}, \infty\right) - \{3\}$$

$$\text{for } x^2-3x+2 > 0 \text{ and } \neq 1$$

$$x \in (-\infty, 1) \cup (2, \infty) - \left\{\frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2}\right\}$$

Combining the two solution sets (taking intersection)

$$x \in \left[-\frac{1}{2}, 1\right) \cup (2, \infty) - \left\{\frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2}\right\}$$

Question105

The number of one-one functions $f : \{a, b, c, d\} \rightarrow \{0, 1, 2, \dots, 10\}$ such that $2f(a) - f(b) + 3f(c) + f(d) = 0$ is
[24-Jun-2022-Shift-1]

Answer: 31

Solution:

$$-1 \leq \frac{x^2 - 5x + 6}{x^2 - 9} \leq 1 \text{ and } x^2 - 3x + 2 > 0, \neq 1$$

$$\frac{(x-3)(2x+1)}{x^2-9} \geq 0 \mid \frac{5(x-3)}{x^2-9} \geq 0$$

The solution to this inequality is

$$x \in \left[\frac{-1}{2}, \infty \right) - \{3\}$$

for $x^2 - 3x + 2 > 0$ and $\neq 1$

$$x \in (-\infty, 1) \cup (2, \infty) - \left\{ \frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2} \right\}$$

Combining the two solution sets (taking intersection)

$$x \in \left[-\frac{1}{2}, 1 \right) \cup (2, \infty) - \left\{ \frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2} \right\}$$

$f(d)$ can't be 9 and 10 as if $f(d) = 9$ or 10 then $f(b) = 2 + 9 = 11$ or $f(b) = 2 + 10 = 12$, which is not possible as here any function's maximum value can be 10 .

∴ Total possible functions when $f(c) = 0$ and $f(a) = 1$ are $= 7$

(2) When $f(c) = 0$ and $f(a) = 2$ then

$$3 \times 0 + 2 \times 2 + f(d) = f(b)$$

$$\Rightarrow 4 + f(d) = f(b)$$

∴ possible value of $f(d) = 1, 3, 4, 5, 6$

∴ Total possible functions in this case $= 5$

(3) When $f(c) = 0$ and $f(a) = 3$ then

$$3 \times 0 + 2 \times 3 + f(d) = f(b)$$

$$\Rightarrow 6 + f(d) = f(b)$$

∴ Possible value of $f(d) = 1, 2, 4$

∴ Total possible functions in this case $= 3$

(4) When $f(c) = 0$ and $f(a) = 4$ then

$$3 \times 0 + 2 \times 4 + f(d) = f(b)$$

$$\Rightarrow 8 + f(d) = f(b)$$

∴ Possible value of $f(d) = 1, 2$

∴ Total possible functions in this case $= 2$

(5) When $f(c) = 0$ and $f(a) = 5$ then

$$3 \times 0 + 2 \times 5 + f(d) = f(b)$$

$$\Rightarrow 10 + f(d) = f(b)$$

Possible value of $f(d)$ can be 0 but $f(c)$ is already zero. So, no value to $f(d)$ can satisfy.

∴ No function is possible in this case.

∴ Total possible functions when $f(c)=0$ and $f(a)=1,2,3$ and 4 are $=7+5+3+2=17$

Case II:

(1) When $f(c)=1$ and $f(a)=0$ then

$$3 \times 1 + 2 \times 0 + f(d) = f(b)$$

$$\Rightarrow 3 + f(d) = f(b)$$

∴ Possible value of $f(d)=2,3,4,5,6,7$

∴ Total possible functions in this case $=6$

(2) When $f(c)=1$ and $f(a)=2$ then

$$3 \times 1 + 2 \times 2 + f(d) = f(b)$$

$$\Rightarrow 7 + f(d) = f(b)$$

$\therefore$ Possible value of $f(d)=0,3$

$\therefore$ Total possible functions in this case =2

(3) When $f(c)=1$ and $f(a)=3$ then

$$3 \times 1 + 2 \times 3 + f(d) = f(b)$$

$$\Rightarrow 9 + f(d) = f(b)$$

$\therefore$ Possible value of $f(d)=0$

$\therefore$ Total possible functions in this case =1

$\therefore$ Total possible functions when $f(c)=1$ and $f(a)=0,2$ and 3 are $=6+2+1=9$

Case III:

(1) When $f(c)=2$ and $f(a)=0$ then

$$3 \times 2 + 2 \times 0 + f(d) = f(b)$$

$$\Rightarrow 6 + f(d) = f(b)$$

$\therefore$ Possible values of $f(d)=1,3,4$

$\therefore$ Total possible functions in this case =3

(2) When $f(c)=2$ and $f(a)=1$ then,

$$3 \times 2 + 2 \times 1 + f(d) = f(b)$$

$$\Rightarrow 8 + f(d) = f(b)$$

$\therefore$ Possible values of $f(d)=0$

$\therefore$ Total possible function in this case =1

$\therefore$ Total possible functions when $f(c)=2$ and $f(a)=0,1$ are $=3+1=4$

Case IV:

(1) When $f(c)=3$ and $f(a)=0$ then

$$3 \times 3 + 2 \times 0 + f(d) = f(b)$$

$$\Rightarrow 9 + f(d) = f(b)$$

$\therefore$ Possible values of $f(d)=1$

$\therefore$ Total one-one functions from four cases

$$=17+9+4+1=31$$

Question106

Let R_1 and R_2 be relations on the set $\{1, 2, \dots, 50\}$ such that

$R_1 = \{(p, p^n) : p \text{ is a prime and } n \geq 0 \text{ is an integer}\}$ and

$R_2 = \{(p, p^n) : p \text{ is a prime and } n = 0 \text{ or } 1\}$

Then, the number of elements in $R_1 - R_2$ is ____

[28-Jun-2022-Shift-1]

Answer: 8

Solution:

$$R_1 - R_2 = \{(2, 2^2), (2, 2^3), (2, 2^4), (2, 2^5), (3, 3^2), (3, 3^3), (5, 5^2), (7, 7^2)\}$$

So number of elements = 8

Question107

Let $R_1 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \leq 13\}$ and

$R_2 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \neq 13\}$. Then on $\mathbb{N}$:

[28-Jun-2022-Shift-2]

Options:

- A. Both R_1 and R_2 are equivalence relations
- B. Neither R_1 nor R_2 is an equivalence relation
- C. R_1 is an equivalence relation but R_2 is not
- D. R_2 is an equivalence relation but R_1 is not

Answer: B

Solution:

$R_1 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \leq 13\}$ and

$R_2 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \neq 13\}$

In R_1 : $\because |2 - 11| = 9 \leq 13$

$\therefore (2, 11) \in R_1$ and $(11, 19) \in R_1$ but $(2, 19) \notin R_1$

$\therefore R_1$ is not transitive

Hence R_1 is not equivalence

In R_2 : $(13, 3) \in R_2$ and $(3, 26) \in R_2$ but $(13, 26) \notin R_2 (\because |13 - 26| = 13)$

$\therefore R_2$ is not transitive

Hence R_2 is not equivalence.

Question108

The probability that a relation R from $\{x, y\}$ to $\{x, y\}$ is both symmetric and transitive, is equal to
[29-Jun-2022-Shift-2]

Options:

A. $\frac{5}{16}$

B. $\frac{9}{16}$

C. $\frac{11}{16}$

D. $\frac{13}{16}$

Answer: A

Solution:

Total no. of relations $= 2^{2 \times 2} = 16$

Fav. relation $= \varnothing, \{(x, x)\}, \{(y, y)\}, \{(x, x)(y, y)\}$
 $\{(x, x), (y, y), (x, y)(y, x)\}$

$$\text{Prob.} = \frac{5}{16}$$

Question109

The number of bijective functions

$f : \{1, 3, 5, 7, \dots, 99\} \rightarrow \{2, 4, 6, 8, \dots, 100\}$, such that

$f(3) \geq f(9) \geq f(15) \geq f(21) \geq \dots \geq f(99)$, is

[25-Jul-2022-Shift-2]

Options:

A. ${}^{50}P_{17}$

B. ${}^{50}P_{33}$

C. $33! \times 17!$

D. $\frac{50!}{2}$

Answer: B

Solution:

As function is one-one and onto, out of 50 elements of domain set 17 elements are following restriction $f(3) > f(9) > f(15) > \dots > f(99)$

So number of ways = ${}^{50}C_{17} \cdot 17!$

= ${}^{50}P_{17}$

Question110

Let $f(x)$ be a quadratic polynomial with leading coefficient 1 such that

$f(0) = p$, $p \neq 0$, and $f(1) = \frac{1}{3}$. If the equations $f(x) = 0$ and

$f \circ f \circ f \circ f(x) = 0$ have a common real root, then $f(-3)$ is equal to____

[25-Jul-2022-Shift-2]

Answer: 25

Solution:

$$\text{Let } f(x) = (x - \alpha)(x - \beta)$$

$$\text{It is given that } f(0) = p \Rightarrow \alpha\beta = p$$

$$\text{and } f(1) = \frac{1}{3} \Rightarrow (1 - \alpha)(1 - \beta) = \frac{1}{3}$$

Now, let us assume that, α is the common root of $f(x) = 0$ and $f \circ f \circ f(x) = 0$

$$f \circ f \circ f(x) = 0$$

$$\Rightarrow f \circ f(0) = 0$$

$$\Rightarrow f(\alpha) = 0$$

So, $f(p)$ is either α or β .

$$(p - \alpha)(p - \beta) = \alpha$$

$$(\alpha\beta - \alpha)(\alpha\beta - \beta) = \alpha \Rightarrow (\beta - 1)(\alpha - 1)\beta = 1 (\because \alpha \neq 0)$$

$$\text{So, } \beta = 3$$

$$(1 - \alpha)(1 - 3) = \frac{1}{3}$$

$$\alpha = \frac{7}{6}$$

$$f(x) = \left(x - \frac{7}{6}\right)(x - 3)$$

$$f(-3) = \left(-3 - \frac{7}{6}\right)(-3 - 3) = 25$$

Question111

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $f(3x) - f(x) = x$. If $f(8) = 7$, then $f(14)$ is equal to:

[26-Jul-2022-Shift-1]

Options:

A. 4

B. 10

C. 11

D. 16

Answer: B

Solution:

$$f(3x) - f(x) = x \dots\dots (1)$$

$$x \rightarrow \frac{x}{3}$$

$$f(x) - f\left(\frac{x}{3}\right) = \frac{x}{3} \dots\dots (2)$$

$$\text{Again } x \rightarrow \frac{x}{3}$$

$$f\left(\frac{x}{3}\right) - f\left(\frac{x}{9}\right) = \frac{x}{3^2} \dots$$

Similarly

$$f\left(\frac{x}{3^{n-2}}\right) - f\left(\frac{x}{3^{n-1}}\right) = \frac{x}{3^{n-1}} \dots (n)$$

Adding all these and applying $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \left(f(3x) - f\left(\frac{x}{3^{n-1}}\right) \right) = x \left(1 + \frac{1}{3} + \frac{1}{3^2} + \dots \right)$$

$$f(3x) - f(0) = \frac{3x}{2}$$

$$\text{Putting } x = \frac{8}{3}$$

$$f(8) - f(0) = 4$$

$$\Rightarrow f(0) = 3$$

$$\text{Putting } x = \frac{14}{3}$$

$$f(14) - 3 = 7 \Rightarrow f(14) = 0$$

Question112

The domain of the function

$$f(x) = \sin^{-1}[2x^2 - 3] + \log_2 \left(\log_{\frac{1}{2}}(x^2 - 5x + 5) \right), \text{ where } [t] \text{ is the greatest}$$

integer function, is :

[27-Jul-2022-Shift-2]

Options:

A. $\left(-\sqrt{\frac{5}{2}}, \frac{5-\sqrt{5}}{2} \right)$

B. $\left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2} \right)$

C. $\left(1, \frac{5-\sqrt{5}}{2} \right)$

D. $\left[1, \frac{5+\sqrt{5}}{2} \right)$

Answer: C

Solution:

$$-1 \leq 2x^2 - 3 < 2$$

$$\text{or } 2 \leq 2x^2 < 5$$

$$\text{or } 1 \leq x^2 < \frac{5}{2}$$

$$x \in \left(-\sqrt{\frac{5}{2}}, -1\right] \cup \left[1, \sqrt{\frac{5}{2}}\right)$$

$$\log_{\frac{1}{2}}(x^2 - 5x + 5) > 0$$

$$0 < x^2 - 5x + 5 < 1$$

$$x^2 - 5x + 5 > 0 \& x^2 - 5x + 4 < 0$$

$$x \in \left(-\infty, \frac{5 - \sqrt{5}}{2}\right) \cup \left(\frac{5 + \sqrt{5}}{2}, \infty\right)$$

$$\& x \in (-\infty, 1) \cup (4, \infty)$$

Taking intersection

$$x \in \left(1, \frac{5 - \sqrt{5}}{2}\right)$$

Question113

The number of functions f , from the set $A = \{x \in \mathbb{N} : x^2 - 10x + 9 \leq 0\}$ to the set $B = \{n^2 : n \in \mathbb{N}\}$ such that $f(x) \leq (x - 3)^2 + 1$, for every $x \in A$, is _____.

[27-Jul-2022-Shift-2]

Answer: 1440

Solution:

$$A = \{x \in \mathbb{N}, x^2 - 10x + 9 \leq 0\}$$

$$= \{1, 2, 3, \dots, 9\}$$

$$B = \{1, 4, 9, 16, \dots\}$$

$$f(x) \leq (x - 3)^2 + 1$$

$$f(1) \leq 5, f(2) \leq 2, \dots \dots \dots f(9) \leq 37$$

$x = 1$ has 2 choices

$x = 2$ has 1 choice

$x = 3$ has 1 choice

$x = 4$ has 1 choice

$x = 5$ has 2 choices

$x = 6$ has 3 choices

$x = 7$ has 4 choices

$x = 8$ has 5 choices

$x = 9$ has 6 choices

$\therefore$ Total functions $= 2 \times 1 \times 1 \times 1 \times 2 \times 3 \times 4 \times 5 \times 6 = 1440$

Question114

Considering only the principal values of the inverse trigonometric functions, the domain of the function $f(x) = \cos^{-1}\left(\frac{x^2 - 4x + 2}{x^2 + 3}\right)$ is :

[28-Jul-2022-Shift-1]

Options:

A. $\left(-\infty, \frac{1}{4}\right]$

B. $\left[-\frac{1}{4}, \infty\right)$

C. $(-1/3, \infty)$

D. $\left(-\infty, \frac{1}{3}\right]$

Answer: B

Solution:

$$-1 \leq \frac{x^2 - 4x + 2}{x^2 + 3} \leq 1$$

$$\Rightarrow -x^2 - 3 \leq x^2 - 4x + 2 \leq x^2 + 3$$

$$\Rightarrow 2x^2 - 4x + 5 \geq 0 \quad -4x \leq 1$$

$$x \in \mathbb{R} \text{ and } x \geq -\frac{1}{4}$$

$$\text{So domain is } \left[-\frac{1}{4}, \infty\right)$$

Question115

Let α , β and γ be three positive real numbers. Let

$f(x) = \alpha x^5 + \beta x^3 + \gamma x$, $x \in \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be such that $g(f(x)) = x$ for

all $x \in \mathbb{R}$. If $a_1, a_2, a_3, \dots, a_n$ be in arithmetic progression with mean zero, then the value of $f \left(g \left(\frac{1}{n} \sum_{i=1}^n f(a_i) \right) \right)$ is equal to:
[28-Jul-2022-Shift-1]

Options:

- A. 0
- B. 3
- C. 9
- D. 27

Answer: A

Solution:

$$f \left(g \left(\frac{1}{n} \sum_{i=1}^n f(a_i) \right) \right)$$

$$\frac{a_1 + a_2 + a_3 + \dots + a_n}{n} = 0$$

$\therefore$ First and last term, second and second last and so on are equal in magnitude but opposite in sign.

$$f(x) = \alpha x^5 + \beta x^3 + \gamma x$$

$$\sum_{i=1}^n f(a_i) = \alpha(a_1^5 + a_2^5 + a_3^5 + \dots + a_n^5) + \beta(a_1^3 + a_2^3 + \dots + a_n^3) + \gamma(a_1 + a_2 + \dots + a_n)$$

$$= 0\alpha + 0\beta + 0\gamma$$

$$= 0$$

$$\therefore f \left(g \left(\frac{1}{n} \sum_{i=1}^n f(a_i) \right) \right) = \frac{1}{n} \sum_{i=1}^n f(a_i) = 0$$

Question116

The number of elements in the set

$$S = \left\{ x \in \mathbb{R} : 2 \cos \left(\frac{x^2 + x}{6} \right) = 4^x + 4^{-x} \right\} \text{ is:}$$

[29-Jul-2022-Shift-2]

Options:

- A. 1

- B. 3
- C. 0
- D. infinite

Answer: A

Solution:

$$2 \cos\left(\frac{x^2+x}{6}\right) = 4^x + 4^{-x}$$

$$\text{L.H.S} \leq 2 \text{ \& R.H.S.} \geq 2$$

$$\text{Hence L.H.S} = 2 \text{ \& R.H.S} = 2$$

$$2 \cos\left(\frac{x^2+x}{6}\right) = 2 \quad 4^x + 4^{-x} = 2$$

Check $x = 0$ Possible hence only one solution.

Question117

The domain of the function $f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ is:

[29-Jul-2022-Shift-2]

Options:

- A. $[1, \infty)$
- B. $[-1, 2]$
- C. $[-1, \infty)$
- D. $(-\infty, 2]$

Answer: C

Solution:

$$f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$$

$$-1 \leq \frac{x^2-3x+2}{x^2+2x+7} \leq 1$$

$$\frac{x^2 - 3x + 2}{x^2 + 2x + 7} \leq 1$$

$$x^2 - 3x + 2 \leq x^2 + 2x + 7$$

$$5x \geq -5$$

$$x \geq -1$$

And $\frac{x^2 - 3x + 2}{x^2 + 2x + 7} \geq -1$

$$x^2 - 3x + 2 \geq -x^2 - 2x - 7$$

$$2x^2 - x + 9 \geq 0$$

$$x \in \mathbb{R}$$

(i) $\cap$ (ii)

Domain $\in [-1, \infty)$

Question118

The total number of functions,

$f : \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4, 5, 6\}$ such that $f(1) + f(2) = f(3)$, is equal to
:
[25-Jul-2022-Shift-1]

Options:

- A. 60
- B. 90
- C. 108
- D. 126

Answer: B

Solution:

Given, $f(1) + f(2) = f(3)$

It means $f(1)$, $f(2)$ and $f(3)$ are dependent on each other. But there is no condition on $f(4)$, so $f(4)$ can be $f(4) = 1, 2, 3, 4, 5, 6$.

For $f(1)$, $f(2)$ and we have to find how many functions possible which will satisfy the condition $f(1) + f(2) = f(3)$

Case 1:

When $f(3) = 2$ then possible values of $f(1)$ and $f(2)$ which satisfy $f(1) + f(2) = f(3)$ is $f(1) = 1$ and $f(2) = 1$.

And $f(4)$ can be $= 1, 2, 3, 4, 5, 6$

$\therefore$ Total possible functions $= 1 \times 6 = 6$

Case 2 :

When $f(3) = 3$ then possible values

(1) $f(1) = 1$ and $f(2) = 2$

(2) $f(1) = 2$ and $f(2) = 1$

And $f(4)$ can be $= 1, 2, 3, 4, 5, 6$.

$\therefore$ Total functions $= 2 \times 6 = 12$

Case 3 :

When $f(3) = 4$ then

(1) $f(1) = 1$ and $f(2) = 3$

(2) $f(1) = 2$ and $f(2) = 2$

(3) $f(1) = 3$ and $f(2) = 1$

And $f(4)$ can be $= 1, 2, 3, 4, 5, 6$

$\therefore$ Total functions $= 3 \times 6 = 18$

Case 4 :

When $f(3) = 5$ then

(1) $f(1) = 1$ and $f(4) = 4$

(2) $f(1) = 2$ and $f(4) = 3$

(3) $f(1) = 3$ and $f(4) = 2$

(4) $f(1) = 4$ and $f(4) = 1$

And $f(4)$ can be $= 1, 2, 3, 4, 5$ and 6

$\therefore$ Total functions $= 4 \times 6 = 24$

Case 5 :

When $f(3) = 6$ then

(1) $f(1) = 1$ and $f(2) = 5$

(2) $f(1) = 2$ and $f(2) = 4$

(3) $f(1) = 3$ and $f(2) = 3$

(4) $f(1) = 4$ and $f(2) = 2$

(5) $f(1) = 5$ and $f(2) = 1$

And $f(4)$ can be $= 1, 2, 3, 4, 5$ and 6

$\therefore$ Total possible functions $= 5 \times 6 = 30$

$\therefore$ Total functions from those 5 cases we get $= 6 + 12 + 18 + 24 + 30 = 90$

Question119

Let $f : \mathbb{N} \rightarrow \mathbb{R}$ be a function such that $f(x + y) = 2f(x)f(y)$ for natural numbers x and y . If $f(1) = 2$, then the value of α for which

$$\sum_{k=1}^{10} f(\alpha + k) = \frac{512}{3}(2^{20} - 1)$$

holds, is :

[25-Jun-2022-Shift-1]

Options:

A. 2

B. 3

C. 4

D. 6

Answer: C

Solution:

Given,

$$f(x+y) = 2f(x)f(y)$$

$$\text{and } f(1) = 2$$

For $x = 1$ and $y = 1$,

$$f(1+1) = 2f(1)f(1)$$

$$\Rightarrow f(2) = 2(f(1))^2 = 2(2)^2 = 2^3$$

For $x = 1, y = 2$

$$f(1+2) = 2f(1)f(2)$$

$$\Rightarrow f(3) = 2 \cdot 2 \cdot 2^3 = 2^5$$

For $x = 1, y = 3$

$$f(1+3) = 2f(1)f(3)$$

$$\Rightarrow f(4) = 2 \cdot 2 \cdot 2^5 = 2^7$$

For $x = 1, y = 4$

$$f(1+4) = 2f(1)f(4)$$

$$\Rightarrow f(5) = 2 \cdot 2 \cdot 2^7 = 2^9 \dots$$

Also given

$$\sum_{k=1}^{10} f(a+k) = \frac{512}{3}(2^{20} - 1)$$

$$\Rightarrow f(\alpha + 1) + f(\alpha + 2) + f(\alpha + 3) + \dots + f(\alpha + 10) = \frac{512}{3}(2^{20} - 1)$$

$$\Rightarrow f(\alpha + 1) + f(\alpha + 2) + f(\alpha + 3) + \dots + f(\alpha + 10) = \frac{2^9((2^2)^{10} - 1)}{2^2 - 1}$$

This represent a G.P with first term $= 2^9$ and common ratio $= 2^2$

$$\therefore \text{First term} = f(\alpha + 1) = 2^9 \dots (2)$$

$$\text{From equation (1), } f(5) = 2^9$$

$\therefore$ From (1) and (2), we get

$$f(\alpha + 1) = 2^9 = f(5)$$

$$\Rightarrow f(\alpha + 1) = f(5)$$

$$\Rightarrow f(\alpha + 1) = f(4 + 1)$$

Comparing both sides we get, $\alpha = 4$

Question120

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \left(2 \left(1 - \frac{x^{25}}{2} \right) (2 + x^{25}) \right)^{\frac{1}{50}}. \text{ If the function}$$

$g(x) = f(f(f(x)) + f(f(x)))$, then the greatest integer less than or equal to $g(1)$ is ____

[25-Jun-2022-Shift-1]

Answer: 2

Solution:

Given,

$$f(x) = \left(2 \left(1 - \frac{x^{25}}{2} \right) (2 + x^{25}) \right)^{\frac{1}{50}}$$

$$\text{and } g(x) = f(f(f(x))) + f(f(x))$$

$$\therefore g(1) = f(f(f(1))) + f(f(1))$$

$$\text{Now, } f(1) = \left(2 \left(1 - \frac{1^{25}}{2} \right) (2 + 1^{25}) \right)^{\frac{1}{50}}$$

$$= \left(2 \left(1 - \frac{1}{2} \right) (2 + 1) \right)^{\frac{1}{50}}$$

$$= (3)^{\frac{1}{50}}$$

$$\therefore f(f(1)) = f\left(3^{\frac{1}{50}}\right)$$

$$= \left(2 \left(1 - \frac{\left(3^{\frac{1}{50}}\right)^{25}}{2} \right) \left(2 + \left(3^{\frac{1}{50}}\right)^{25} \right) \right)^{\frac{1}{50}}$$

$$= \left(2 \left(1 - \frac{3^{\frac{1}{2}}}{2} \right) \left(2 + 3^{\frac{1}{2}} \right) \right)^{\frac{1}{50}}$$

$$= \left(2 \times \left(\frac{2 - \sqrt{3}}{2} \right) (2 + \sqrt{3}) \right)^{\frac{1}{50}}$$

$$= [(2 - \sqrt{3})(2 + \sqrt{3})]^{\frac{1}{50}}$$

$$= (4 - 3)^{\frac{1}{50}}$$

$$= 1^{\frac{1}{50}} = 1$$

$$\text{Now, } f(f(f(1))) = f(1) = 3^{\frac{1}{50}}$$

$$\therefore g(1) = f(f(f(1))) + f(f(1))$$

$$= 3^{\frac{1}{50}} + 1$$

Now, greatest integer less than or equal to $g(1)$

$$= [g(1)]$$

$$= \left[3^{\frac{1}{50}} + 1 \right]$$

$$= \left[3^{\frac{1}{50}} \right] + [1]$$

$$= [1.02] + 1$$

$$= 1 + 1 = 2$$

Question 121

Let $f(x) = \frac{x-1}{x+1}$, $x \in \mathbb{R} - \{0, -1, 1\}$. If $f^{n+1}(x) = f(f^n(x))$ for all $n \in \mathbb{N}$, then $f^6(6) + f^7(7)$ is equal to :
[26-Jun-2022-Shift-1]

Options:

A. $\frac{7}{6}$

B. $-\frac{3}{2}$

C. $\frac{7}{12}$

D. $-\frac{11}{12}$

Answer: B

Solution:

Given,

$$f(x) = \frac{x-1}{x+1}$$

Also given,

$$f^{n+1}(x) = f(f^n(x)) \dots (1)$$

$\therefore$ For $n = 1$

$$f^{1+1}(x) = f(f^1(x))$$

$$\Rightarrow f^2(x) = f(f(x))$$

$$= f\left(\frac{x-1}{x+1}\right)$$

$$= \frac{\frac{x-1}{x+1} - 1}{\frac{x-1}{x+1} + 1}$$

$$= \frac{\frac{x-1-x-1}{x+1}}{\frac{x-1+x+1}{x+1}}$$

$$= \frac{-2}{2x} = -\frac{1}{x}$$

From equation (1), when $n = 2$ $f^{2+1}(x) = f(f^2(x))$

$$\Rightarrow f^3(x) = f(f^2(x))$$

$$= f\left(-\frac{1}{x}\right)$$

$$= \frac{-\frac{1}{x} - 1}{-\frac{1}{x} + 1}$$

$$= \frac{\frac{-1-x}{-1}}{\frac{-1+x}{x}}$$

$$= \frac{-1-x}{-1+x} = \frac{-(x+1)}{x-1}$$

Similarly,

$$f^4(x) = f(f^3(x))$$

$$= f\left(\frac{-(x+1)}{x-1}\right)$$

$$= \frac{\frac{-(x+1)}{x-1} - 1}{\frac{-(x+1)}{x-1} + 1}$$

$$= \frac{\frac{x-1-x+1}{x-1}}{\frac{-x-1+x-1}{x-1}}$$

$$= \frac{-2x}{-2} = x$$

$$\therefore f^5(x) = f(f^4(x))$$

$$= f(x)$$

$$= \frac{x-1}{x+1}$$

$$f^6(x) = f(f^5(x))$$

$$= f\left(\frac{x-1}{x+1}\right)$$

$$= -\frac{1}{x} \text{ (Already calculated earlier)}$$

$$f^7(x) = f(f^6(x))$$

$$= f\left(-\frac{1}{x}\right)$$

$$= \frac{-\frac{1}{x} - 1}{-\frac{1}{x} + 1}$$

$$= \frac{-(x+1)}{x-1}$$

$$\therefore f^6(6) = -\frac{1}{6}$$

$$\text{and } f^7(7) = \frac{-(7+1)}{7-1} = -\frac{8}{6}$$

$$\text{So, } f^6(6) + f^7(7)$$

$$= -\frac{1}{6} - \frac{8}{6}$$

$$= -\frac{3}{2}$$

Question122

The range of the function,

$$f(x) = \log_{\sqrt{5}} \left(3 + \cos \left(\frac{3\pi}{4} + x \right) + \cos \left(\frac{\pi}{4} + x \right) + \cos \left(\frac{\pi}{4} - x \right) - \cos \left(\frac{3\pi}{4} - x \right) \right) \text{ is}$$

[2021, 01 Sep. Shift-II]

Options:

A. $(0, \sqrt{5})$

B. $[-2, 2]$

C. $\left[\frac{1}{\sqrt{5}}, \sqrt{5} \right]$

D. $[0, 2]$

Answer: D

Solution:

$$\begin{aligned} f(x) &= \log_{\sqrt{5}} \left(3 + \cos \left(\frac{3\pi}{4} + x \right) + \cos \left(\frac{\pi}{4} + x \right) \right. \\ &\quad \left. + \cos \left(\frac{\pi}{4} - x \right) - \cos \left(\frac{3\pi}{4} - x \right) \right) \\ &= \log_{\sqrt{5}} (3 - \sqrt{2} \sin x + \sqrt{2} \cos x) \\ \therefore -2 &\leq -\sqrt{2} \sin x + \sqrt{2} \cos x \leq 2 \end{aligned}$$

$$\Rightarrow 1 \leq 3 - \sqrt{2} \sin x + \sqrt{2} \cos x \leq 5$$

$$\Rightarrow \log_{\sqrt{5}} 1 \leq \log_{\sqrt{5}} (3 - \sqrt{2} \sin x + \sqrt{2} \cos x)$$

$$\Rightarrow 0 \leq f(x) \leq 2$$

$$\Rightarrow f(x) \in [0, 2]$$

Question123

The domain of the function $f(x) = \sin^{-1} \left(\frac{3x^2 + x - 1}{(x-1)^2} \right) + \cos^{-1} \left(\frac{x-1}{x+1} \right)$ is
[2021, 31 Aug. Shift-II]

Options:

- A. $\left[0, \frac{1}{4} \right]$
- B. $[-2, 0] \cup \left[\frac{1}{4}, \frac{1}{2} \right]$
- C. $\left[\frac{1}{4}, \frac{1}{2} \right] \cup \{0\}$
- D. $\left[0, \frac{1}{2} \right]$

Answer: C

Solution:

$$f(x) = \sin^{-1} \left(\frac{3x^2 + x - 1}{(x-1)^2} \right) + \cos^{-1} \left(\frac{x-1}{x+1} \right)$$

$$-1 \leq \frac{x-1}{x+1} \leq 1 \Rightarrow -1 - 1 \leq \frac{x-1}{x+1} - 1 \leq 1 - 1$$

$$\Rightarrow -2 \leq \frac{-2}{x+1} \leq 0 \Rightarrow 0 \leq \frac{1}{x+1} \leq 1$$

$$\Rightarrow 1 \leq x+1 < \infty$$

$$\Rightarrow 0 \leq x < \infty$$

$$\Rightarrow x \in [0, \infty)$$

$$\text{and } -1 \leq \frac{3x^2 + x - 1}{(x-1)^2} \leq 1$$

$$\Rightarrow -(x-1)^2 \leq 3x^2 + x - 1 \leq (x-1)^2, x \neq 1$$

$$\Rightarrow -(x^2 - 2x + 1) \leq 3x^2 + x - 1$$

$$\text{and } 3x^2 + x - 1 \leq x^2 - 2x + 1$$

$$\Rightarrow 4x^2 - x \geq 0$$

$$\text{and } 2x^2 + 3x - 2 \leq 0$$

$$\Rightarrow x(4x-1) \geq 0$$

$$\text{and } (x+2)(2x-1) \leq 0$$

$$\Rightarrow x \in (-\infty, 0] \cup \left[\frac{1}{4}, \infty \right)$$

$$\text{and } x \in \left[-2, \frac{1}{2} \right]$$

$$\Rightarrow x \in (-2, 0] \cup \left[\frac{1}{4}, \frac{1}{2} \right]$$

Domain of f in Eq. (i) $\cap$ Eq. (ii)

$$\therefore x \in \{0\} \cup \left[\frac{1}{4}, \frac{1}{2} \right]$$

Question124

Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function such that $f(m+n) = f(m) + f(n)$ for every $m, n \in \mathbb{N}$. If $f(6) = 18$, then $f(2) \cdot f(3)$ is equal to
[2021, 31 Aug. Shift-11]

Options:

A. 6

B. 54

C. 18

D. 36

Answer: B

Solution:

$$f(m+n) = f(m) + f(n), m, n \in \mathbb{N}$$

$$\therefore f(3+3) = f(3) + f(3)$$

$$\Rightarrow f(6) = 2f(3) = 18 \quad [\because f(6) = 18]$$

$$\begin{aligned} \text{Also } f(3) &= f(2+1) = f(2) + f(1) \\ &= f(1+1) + f(1) \end{aligned}$$

$$\begin{aligned}
 f(3) &= f(1) + f(1) + f(1) \\
 \Rightarrow 9 &= 3f(1) \Rightarrow f(1) = 3 \\
 \therefore f(2) &= f(1+1) = f(1) + f(1) = 3 + 3 = 6 \\
 \text{Hence, } f(2) \cdot f(3) &= 6 \cdot 9 = 54
 \end{aligned}$$

Question125

The domain of the function $\operatorname{cosec}^{-1}\left(\frac{1+x}{x}\right)$ is
[2021, 26 Aug. Shift-II]

Options:

A. $\left(-1, -\frac{1}{2}\right] \cup (0, \infty)$

B. $\left[-\frac{1}{2}, 0\right) \cup [1, \infty)$

C. $\left(-\frac{1}{2}, \infty\right) - \{0\}$

D. $\left[-\frac{1}{2}, \infty\right) - \{0\}$

Answer: D

Solution:

$$f(x) = \operatorname{cosec}^{-1}\left(\frac{1+x}{x}\right) \quad \left| \frac{1+x}{x} \right| \geq 1$$

Clearly, $x \neq 0$

$$\left| \frac{1+x}{x} \right|^2 \geq |x|^2$$

$$1 + x^2 + 2x \geq x^2$$

$$2x + 1 \geq 0$$

$$x \geq -\frac{1}{2}$$

So,

$$x \in \left[-\frac{1}{2}, \infty\right) - \{0\}$$

Question126

Which of the following is not correct for relation R on the set of real numbers ?

[2021, 31 Aug. Shift-1]

Options:

A. $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric.

B. $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$ is symmetric and transitive.

C. $(x, y) \in R \Leftrightarrow |x| - |y| \leq 1$ is reflexive but not symmetric.

D. $(x, y) \in R \Leftrightarrow |x - y| \leq 1$ is reflexive and symmetric.

Answer: B

Solution:

According to the question, let's consider option (b) (2, 3) and (3, 4) satisfy $0 < |x - y| \leq 1$ but (2, 4) does not satisfy it.

Question127

Let N be the set of natural numbers and a relation R on N be defined by $R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$. Then the relation R is

[2021, 27 July Shift-11]

Options:

A. symmetric but neither reflexive nor transitive.

B. reflexive but neither symmetric nor transitive.

C. reflexive and symmetric, but not transitive.

D. an equivalence relation.

Answer: B

Solution:

Given, relation R on N is defined by

$$R = \{(x, y) \in N \times N : x^3 - 3x^2 - xy^2 + 3y^3 = 0\}$$

$$x^3 - 3x^2y - xy^2 + 3y^3 = 0$$

$$\Rightarrow x^3 - xy^2 - 3x^2y + 3y^3 = 0$$

$$\Rightarrow x(x^2 - y^2) - 3y(x^2 - y^2) = 0$$

$$\Rightarrow (x - 3y)(x^2 - y^2) = 0$$

$$\Rightarrow (x - 3y)(x - y)(x + y) = 0$$

$$\text{Now, } x - x = 0$$

$$\Rightarrow x = x, \forall (x, x) \in N \times N$$

So, R is a reflexive relation.

But not symmetric and transitive relation because,

$(3, 1)$ satisfies but $(1, 3)$ does not. Also, $(3, 1)$ and

$(1, -1)$ satisfies but $(3, -1)$ does not.

Hence, relation R is reflexive but neither symmetric nor transitive.

Question128

Define a relation R over a class of $n \times n$ real matrices A and B as

" ARB , if there exists a non-singular matrix P such that $PAP^{-1} = B$ '.

Then which of the following is true ?

[2021, 18 March Shift-II]

Options:

A. R is symmetric, transitive but not reflexive.

B. R is reflexive, symmetric but not transitive.

C. R is an equivalence relation.

D. R is reflexive, transitive but not symmetric.

Answer: C

Solution:

For reflexive relation, $\forall (A, A) \in R$ for matrix P .

$$\Rightarrow A = PAP^{-1} \text{ is true for } P = 1$$

So, R is reflexive relation.

For symmetric relation,

Let $(A, B) \in R$ for matrix P .

$$\Rightarrow A = PBP^{-1} \text{ After pre-multiply by } P^{-1} \text{ and post-multiply by } P_1$$

we get

$$P^{-1}AP = B$$

So, $(B, A) \in R$ for matrix P^{-1} .
 So, R is a symmetric relation.
 For transitive relation,
 Let ARB and BRC
 So, $A = PBP^{-1}$ and $B = PCP^{-1}$
 Now, $A = P(PCP^{-1})P^{-1}$
 $\Rightarrow A = (P)^2 C (P^{-1})^2 \Rightarrow A = (P)^2 \cdot C \cdot (P^2)^{-1}$
 $\therefore (A, C) \in R$ for matrix P^2 .
 $\therefore R$ is transitive relation.
 Hence, R is an equivalence relation.

Question 129

Let $A = \{2, 3, 4, 5, \dots, 30\}$ and $' \sim '$ be an equivalence relation on $A \times A$, defined by $(a, b) \sim (c, d)$, if and only if $ad = bc$. Then, the number of ordered pairs, which satisfy this equivalence relation with ordered pair $(4, 3)$ is equal to
[2021, 16 March Shift-II]

Options:

- A. 5
- B. 6
- C. 8
- D. 7

Answer: D

Solution:

$$A = \{2, 3, 4, 5, \dots, 30\}$$

$$a = bc$$

$$\therefore (a, b)R(4, 3)$$

$$\Rightarrow 3a = 4b$$

$$\Rightarrow a = \left(\frac{4}{3}\right)b$$

b must be a multiple of 3, b can be $(3, 6, 9, \dots, 30)$.

Also, a must be less than or equal to 30.

$$(a, b) = (4, 3), (8, 6), (12, 9), (16, 12), (20, 15)$$

$$(24, 18), (28, 21)$$

$\Rightarrow 7$ ordered pairs

Question130

Let $R = \{(P, O) \mid P \text{ and } Q \text{ are at the same distance from the origin}\}$ be a relation, then the equivalence class of $(1, -1)$ is the set
[2021, 26 Feb. Shift-1]

Options:

A. $S = \{(x, y) \mid x^2 + y^2 = 4\}$

B. $S = \{(x, y) \mid x^2 + y^2 = 1\}$

C. $S = \{(x, y) \mid x^2 + y^2 = \sqrt{2}\}$

D. $S = \{(x, y) \mid x^2 + y^2 = 2\}$

Answer: D

Solution:

Let $P(a, b)$ and $Q(c, d)$ are any two points.

Given, $OP = OQ$

i.e. $\sqrt{a^2 + b^2} = \sqrt{c^2 + d^2}$

Squaring on both sides,

$$R = \{((a, b), (c, d)) : a^2 + b^2 = c^2 + d^2\}$$

$R(x, y), S(1, -1), OR = OS$ (equivalence class)

This gives $OR = \sqrt{x^2 + y^2}$ and $OS = \sqrt{2}$

$$1 \Rightarrow \sqrt{x^2 + y^2} = \sqrt{2}$$

$$\Rightarrow x^2 + y^2 = 2 \quad (\text{Squaring on both sides})$$

$$\therefore S = \{(x, y) : x^2 + y^2 = 2\}$$

Question131

Let $S = \{1, 2, 3, 4, 5, 6, 7\}$. Then the number of possible functions $f : S \rightarrow S$ such that $f(m \cdot n) = f(m) \cdot f(n)$ for every $m, n \in S$ and $m \cdot n \in S$ is equal to
[2021, 27 July Shift-I]

Answer: 490

Solution:

$$S = \{1, 2, 3, 4, 5, 6, 7\}$$

$$f : S \rightarrow S$$

$$f(m \cdot n) = f(m)f(n)$$

$$m, n \in S \Rightarrow mn \in S$$

$$\text{If } mn \in S \Rightarrow mn \leq 7$$

$$\text{So, } (1 \cdot 1, 1 \cdot 2, \dots, 1 \cdot 7) \leq 7$$

$$(2 \cdot 2, 2 \cdot 3) \leq 7$$

$$\text{When } m = 1, f(n) = f(1) \cdot f(n) \Rightarrow f(1) = 1$$

$$\text{When } m = n = 2,$$

$$f(4) = f(2)f(2) = \begin{cases} f(2) = 1 \Rightarrow f(4) = 1 & \text{or} \\ f(2) = 2 \Rightarrow f(4) = 4. \end{cases}$$

$$\text{When, } m = 2, n = 3$$

$$f(6) = f(2)f(3) \begin{cases} \text{When, } f(2) = 1 \\ f(3) = 1 \text{ to } 7 \\ \text{When, } f(2) = 2 \\ f(3) = 1 \text{ or } 2 \text{ or } 3. \end{cases}$$

And $f(5), f(7)$ can take any value (1-7) [$\because f(5) = f(1) \cdot f(5) \leq 7$ and $f(7) = f(1) \cdot f(7) \leq 7$] The possible combination is

$$1 \mid f(1) = 1 \quad f(1) = 1$$

$$f(2) = 1 \quad f(2) = 2$$

$$f(3) = (1-7) \quad f(3) = (1-3)$$

$$f(4) = 1 \quad f(4) = 4$$

$$f(5) = (1-7) \quad f(5) = (1-7)$$

$$f(6) = f(3) \quad f(6) = f(3)$$

$$f(7) = (1-7) \quad f(7) = (1-7)$$

$$\text{So, total} = (1 \times 1 \times 7 \times 1 \times 7 \times 1 \times 7)$$

$$+ (1 \times 1 \times 3 \times 1 \times 7 \times 1 \times 7)$$

$$= 490$$

Question132

If $[x]$ be the greatest integer less than or equal to x , then $\sum_{n=8}^{100} \left[\frac{(-1)^n n}{2} \right]$ is equal to
[25 July 2021, Shift-III]

Options:

A. 0

- B. 4
- C. -2
- D. 2

Answer: D

Solution:

We have,

$$\sum_{n=8}^{100} \left[\frac{(-1)^n n}{2} \right] (\because [x] \text{ is the greatest integer function})$$

Substitute the values of n

$$\begin{aligned} &= [4] + [-4.5] + [5] + [-5.5] \\ &\quad + \dots + [-49.5] + [50] \\ &= 4 - 5 + 5 - 6 + \dots - 50 + 50 \\ &= 4 \end{aligned}$$

Question133

If the domain of the function $f(x) = \frac{\cos^{-1} \sqrt{x^2 - x + 1}}{\sqrt{\sin^{-1} \left(\frac{2x-1}{2} \right)}}$ is the interval (α, β) ,

then $\alpha + \beta$ is equal to
[2021, 22 July Shift-II]

Options:

- A. $\frac{3}{2}$
- B. 2
- C. $\frac{1}{2}$
- D. 1

Answer: A

Solution:

$$f(x) = \frac{\cos^{-1} \sqrt{x^2 - x + 1}}{\sqrt{\sin^{-1} \left(\frac{2x-1}{2} \right)}}$$

$$\Rightarrow x \in \mathbb{R}, \quad x(x-1) \leq 0$$

$$x^2 - x + 1 \geq 0 \text{ and } x^2 - x + 1 \leq 1$$

$$0 \leq x \leq 1 \quad \dots\dots\dots (i) \Rightarrow 0 < \sin^{-1} \left(\frac{2x-1}{2} \right) < \frac{\pi}{2}$$

$$\Rightarrow 0 < \frac{2x-1}{2} < 1$$

$$\Rightarrow \frac{1}{2} < x < \frac{3}{2} \quad \dots\dots\dots (ii)$$

$$(A) \cap (B) = x \in \left(\frac{1}{2}, 1 \right]$$

$$\therefore \alpha + \beta = \frac{3}{2}$$

Question 134

Let $[x]$ denote the greatest integer $\leq x$, where $x \in \mathbb{R}$. If the domain of the real valued function

$$f(x) = \sqrt{\frac{[x]-2}{[x]-3}}$$
 is $(-\infty, a) \cup [b, c)$

$\cup [u, \infty)$, $a < b < c$, then the value of $a + b + c$ is

[2021, 20 July Shift-I]

Options:

A. 8

B. 1

C. -2

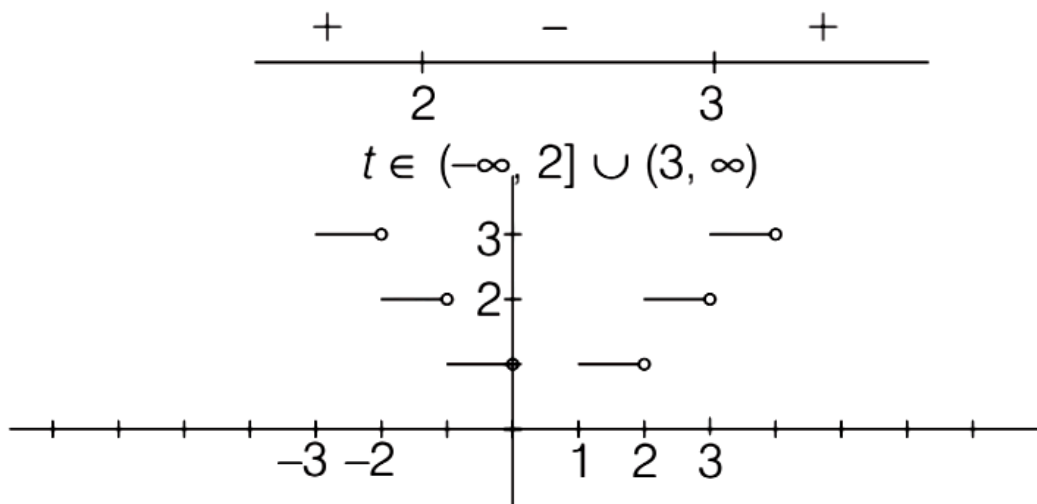
D. -3

Answer: C

Solution:

$$f(x) = \sqrt{\frac{[x]-2}{[x]-3}} \geq 0$$

$$\text{Let } [x] = t$$



$$| [x] | = 3 \Rightarrow x \in [-3, -2) \cup [3, 4)$$

Domain of $x = [-\infty, -3) \cup [-2, 3) \cup [4, \infty)$

$$a = -3$$

$$b = -2$$

$$c = 3$$

$$\therefore a + b + c = -3 + (-2) + 3 = -2$$

Question135

The real valued function $f(x) = \frac{\operatorname{cosec}^{-1} x}{\sqrt{x - [x]}}$, where $[x]$ denotes the greatest integer less than or equal to x , is defined for all x belonging to
[2021, 18 March Shift-I]

Options:

- A. all reals except integers
- B. all non-integers except the interval $[-1, 1]$
- C. all integers except 0, -1, 1
- D. all reals except the interval $[-1, 1]$

Answer: B

Solution:

$$\text{Given, } f(x) = \frac{\operatorname{cosec}^{-1} x}{\sqrt{x - [x]}}$$

$$\Rightarrow f(x) = \frac{\operatorname{cosec}^{-1} x}{\sqrt{\{x\}}}$$

For $f(x)$ to be defined,

$$\begin{cases} |x| \geq 1 \\ \{x\} > 0. \end{cases} \Rightarrow \begin{cases} x \leq -1 \text{ or } x \geq 1 \\ x \neq \text{integers} \end{cases}$$

i.e. $x \in (-\infty, -1] \cup [1, \infty) - \{\text{integers}\}$

i.e. all non-integers except the interval $[-1, 1]$

(here, -1 and 1 are included in except case, because of -1 and 1 are integers).

Question 136

If the functions are defined as $f(x) = \sqrt{x}$ and $g(x) = \sqrt{1-x}$, then what is the common domain of the following functions?

$f + g, f - g, f/g, g/f, g - f$, where

$$(f \pm g)(x) = f(x) \pm g(x), (f/g)(x) = \frac{f(x)}{g(x)}$$

[2021, 18 March, Shift-1]

Options:

A. $0 \leq x \leq 1$

B. $0 \leq x < 1$

C. $0 < x < 1$

D. $0 < x \leq 1$

Answer: C

Solution:

$$\text{Given, } f(x) = \sqrt{x} \text{ and } g(x) = \sqrt{1-x}$$

$$\therefore \text{Domain of } f(x) = D_1 \text{ is } x \geq 0$$

$$\text{i.e. } D_1 : x \in (0, \infty)$$

$$\text{and domain of } g(x) = D_2 \text{ is } 1 - x \geq 0 \Rightarrow x \leq 1$$

$$\text{i.e. } D_2 : x \in (-\infty, 1]$$

As, we know that, the domain of $f + g, f - g, g - f$ will be $D_1 \cap D_2$ as well as the domain for $\frac{f}{g}$ is $D_1 \cap D_2$ except all

those value(s) of x , such that $g(x) = 0$.

Similarly, for $\frac{g}{f}$ is $D_1 \cap D_2$ but $f(x) \neq 0$.

Hence, common domain for $(f + g)$, $(f - g)$, $\left(\frac{f}{g}\right)$, $\left(\frac{g}{f}\right)$ and $(g - f)$ will be $0 < x < 1$

Question 137

A function $f(x)$ is given by $f(x) = \frac{5^x}{5^x + 5}$, then the sum of the series

$$f\left(\frac{1}{20}\right) + f\left(\frac{2}{20}\right) + f\left(\frac{3}{20}\right) + \dots + f\left(\frac{39}{20}\right)$$

is equal to

[2021, 25 Feb. Shift-II]

Options:

A. $\frac{29}{2}$

B. $\frac{49}{2}$

C. $\frac{39}{2}$

D. $\frac{19}{2}$

Answer: C

Solution:

Given, $f(x) = \frac{5^x}{5^x + 5}$, then,

$$\begin{aligned} f(2-x) &= \frac{5^{2-x}}{5^{2-x} + 5} \\ &= \frac{5}{5^x + 5} \end{aligned}$$

This gives, $f(x) + f(2-x) = \frac{5^x + 5}{5^x + 5} = 1 \Rightarrow f\left(\frac{1}{20}\right) + f\left(2 - \frac{1}{20}\right) = f\left(\frac{1}{20}\right) + f\left(\frac{39}{20}\right) = 1$

Similarly,

$$cf\left(\frac{2}{20}\right) + f\left(\frac{38}{20}\right) = 1 \text{ and so on,}$$

$$\begin{aligned}
&\therefore f\left(\frac{1}{20}\right) + f\left(\frac{2}{20}\right) + \dots + f\left(\frac{38}{20}\right) + f\left(\frac{39}{20}\right) \\
&= 1 + 1 + \dots + 1 + f\left(\frac{20}{20}\right) \\
&= 19 + f(1) = 19 + \frac{1}{2} = \frac{39}{2}
\end{aligned}$$

Question 138

If $a + \alpha = 1$, $b + \beta = 2$ and $af(x) + \alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x}$, $x \neq 0$, then the value

of expression $\frac{f(x) + f\left(\frac{1}{x}\right)}{x + \frac{1}{x}}$ is

[2021, 24 Feb. Shift-II]

Answer: 2

Solution:

Given, $a + \alpha = 1$

$b + \beta = 2$

$$\therefore a \cdot f(x) + \alpha \cdot f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x} \quad \dots\dots\dots (i)$$

Replace x by $\frac{1}{x}$,

$$af\left(\frac{1}{x}\right) + af(x) = \frac{b}{x} + \beta x$$

Adding Eqs. (i) and (ii), we get

$$(a + \alpha) \left[f(x) + f\left(\frac{1}{x}\right) \right] = \left(x + \frac{1}{x} \right) (b + \beta) \quad \dots\dots\dots (ii)$$

$$\Rightarrow \frac{f(x) + f\left(\frac{1}{x}\right)}{x + \frac{1}{x}} = \frac{b + \beta}{a + \alpha} = \frac{2}{1} = 2$$

Question 139

Let $f(x) = \sin^{-1}x$ and

$$g(x) = \frac{x^2 - x - 2}{2x^2 - x - 6}$$

If $g(2) = \lim_{x \rightarrow 2} g(x)$, then the domain of the function $f \circ g$ is

[2021, 26 Feb. Shift-II]

Options:

A. $(-\infty, -2] \cup \left[-\frac{3}{2}, \infty\right)$

B. $(-\infty, -2] \cup [-1, \infty)$

C. $(-\infty, -2] \cup \left[-\frac{4}{3}, \infty\right)$

D. $(-\infty, -1] \cup [2, \infty)$

Answer: C

Solution:

Given, $g(x) = \frac{x^2 - x - 2}{2x^2 - x - 6}$, $f(x) = \sin^{-1}x$

$$f(g(x)) = \sin^{-1}(g(x))$$

$$f \circ g(x) = \sin^{-1}\left(\frac{x^2 - x - 2}{2x^2 - x - 6}\right)$$

For the domain of $f \circ g(x)$,

$$|g(x)| \leq 1$$

$$[\because \text{Domain of } f(x) \text{ is } [-1, 1]]$$

$$\Rightarrow \left| \frac{x^2 - x - 2}{2x^2 - x - 6} \right| \leq 1$$

$$\Rightarrow \left| \frac{(x+1)(x-2)}{(2x+3)(x-2)} \right| \leq 1$$

$$\Rightarrow \left| \frac{x+1}{2x+3} \right| \leq 1$$

$$\Rightarrow -1 \leq \frac{x+1}{2x+3} \leq 1$$

$$\Rightarrow \left(\frac{x+1}{2x+3} \right)^2 \leq 1$$

$$\Rightarrow (x+1)^2 \leq (2x+3)^2$$

$$\Rightarrow 3x^2 + 10x + 8 \geq 0$$

$$\Rightarrow (3x+y)(x+2) \geq 0$$

This implies,

$$x \in (-\infty, -2] \cup \left[-\frac{4}{3}, \infty\right)$$

Question 140

Let $g : \mathbb{N} \rightarrow \mathbb{N}$ be defined as

$$g(3n+1) = 3n+2$$

$$g(3n+2) = 3n+3,$$

$$g(3n+3) = 3n+1, \text{ for all } n \geq 0.$$

Then which of the following statements is true?

[2021, 25 July Shift-1]

Options:

A. There exists an onto function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f \circ g = f$

B. There exists a one-one function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f \circ g = f$

C. $g \circ g \circ g = g$

D. There exists a function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $g \circ f = f$

Answer: A

Solution:

$$g(3n+1) = 3n+2$$

$$g(3n+2) = 3n+3$$

$$g(3n+3) = 3n+1, \text{ for all } n \geq 0$$

$$g : \mathbb{N} \rightarrow \mathbb{N}$$

$$g(1) = 2, g(4) = 5, g(7) = 8$$

$$g(2) = 3, g(5) = 6, g(8) = 9$$

$$g(3) = 1, g(6) = 4, g(9) = 7$$

$$\Rightarrow f[g(1)] = f(1)$$

$$\Rightarrow f(2) = f(1)$$

Clearly, it is not a one - one function.

$$\text{Now, } f[g(2)] = f(2)$$

$$f(3) = f(2)$$

$$\text{And, } f[g(3)] = f(3)$$

$$f(1) = f(3)$$

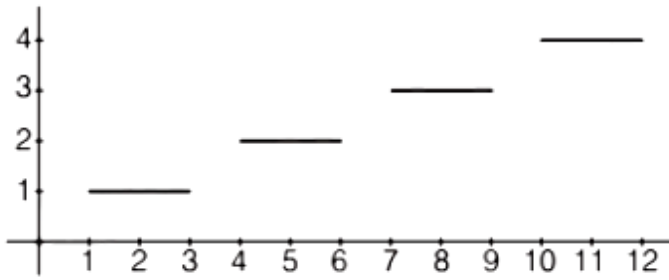
$$\text{Similarly, } f[g(4)] = f(4)$$

$$f(5) = f(4)$$

And, so on

$$f(1) = f(2) = f(3)$$

$$f(4) = f(5) = f(6)$$



Now, there can be a possibility such that

So, $f(x)$ can be onto function.

$$\text{When } f(1) = f(2) = f(3) = 1$$

$$f(4) = f(5) = f(6) = 2$$

and so on.

Question141

Consider function $f : A \rightarrow B$ and $g : B \rightarrow C$ ($A, B, C \subseteq \text{eqR}$) such that $(g \circ f)^{-1}$ exists, then
[2021, 25 July Shift-II]

Options:

- A. f and g both are one-one
- B. f and g both are onto
- C. f is one-one and g is onto
- D. f is onto and g is one-one

Answer: C

Solution:

Given functions, $f : A \rightarrow B$ and $g : B \rightarrow C$ ($A, B, C \subseteq \text{eqR}$)

$\therefore (g \circ f)^{-1}$ exists $\Rightarrow g \circ f$ is a bijective function.

$\Rightarrow 'f'$ must be 'one-one' and ' g ' must be 'onto' function.

Question142

Let $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Then, the number of bijective functions $F : A \rightarrow A$ such that $f(1) + f(2) = 3 - f(3)$ is equal to

[2021, 22 July Shift-III]

Answer: 720

Solution:

$$f(1) + f(2) = 3 - f(3)$$

$$A = \{0, 1, 2, 3, 4, 5, 6, 7\}$$

$$f : A \rightarrow A$$

$$\text{So, } f(1) + f(2) + f(3) = 3$$

$0 + 1 + 2 = 3$ is the only possibility.

So, $f(0)$ can be either 0 or 1 or 2 .

Similarly, $f(1)$ and $f(2)$ can be 0, 1 and 2 .

and $\{3, 4, 5, 6, 7\} \rightarrow \{3, 4, 5, 6, 7\}$

They have $5!$ choices.

And $\{0, 1, 2\}$

They have $3!$ choices.

Number of bijective functions

$$= 3! \times 5! = 720$$

Question 143

Let $f : \mathbb{R} - \left\{ \frac{\alpha}{6} \right\} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \frac{5x + 3}{6x - \alpha}$$

Then, the value of α for which $(f \circ f)(x) = x$, for all $x \in \mathbb{R} - \left\{ \frac{\alpha}{6} \right\}$ is

[2021, 20 July Shift-II]

Options:

A. No such α exists

B. 5

C. 8

D. 6

Answer: B

Solution:

$$f(x) = \frac{5x+3}{6x-\alpha}$$

$$\text{Now, } f \circ f(x) = f\left(\frac{5x+3}{6x-\alpha}\right)$$

$$= \frac{5\left(\frac{5x+3}{6x-\alpha}\right) + 3}{6\left(\frac{5x+3}{6x-\alpha}\right) - \alpha}$$

$$= \frac{5(5x+3) + 3(6x-\alpha)}{6(5x+3) - \alpha(6x-\alpha)}$$

$$= \frac{5(5x+3) + 3(6x-\alpha)}{6(5x+3) - \alpha(6x-\alpha)}$$

$$\text{Given, } f \circ f(x) = x$$

$$\Rightarrow \frac{5(5x+3) + 3(6x-\alpha)}{6(5x+3) - \alpha(6x-\alpha)} = x$$

$$\Rightarrow 25x + 15 + 18x - 3\alpha$$

$$= 30x^2 + 18x - 6\alpha x^2 + \alpha^2 x$$

$$\Rightarrow x^2(30 - 6\alpha) - x(\alpha^2 - 25) + 3\alpha - 15 = 0$$

Comparing coefficients,

$$30 - 6\alpha = 0$$

$$\Rightarrow 6\alpha = 30$$

$$\Rightarrow \alpha = 5$$

Question 144

Let $f : \mathbb{R} - \{3\} \rightarrow \mathbb{R} - \{1\}$ be defined by $f(x) = \frac{x-2}{x-3}$. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be given as $g(x) = 2x - 3$. Then, the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to

[2021, 18 March Shift-II]

Options:

A. 7

B. 2

C. 5

D. 3

Answer: C

Solution:

$$\text{Given, } f(x) = \frac{x-2}{x-3}$$

$$g(x) = 2x - 3$$

$$\text{Let } y = f(x) = \frac{x-2}{x-3}$$

$$\Rightarrow xy - 3y = x - 2 \Rightarrow xy - x = 3y - 2$$

$$\Rightarrow x(y-1) = 3y-2$$

$$\Rightarrow x = \frac{3y-2}{y-1}$$

$$\Rightarrow f^{-1}(y) = \frac{3y-2}{y-1}$$

$$\Rightarrow f^{-1}(x) = \frac{3x-2}{x-1}$$

$$\text{Similarly, } g^{-1}(x) = \frac{x+3}{2}$$

$$\text{Given, } f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$$

$$\Rightarrow \frac{3x-2}{x-1} + \frac{x+3}{2} = \frac{13}{2}$$

$$\Rightarrow x^2 + 8x - 7 = 13(x-1)$$

$$\Rightarrow x^2 - 5x + 6 = 0$$

$$\Rightarrow (x-2)(x-3) = 0$$

$$\Rightarrow x = 2, 3$$

$$\therefore \text{Sum} = 2 + 3 = 5$$

Question145

**The inverse of $y = 5^{\log x}$ is
[2021, 17 March Shift-I]**

Options:

A. $x = 5^{\log y}$

B. $x = y^{\log 5}$

C. $x = y^{\frac{1}{\log 5}}$

D. $x = 5^{\frac{1}{\log y}}$

Answer: C

Solution:

$$y = 5^{\log x}$$

Taking log on both sides,

$$\Rightarrow \frac{\log y}{\log 5} = \log x \cdot \log 5$$

$$\Rightarrow \frac{1}{\log 5} = \frac{\log x}{\log y}$$

$$\frac{1}{\log 5} = \log_y x$$

$$x = y^{\frac{1}{\log 5}}$$

Question 146

Let $A = \{1, 2, 3, \dots, 10\}$ and $f : A \rightarrow A$ be defined as defined as

$$f(x) = \begin{cases} x+1 & \text{if } x \text{ is odd} \\ x & \text{if } x \text{ is even} \end{cases}.$$

Then, the number of possible functions $g : A \rightarrow A$, such that $g \circ f = f$ is
[2021, 26 Feb. Shift-II]

Options:

A. 10^5

B. ${}^{10}C_5$

C. 5^5

D. $5!$

Answer: A

Solution:

$$f(x) = \begin{cases} x+1 & \text{if } x \text{ is odd} \\ x & \text{if } x \text{ is even} \end{cases}.$$

Given, $g : A \rightarrow A$ such that,

$$g(f(x)) = f(x)$$

When x is even, then

$$g(x) = x$$

When x is odd, then

$$g(x+1) = x+1$$

This implies,

$$g(x) = x, x \text{ is even.}$$

$\Rightarrow$ If x is odd, then $g(x)$ can take any value in set A .

So, number of $g(x) = 10^5$

Question 147

Let $f, g : \mathbb{N} \rightarrow \mathbb{N}$, such that $f(n+1) = f(n) + f(1) \forall n \in \mathbb{N}$ and g be any arbitrary function. Which of the following statements is not true? [2021, 25 Feb. Shift-1]

Options:

A. if $f \circ g$ is one-one, then g is one-one.

B. if f is onto, then $f(n) = n, \forall n \in \mathbb{N}$.

C. f is one-one.

D. if g is onto, then $f \circ g$ is one-one.

Answer: D

Solution:

Given, $f(n+1) = f(n) + f(1), \forall n \in \mathbb{N}$

$$\Rightarrow f(n+1) - f(n) = f(1)$$

It is an AP with common difference $= f(1)$

Also, general term

$$11 = T_n = f(1) + (n-1)f(1) = nf(1)$$

$$\Rightarrow f(n) = nf(1)$$

Clearly, $f(n)$ is one-one.

For $f \circ g$ to be one-one, g must be one-one.

For f to be onto, $f(n)$ should take all the values of natural numbers.

As, $f(x)$ is increasing, $f(1) = 1$

$$\Rightarrow f(n) = n$$

If g is many-one, then $f \circ g$ is many one.

So, if g is onto, then $f \circ g$ is one-one.

Question 148

Let x denote the total number of one-one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one-one functions from the set A to the set $A \times B$. Then, [2021, 25 Feb. Shift-II]

Options:

A. $2y = 91x$

B. $2y = 273x$

C. $y = 91x$

D. $y = 273x$

Answer: A

Solution:

$$x = \{ f : A \rightarrow B, f \text{ is one - one } \}$$

$$y = \{ g : A \rightarrow A \times B, g \text{ is one one } \}$$

Number of elements in $A = 3$ i.e. $|A| = 3$

Similarly, $|B| = 5$

$$\text{Then, } |A \times B| = |A| \times |B| = 3 \times 5 = 15$$

Now, number of one-one function from A to B will be

$${}^5P_3 = \frac{5!}{(5-3)!} = \frac{5!}{2!} = 5 \times 4 \times 3 = 60$$

$$\therefore x = 60$$

Now, number of one-one function from A

$$\text{to } A \times B \text{ will be } = {}^{15}P_3 = \frac{15!}{(15-3)!} = \frac{15!}{12!}$$

$$= 15 \times 14 \times 13 = 2730$$

$$\therefore y = 2730 \text{ i.e. } \therefore y = 2730$$

$$\text{Thus, } 2 \times (2730) = 91 \times (60)$$

$$\Rightarrow 2y = 91x$$

Question 149

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 2x - 1$ and $g : \mathbb{R} - \{1\} \rightarrow \mathbb{R}$ be

defined as $g(x) = \frac{x - \frac{1}{2}}{x - 1}$. Then the composition function $f(g(x))$ is :

24 Feb 2021 Shift 1

Options:

- A. onto but not one-one
- B. both one-one and onto
- C. one-one but not onto
- D. neither one-one nor onto

Answer: C

Solution:

$$f(g(x)) = 2g(x) - 1 = 2 \left(\frac{2x-1}{2(x-1)} \right) - 1 = \frac{x}{x-1} = 1 + \frac{1}{x-1}$$

Range of $f(g(x)) = \mathbb{R} - \{1\}$

Range of $f(g(x))$ is not onto

& $f(g(x))$ is one-one

So, $f(g(x))$ is one-one but not onto.

Question 150

Let R_1 and R_2 be two relations defined as follows :

$$R_1 = \{(a, b) \in \mathbb{R}^2 : a^2 + b^2 \in \mathbb{Q}\} \text{ and } R_2 = \{(a, b) \in \mathbb{R}^2 : a^2 + b^2 \notin \mathbb{Q}\},$$

where $\mathbb{Q}$ is the set of all rational numbers. Then :

[Sep. 03, 2020 (II)]

Options:

- A. Neither R_1 nor R_2 is transitive.
- B. R_2 is transitive but R_1 is not transitive.
- C. R_1 is transitive but R_2 is not transitive.
- D. R_1 and R_2 are both transitive.

Answer: A

Solution:

(a) For R_1 let $a = 1 + \sqrt{2}$, $b = 1 - \sqrt{2}$, $c = 8^{1/4}$

$$aR_1b \Rightarrow a^2 + b^2 = (1 + \sqrt{2})^2 + (1 - \sqrt{2})^2 = 6 \in \mathbb{Q}$$

$$bR_1c \Rightarrow b^2 + c^2 = (1 - \sqrt{2})^2 + (8^{1/4})^2 = 3 \in \mathbb{Q}$$

$$aR_1c \Rightarrow a^2 + c^2 = (1 + \sqrt{2})^2 + (8^{1/4})^2 = 3 + 4\sqrt{2} \notin \mathbb{Q}$$

$\therefore R_1$ is not transitive.

For R_2 let $a = 1 + \sqrt{2}$, $b = \sqrt{2}$, $c = 1 - \sqrt{2}$

$$aR_2b \Rightarrow a^2 + b^2 = (1 + \sqrt{2})^2 + (\sqrt{2})^2 = 5 + 2\sqrt{2} \notin \mathbb{Q}$$

$$bR_2c \Rightarrow b^2 + c^2 = (\sqrt{2})^2 + (1 - \sqrt{2})^2 = 5 - 2\sqrt{2} \notin \mathbb{Q}$$

$$aR_2c \Rightarrow a^2 + c^2 = (1 + \sqrt{2})^2 + (1 - \sqrt{2})^2 = 6 \in \mathbb{Q}$$

$\therefore R_2$ is not transitive.

Question151

The domain of the function $f(x) = \sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$ is $(-\infty, -a] \cup [a, \infty)$.

Then a is equal to :

[Sep. 02, 2020 (I)]

Options:

A. $\frac{\sqrt{17}}{2}$

B. $\frac{\sqrt{17}-1}{2}$

C. $\frac{1+\sqrt{17}}{2}$

D. $\frac{\sqrt{17}}{2} + 1$

Answer: C

Solution:

$$\because f(x) = \sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$$

$$\therefore -1 \leq \frac{|x|+5}{x^2+1} \leq 1$$

$$\Rightarrow |x| + 5 \leq x^2 + 1$$

$$[\because x^2 + 1 \neq 0]$$

$$\Rightarrow x^2 - |x| - 4 \geq 0$$

$$\Rightarrow \left(|x| - \frac{1 - \sqrt{17}}{2} \right) \left(|x| - \frac{1 + \sqrt{17}}{2} \right) \geq 0$$

$$\Rightarrow x \in \left(-\infty, -\frac{1 + \sqrt{17}}{2} \right) \cup \left[\frac{1 + \sqrt{17}}{2}, \infty \right)$$

$$\therefore a = \frac{1 + \sqrt{17}}{2}$$

Question152

If $R = \{(x, y) : x, y \in \mathbb{Z}, x^2 + 3y^2 \leq 8\}$ is a relation on the set of integers $\mathbb{Z}$, then the domain of R^{-1} is :
[Sep. 02, 2020 (I)]

Options:

- A. $\{-2, -1, 1, 2\}$
- B. $\{0, 1\}$
- C. $\{-2, -1, 0, 1, 2\}$
- D. $\{-1, 0, 1\}$

Answer: D

Solution:

$$\text{Since, } R = \{(x, y) : x, y \in \mathbb{Z}, x^2 + 3y^2 \leq 8\}$$

$$\therefore R = \{(1, 1), (2, 1), (1, -1), (0, 1), (1, 0)\}$$

$$\Rightarrow D_{R^{-1}} = \{-1, 0, 1\}$$

Question153

Let $[t]$ denote the greatest integer $\leq t$. Then the equation in x , $[x]^2 + 2[x + 2] - 7 = 0$ has :
[Sep. 04, 2020 (I)]

Options:

- A. exactly two solutions
- B. exactly four integral solutions
- C. no integral solution
- D. infinitely many solutions

Answer: D

Solution:

The given equation $[x]^2 + 2[x] + 4 - 7 = 0$
 $\Rightarrow [x]^2 + 2[x] - 3 = 0$
 $\Rightarrow [x]^2 + 3[x] - [x] - 3 = 0$
 $\Rightarrow ([x] + 3)([x] - 1) = 0 \Rightarrow [x] = 1 \text{ or } -3 \Rightarrow x \in [-3, -2) \cup [1, 2)$
 $\therefore$ The equation has infinitely many solutions.

Question154

**Let $f(x)$ be a quadratic polynomial such that $f(-1) + f(2) = 0$. If one of the roots of $f(x) = 0$ is 3, then its other root lies in:
[Sep. 02, 2020 (II)]**

Options:

- A. (-1,0)
- B. (1,3)
- C. (-3,-1)
- D. (0,1)

Answer: A

Solution:

Let $f(x) = ax^2 + bx + c$

Given : $f(-1) + f(2) = 0$

$$a - b + c + 4a + 2b + c = 0$$

$$\Rightarrow 5a + b + 2c = 0 \dots\dots(i)$$

and $f(3) = 0 \Rightarrow 9a + 3b + c = 0 \dots\dots(ii)$

From equations (i) and (ii),

$$\frac{a}{1-6} = \frac{b}{18-5} = \frac{c}{15-9} \Rightarrow \frac{a}{-5} = \frac{b}{13} = \frac{c}{6}$$

Product of roots, $\alpha\beta = \frac{c}{a} = \frac{-6}{5}$ and $\alpha = 3$

$$\Rightarrow \beta = \frac{-2}{5} \in (-1, 0)$$

Question155

Let $f(1, 3) \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{x[x]}{1+x^2}$ where $[x]$ denotes the greatest integer $\leq x$. Then the range of f is:
[Jan. 8, 2020 (II)]

Options:

A. $\left(\frac{2}{5}, \frac{3}{5}\right) \cup \left(\frac{3}{4}, \frac{4}{5}\right)$

B. $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right)$

C. $\left(\frac{2}{5}, \frac{4}{5}\right)$

D. $\left(\frac{3}{5}, \frac{4}{5}\right)$

Answer: B

Solution:

$$f(x) \begin{cases} \frac{x}{x^2+1} & x \in (1, 2) \\ \frac{2x}{x^2+1} & x \in [2, 3). \end{cases}$$

$$f'(x) \begin{cases} \frac{1-x^2}{1+x^2} & x \in (1, 2) \\ \frac{1-2x^2}{1+x^2} & x \in [2, 3). \end{cases}$$

$\therefore f(x)$ is a decreasing function

$$\therefore y \in \left(\frac{2}{5}, \frac{1}{2} \right) \cup \left(\frac{6}{10}, \frac{4}{5} \right]$$

$$\Rightarrow y \in \left(\frac{2}{5}, \frac{1}{2} \right) \cup \left(\frac{3}{5}, \frac{4}{5} \right]$$

Question156

Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$. Then the number of elements in the set $C = \{f : A \rightarrow B \mid 2 \in f(A) \text{ and } f \text{ is not one-one}\}$ is _____.
[NA Sep. 05,2020 (II)]

Answer: 19

Solution:

The desired functions will contain either one element or two elements in its codomain of which '2' always belongs to $f(A)$.

$\therefore$ The set B can be $\{2\}, \{1, 2\}, \{2, 3\}, \{2, 4\}$

Total number of functions = $1 + (2^3 - 2)3 = 19$

Question157

The inverse function of $f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$, $x \in (-1, 1)$, is _____.

[Jan. 8, 2020 (I)]

Options:

A. $\frac{1}{4} \log_e \left(\frac{1+x}{1-x} \right)$

B. $\frac{1}{4} (\log_8 e) \log_e \left(\frac{1-x}{1+x} \right)$

C. $\frac{1}{4} \log_e \left(\frac{1-x}{1+x} \right)$

D. $\frac{1}{4} (\log_8 e) \log_e \left(\frac{1+x}{1-x} \right)$

Answer: A

Solution:

$$y = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$$

$$\frac{1+y}{1-y} = \frac{8^{2x}}{8^{-2x}} \Rightarrow 8^{4x} = \frac{1+y}{1-y}$$

$$\Rightarrow 4x = \log_8 \left(\frac{1+y}{1-y} \right)$$

$$\Rightarrow x = \frac{1}{4} \log_8 \left(\frac{1+y}{1-y} \right)$$

$$\therefore f^{-1}(x) = \frac{1}{4} \log_8 \left(\frac{1+x}{1-x} \right)$$

Question 158

If $g(x) = x^2 + x - 1$ and $(g \circ f)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to:

[Jan. 7, 2020 (I)]

Options:

A. $\frac{3}{2}$

B. $-\frac{1}{2}$

C. $\frac{1}{2}$

D. $-\frac{3}{2}$

Answer: B

Solution:

$$(g \circ f)(x) = g(f(x)) = f^2(x) + f(x) - 1$$

$$g\left(f\left(\frac{5}{4}\right)\right) = 4\left(\frac{5}{4}\right)^2 - 10 \cdot \frac{5}{4} + 5 = -\frac{5}{4}$$

$$[\because g(f(x)) = 4x^2 - 10x + 5]$$

$$g\left(f\left(\frac{5}{4}\right)\right) = f^2\left(\frac{5}{4}\right) + f\left(\frac{5}{4}\right) - 1$$

$$-\frac{5}{4} = f^2\left(\frac{5}{4}\right) + f\left(\frac{5}{4}\right) - 1$$

$$f^2\left(\frac{5}{4}\right) + f\left(\frac{5}{4}\right) + \frac{1}{4} = 0$$

$$\left(f\left(\frac{5}{4}\right) + \frac{1}{2}\right)^2 = 0$$

$$f\left(\frac{5}{4}\right) = -\frac{1}{2}$$

Question 159

For a suitably chosen real constant a , let a function, $f : \mathbb{R} - \{-a\} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{a-x}{a+x}$. Further suppose that for any real number

$x \neq -a$ and $f(x) \neq -a$, $(f \circ f)(x) = x$. Then $f\left(-\frac{1}{2}\right)$ is equal to:

[Sep. 06, 2020 (II)]

Options:

A. $\frac{1}{3}$

B. $-\frac{1}{3}$

C. -3

D. 3

Answer: D

Solution:

$$f(f(x)) = \frac{a - \left(\frac{a-x}{a+x}\right)}{a + \left(\frac{a-x}{a+x}\right)} = x$$
$$\Rightarrow \frac{a-ax}{1+x} = f(x) \Rightarrow \frac{a(1-x)}{1+x} = \frac{a-x}{a+x} \Rightarrow a = 1$$
$$\therefore f(x) = \frac{1-x}{1+x} \Rightarrow f\left(-\frac{1}{2}\right) = 3$$

Question 160

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{x}{1+x^2}$, $x \in \mathbb{R}$. Then the range of f is :

[Jan. 11, 2019 (I)]

Options:

A. $\left[-\frac{1}{2}, \frac{1}{2}\right]$

B. $\mathbb{R} - [-1, 1]$

C. $\mathbb{R} - \left[-\frac{1}{2}, \frac{1}{2}\right]$

D. $(-1, 1) - \{0\}$

Answer: A

Solution:

$$f(x) = \frac{x}{1+x^2}, x \in \mathbb{R}$$

$$\text{Let, } y = \frac{x}{1+x^2}$$

$$\Rightarrow yx^2 - x + y = 0 \Rightarrow x = \frac{1 \pm \sqrt{1-4y^2}}{2}$$

$$\Rightarrow 1 - 4y^2 \geq 0$$

$$\Rightarrow 1 \geq 4y^2$$

$$\Rightarrow |y| \leq \frac{1}{2}$$

$$\Rightarrow -\frac{1}{2} \leq y \leq \frac{1}{2}$$

$$\Rightarrow \text{The range of } f \text{ is } \left[-\frac{1}{2}, \frac{1}{2}\right]$$

Question161

The domain of the definition of the function $f(x) = \frac{1}{4-x^2} + \log_{10}(x^3 - x)$ is:

[April. 09, 2019 (II)]

Options:

A. $(-1,0) \cup (1, 2) \cup (3, \infty)$

B. $(-2,-1) \cup (-1, 0) \cup (2, \infty)$

C. $(-1,0) \cup (1, 2) \cup (2, \infty)$

D. $(1,2) \cup (2, \infty)$

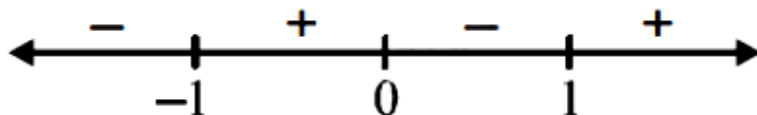
Answer: C

Solution:

To determine domain, denominator $\neq 0$ and $x^3 - x > 0$

i.e., $4 - x^2 \neq 0 \Rightarrow x \neq \pm 2$ (1)

and $x(x-1)(x+1) > 0$



$x \in (-1, 0) \cup (1, \infty)$ (2)

Hence domain is intersection of (1)&(2).

i.e., $x \in (-1, 0) \cup (1, 2) \cup (2, \infty)$

Question162

If $f(x) = \log_e \left(\frac{1-x}{1+x} \right)$, $|x| < 1$, then $f \left(\frac{2x}{1+x^2} \right)$ is equal to

[April 8, 2019 (I)]

Options:

A. $2f(x)$

B. $2f(x^2)$

C. $(f(x))^2$

D. $-2f(x)$

Answer: A

Solution:

$$f(x) = \log \left(\frac{1-x}{1+x} \right), |x| < 1$$

$$f \left(\frac{2x}{1+x^2} \right) = \log \left(\frac{1 - \frac{2x}{1+x^2}}{1 + \frac{2x}{1+x^2}} \right)$$

$$= \log \left(\frac{1+x^2-2x}{1+x^2+2x} \right) = \log \left(\frac{1-x}{1+x} \right)^2$$

$$= 2 \log \left(\frac{1-x}{1+x} \right) = 2f(x)$$

Question 163

Let $f(x) = a^x$ ($a > 0$) be written as $f(x) = f_1(x) + f_2(x)$, where $f_1(x)$ is an even function and $f_2(x)$ is an odd function. Then $f_1(x+y) + f_1(x-y)$ equals:

[April. 08, 2019 (II)]

Options:

A. $2f_1(x)f_1(y)$

B. $2f_1(x+y)f_1(x-y)$

C. $2f_1(x)f_2(y)$

D. $2f_1(x+y)f_2(x-y)$

Answer: A

Solution:

Given function can be written as

$$f(x) = a^x = \left(\frac{a^x + a^{-x}}{2} \right) + \left(\frac{a^x - a^{-x}}{2} \right)$$

where $f_1(x) = \frac{a^x + a^{-x}}{2}$ is even function

$f_2(x) = \frac{a^x - a^{-x}}{2}$ is odd function

$$\Rightarrow f_1(x+y) + f_1(x-y)$$

$$= \left(\frac{a^{x+y} + a^{-x-y}}{2} \right) + \left(\frac{a^{x-y} + a^{-x+y}}{2} \right)$$

$$= \frac{1}{2} [a^x(a^y + a^{-y}) + a^{-x}(a^y + a^{-y})]$$

$$= \frac{(a^x + a^{-x})(a^y + a^{-y})}{2} = 2f_1(x) \cdot f_1(y)$$

Question164

Let a function $f : (0, \infty) \rightarrow (0, \infty)$ be defined by $f(x) = \left| 1 - \frac{1}{x} \right|$. Then f is :

[Jan. 11, 2019 (II)]

Options:

A. not injective but it is surjective

B. injective only

C. neither injective nor surjective

D. (Bonus)

Answer: D

Solution:

$$f : (0, \infty) \rightarrow (0, \infty)$$

$$f(x) = \left| 1 - \frac{1}{x} \right| \text{ is not a function}$$

$$\because f(1) = 0 \text{ and } 1 \in \text{domain but } 0 \notin \text{co-domain}$$

Hence, $f(x)$ is not a function.

Question165

**The number of functions f from $\{1, 2, 3, \dots, 20\}$ onto $\{1, 2, 3, \dots, 20\}$ such that $f(k)$ is a multiple of 3, whenever k is a multiple of 4 is :
[Jan. 11, 2019 (II)]**

Options:

A. $6^5 \times (15)!$

B. $5! \times 6!$

C. $(15)! \times 6!$

D. $5^6 \times 15$

Answer: C

Solution:

Domain and codomain = $\{1, 2, 3, \dots, 20\}$.

There are five multiple of 4 as 4, 8, 12, 16 and 20 .

and there are 6 multiple of 3 as 3, 6, 9, 12, 15, 18 .

Since, when ever k is multiple of 4 then $f(k)$ is multiple of 3 then total number of arrangement
 $= {}^6C_5 \times 5! = 6!$

Remaining 15 elements can be arranged in $15!$ ways.

Since, for every input, there is an output

$\Rightarrow$ function $f(k)$ is onto

$\therefore$ Total number of arrangement = $15! \cdot 6!$

Question166

Let $\mathbb{N}$ be the set of natural numbers and two functions f and g be

defined as $f, g : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(n) = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases}$

and $g(n) = n - (-1)^n$. Then $f \circ g$ is:
[Jan. 10, 2019 (II)]

Options:

- A. onto but not one-one.
- B. one-one but not onto.
- C. both one-one and onto.
- D. neither one-one nor onto.

Answer: A

Solution:

$$f(n) = \begin{cases} \frac{n+1}{2}, & \text{if } n \text{ is odd} \\ \frac{n}{2}, & \text{if } n \text{ is even} \end{cases}$$

$$g(n) = \begin{cases} 2, & n = 1 \\ 1, & n = 2 \\ 4, & n = 3 \\ 3, & n = 4 \\ 6, & n = 5 \\ 5, & n = 6 \end{cases}$$

Then,

$$f(g(n)) = \begin{cases} \frac{g(n)+1}{2}, & \text{if } g(n) \text{ is odd} \\ \frac{g(n)}{2}, & \text{if } g(n) \text{ is even} \end{cases}$$

$$f(g(n)) = \begin{cases} 1, & n = 1 \\ 1, & n = 2 \\ 2, & n = 3 \\ 2, & n = 4 \\ 3, & n = 5 \\ 3, & n = 6 \\ \vdots & \vdots \\ \vdots & \vdots \\ \vdots & \vdots \end{cases}$$

$\Rightarrow$ fog is onto but not one - one

Question167

Let $A = \{x \in \mathbb{R} : x \text{ is not a positive integer}\}$. Define a function $f : A \rightarrow \mathbb{R}$ as $f(x) = \frac{2x}{x-1}$, then f is:
[Jan. 09, 2019 (II)]

Options:

- A. not injective
- B. neither injective nor surjective
- C. surjective but not injective
- D. injective but not surjective

Answer: D

Solution:

As $A = \{x \in \mathbb{R} : x \text{ is not a positive integer}\}$

A function $f : A \rightarrow \mathbb{R}$ given by $f(x) = \frac{2x}{x-1}$

$$f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$$

So, f is one-one.

As $f(x) \neq 2$ for any $x \in A \Rightarrow f$ is not onto.

Hence f is injective but not surjective.

Question168

For $x \in \left(0, \frac{3}{2}\right)$, let $f(x) = \sqrt{x}$, $g(x) = \tan x$ and $h(x) = \frac{1-x^2}{1+x^2}$

If $\phi(x) = ((h \circ f) \circ g)(x)$, then $\phi\left(\frac{\pi}{3}\right)$ is equal to

[April 12, 2019 (I)]

Options:

A. $\tan \frac{\pi}{12}$

B. $\tan \frac{11\pi}{12}$

C. $\tan \frac{7\pi}{12}$

D. $\tan \frac{5\pi}{12}$

Answer: B

Solution:

$$\because \phi(x) = ((h \circ f) \circ g)(x)$$

$$\because \phi\left(\frac{\pi}{3}\right) = h\left(f\left(g\left(\frac{\pi}{3}\right)\right)\right) = h(f(\sqrt{3})) = h(3^{1/4})$$

$$= \frac{1-\sqrt{3}}{1+\sqrt{3}} = -\frac{1}{2}(1+3-2\sqrt{3}) = \sqrt{3}-2 = -(-\sqrt{3}+2)$$

$$= -\tan 15^\circ = \tan(180^\circ - 15^\circ) = \tan\left(\pi - \frac{\pi}{12}\right) = \tan \frac{11\pi}{12}$$

Question 169

Let $f(x) = x^2$, $x \in \mathbb{R}$. For any $A \subseteq \mathbb{R}$, define $g(A) = \{x \in \mathbb{R} : f(x) \in A\}$.

If $S = [0, 4]$, then which one of the following statements is not true?

[April 10, 2019 (I)]

Options:

A. $g(f(S)) \neq S$

B. $f(g(S)) = S$

C. $g(f(S)) = g(S)$

D. $f(g(S)) \neq f(S)$

Answer: C

Solution:

$$\begin{aligned} f(x) &= x^2; x \in \mathbb{R} \\ g(A) &= \{x \in \mathbb{R} : f(x) \in A\} \quad S = [0, 4] \\ g(S) &= \{x \in \mathbb{R} : f(x) \in S\} \\ &= \{x \in \mathbb{R} : 0 \leq x^2 \leq 4\} = \{x \in \mathbb{R} : -2 \leq x \leq 2\} \\ \therefore g(S) &\neq S \therefore f(g(S)) \neq f(S) \\ g(f(S)) &= \{x \in \mathbb{R} : f(x) \in f(S)\} \\ &= \{x \in \mathbb{R} : x^2 \in S^2\} = \{x \in \mathbb{R} : 0 \leq x^2 \leq 16\} \\ &= \{x \in \mathbb{R} : -4 \leq x \leq 4\} \\ \therefore g(f(S)) &\neq g(S) \\ \therefore g(f(S)) &= g(S) \text{ is incorrect.} \end{aligned}$$

Question 170

For $x \in \mathbb{R} - \{0, 1\}$, let $f_1(x) = \frac{1}{x}$, $f_2(x) = 1 - x$ and $f_3(x) = \frac{1}{1-x}$ be three given functions. If a function, $J(x)$ satisfies $(f_2 \circ J \circ f_1)(x) = f_3(x)$ then $J(x)$ is equal to:
[Jan. 09, 2019 (I)]

Options:

A. $f_3(x)$

B. $\frac{1}{x}f_3(x)$

C. $f_2(x)$

D. $f_1(x)$

Answer: A

Solution:

The given relation is

$$(f_2 \circ J \circ f_1)(x) = f_3(x) = \frac{1}{1-x}$$

$$\Rightarrow (f_2 \circ J)(f_1(x)) = \frac{1}{1-x}$$

$$\Rightarrow (f_2 \circ J)\left(\frac{1}{x}\right) = \frac{1}{1 - \frac{1}{\frac{1}{x}}} = \frac{\frac{1}{x}}{\frac{1}{x} - 1} \left[\because f_1(x) = \frac{1}{x} \right]$$

$$\Rightarrow (f_2 \circ J)(x) = \frac{x}{x-1} \left[\frac{1}{x} \text{ is replaced by } x \right]$$

$$\Rightarrow f_2(J(x)) = \frac{x}{x-1}$$

$$\Rightarrow 1 - J(x) = \frac{x}{x-1} \left[\because f_2(x) = 1 - x \right]$$

$$\therefore J(x) = 1 - \frac{x}{x-1} = \frac{1}{1-x} = f_3(x)$$

Question 171

Let N denote the set of all natural numbers. Define two binary relations on N as $R_1 = \{(x, y) \in N \times N : 2x + y = 10\}$ and $R_2 = \{(x, y) \in N \times N : x + 2y = 10\}$. Then
[Online April 16, 2018]

Options:

- A. Both R_1 and R_2 are transitive relations
- B. Both R_1 and R_2 are symmetric relations
- C. Range of R_2 is $\{1, 2, 3, 4\}$
- D. Range of R_1 is $\{2, 4, 8\}$

Answer: C

Solution:

Here, $R_1 = \{(x, y) \in N \times N : 2x + y = 10\}$ and

$R_2 = \{(x, y) \in N \times N : x + 2y = 10\}$

For R_1 ; $2x + y = 10$ and $x, y \in N$

So, possible values for x and y are:

$x = 1, y = 8$ i.e. $(1, 8)$;

$x = 2, y = 6$ i.e. $(2, 6)$;

$x = 3, y = 4$ i.e. $(3, 4)$ and $x = 4, y = 2$ i.e. $(4, 2)$.

$$R_1 = \{(1, 8), (2, 6), (3, 4), (4, 2)\}$$

Therefore, Range of R_1 is $\{2, 4, 6, 8\}$

R_1 is not symmetric

Also, R_1 is not transitive because $(3, 4), (4, 2) \in R_1$ but $(3, 2) \notin R_1$

Thus, options A, B and D are incorrect.

For R_2 ; $x + 2y = 10$ and $x, y \in \mathbb{N}$

So, possible values for x and y are: $x = 8, y = 1$ i.e. $(8, 1)$;

$x = 6, y = 2$ i.e. $(6, 2)$;

$x = 4, y = 3$ i.e. $(4, 3)$ and

$x = 2, y = 4$ i.e. $(2, 4)$

$$R_2 = \{(8, 1), (6, 2), (4, 3), (2, 4)\}$$

Therefore, Range of R_2 is $\{1, 2, 3, 4\}$

R_2 is not symmetric and transitive.

Question 172

Consider the following two binary relations on the set

$A = \{a, b, c\}$: $R_1 = \{(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)\}$ and

$R_2 = \{(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)\}$. Then

[Online April 15, 2018]

Options:

A. R_2 is symmetric but it is not transitive

B. Both R_1 and R_2 are transitive

C. Both R_1 and R_2 are not symmetric

D. R_1 is not symmetric but it is transitive

Answer: A

Solution:

Both R_1 and R_2 are symmetric as

For any $(x, y) \in R_1$, we have

$(y, x) \in R_1$ and similarly for R_2

Now, for R_2 , $(b, a) \in R_2$, $(a, c) \in R_2$ but $(b, c) \notin R_2$

Similarly, for R_1 , $(b, c) \in R_1$, $(c, a) \in R_1$ but $(b, a) \notin R_1$

Therefore, neither R_1 nor R_2 is transitive.

Question173

Let $f : A \rightarrow B$ be a function defined as $f(x) = \frac{x-1}{x-2}$, where $A = \mathbb{R} - \{2\}$ and $B = \mathbb{R} - \{1\}$. Then f is
[Online April 15, 2018]

Options:

A. invertible and $f^{-1}(y) = \frac{2y+1}{y-1}$

B. invertible and $f^{-1}(y) = \frac{3y-1}{y-1}$

C. no invertible

D. invertible and $f^{-1}(y) = \frac{2y-1}{y-1}$

Answer: D

Solution:

Suppose $y = f(x)$

$$\Rightarrow y = \frac{x-1}{x-2}$$

$$\Rightarrow yx - 2y = x - 1$$

$$\Rightarrow (y-1)x = 2y-1$$

$$\Rightarrow x = f^{-1}(y) = \frac{2y-1}{y-1}$$

As the function is invertible on the given domain and its inverse can be obtained as above.

Question174

The function $f : \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as $f(x) = \frac{x}{1+x^2}$, is:
[2017]

Options:

A. neither injective nor surjective

B. invertible

C. injective but not surjective

D. surjective but not injective

Answer: D

Solution:

We have $f : \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$,

$$f(x) = \frac{x}{1+x^2} \quad \forall x \in \mathbb{R}$$

$$\Rightarrow f'(x) = \frac{(1+x^2) \cdot 1 - x \cdot 2x}{(1+x^2)^2} = \frac{-(x+1)(x-1)}{(1+x^2)^2}$$

sign of $f'(x)$

$\Rightarrow f'(x)$ changes sign in different intervals.

$\therefore$ Not injective Now $y = \frac{x}{1+x^2}$

$$\Rightarrow y + yx^2 = x$$

$$\Rightarrow yx^2 - x + y = 0$$

$$\text{For } y \neq 0, D = 1 - 4y^2 \geq 0$$

$$\Rightarrow y \in \left[-\frac{1}{2}, \frac{1}{2}\right] - \{0\}$$

$$\text{For } y = 0 \Rightarrow x = 0$$

$$\therefore \text{Range is } \left[-\frac{1}{2}, \frac{1}{2}\right]$$

$\Rightarrow$ Surjective but not injective

Question 175

The function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x - 5 \left\lfloor \frac{x}{5} \right\rfloor$, where $\mathbb{N}$ is set of natural numbers and $[x]$ denotes the greatest integer less than or equal to x , is:

[Online April 9, 2017]

Options:

A. one-one and onto.

B. one-one but not onto.

C. onto but not one-one.

D. neither one-one nor onto.

Answer: D

Solution:

$$\left. \begin{array}{l} f(1) = 1 - 5[1/5] = 1 \\ f(6) = 6 - 5[6/5] = 1 \end{array} \right\} \rightarrow \text{Many one}$$

$f(10) = 10 - 5(2) = 0$ which is not in co-domain.
Neither one-one nor onto.

Question176

Let $f(x) = 2^{10} \cdot x + 1$ and $g(x) = 3^{10} \cdot x - 1$. If $(f \circ g)(x) = x$, then x is equal to:

[Online April 8, 2017]

Options:

A. $\frac{3^{10} - 1}{3^{10} - 2^{-10}}$

B. $\frac{2^{10} - 1}{2^{10} - 3^{-10}}$

C. $\frac{1 - 3^{-10}}{2^{10} - 3^{-10}}$

D. $\frac{1 - 2^{-10}}{3^{10} - 2^{-10}}$

Answer: D

Solution:

$$\begin{aligned} f(g(x)) &= x \\ \Rightarrow f(3^{10}x - 1) &= 2^{10}(3^{10} \cdot x - 1) + 1 = x \\ \Rightarrow 2^{10}(3^{10}x - 1) + 1 &= x \\ \Rightarrow x(2^{10} \cdot 3^{10} - 1) &= 2^{10} - 1 \\ \Rightarrow x &= \frac{2^{10} - 1}{2^{10} \cdot 3^{10} - 1} = \frac{1 - 2^{-10}}{3^{10} - 2^{-10}} \end{aligned}$$

Question177

For $x \in \mathbb{R}$, $x \neq 0$, let $f_0(x) = \frac{1}{1-x}$ and $f_{n+1}(x) = f_0(f_n(x))$ $n = 0, 1, 2, \dots$

Then the value of $f_{100}(3) + f_1\left(\frac{2}{3}\right) + f_2\left(\frac{3}{2}\right)$ is equal to :

[Online April 9, 2016]

Options:

A. $\frac{8}{3}$

B. $\frac{4}{3}$

C. $\frac{5}{3}$

D. $\frac{1}{3}$

Answer: C

Solution:

$$f_1(x) = f_{0+1}(x) = f_0(f_0(x)) = \frac{1}{1 - \frac{1}{1-x}} = \frac{x-1}{x}$$

$$f_2(x) = f_{1+1}(x) = f_0(f_1(x)) = \frac{1}{1 - \frac{x-1}{x}} = x$$

$$f_3(x) = f_{2+1}(x) = f_0(f_2(x)) = f_0(x) = \frac{1}{1-x}$$

$$f_4(x) = f_{3+1}(x) = f_0(f_3(x)) = \frac{x-1}{x}$$

$$\therefore f_0 = f_3 = f_6 = \dots = \frac{1}{1-x}$$

$$f_1 = f_4 = f_7 = \dots = \frac{x-1}{x}$$

$$f_2 = f_5 = f_8 = \dots = x$$

$$f_{100}(3) = \frac{3-1}{3} = \frac{2}{3} f_1\left(\frac{2}{3}\right) = \frac{\frac{2}{3}-1}{\frac{2}{3}} = -\frac{1}{2}$$

$$f_2\left(\frac{3}{2}\right) = \frac{3}{2}$$

$$\therefore f_{100}(3) + f_1\left(\frac{2}{3}\right) + f_2\left(\frac{3}{2}\right) = \frac{5}{3}$$

Question 178

Let $A = \{x_1, x_2, \dots, x_7\}$ and $B = \{y_1, y_2, y_3\}$ be two sets containing seven and three distinct elements respectively. Then the total number of functions $f : A \rightarrow B$ that are onto, if there exist exactly three elements x in A such that $f(x) = y_2$, is equal to

(Online April 11, 2015)

Options:

A. $14 \cdot {}^7C_3$

B. $16 \cdot {}^7C_3$

C. $14 \cdot {}^7C_2$

D. $12 \cdot {}^7C_2$

Answer: A

Solution:

Number of onto function such that exactly three elements in $x \in A$ such that $f(x) = \frac{1}{2}$ is equal to
 $= {}^7C_3 \cdot \{2^4 - 2\} = 14 \cdot {}^7C_3$

Question179

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{|x|-1}{|x|+1}$ then f is:

[Online April 19, 2014]

Options:

A. both one-one and onto

B. one-one but not onto

C. onto but not one-one

D. neither one-one nor onto.

Answer: C

Solution:

Solution:

$$f(x) = \frac{|x| - 1}{|x| + 1}$$

for one-one function if $f(x_1) = f(x_2)$ then

x_1 must be equal to x_2

Let $f(x_1) = f(x_2)$

$$\frac{|x_1| - 1}{|x_1| + 1} = \frac{|x_2| - 1}{|x_2| + 1} \quad |x_1| |x_2| + |x_1| - |x_2| - 1 = |x_1| |x_2| - |x_1| + |x_2| - 1$$

$$\Rightarrow |x_1| - |x_2| = |x_2| - |x_1|$$

$$2|x_1| = 2|x_2|$$

$$|x_1| = |x_2|$$

$$x_1 = x_2, x_1 = -x_2$$

here x_1 has two values therefore function is many one function.

For onto : $f(x) = \frac{|x| - 1}{|x| + 1}$

for every value of $f(x)$ there is a value of x in domain set.

If $f(x)$ is negative then $x = 0$

for all positive value of $f(x)$, domain contain atleast one element. Hence $f(x)$ is onto function.

Question180

Let P be the relation defined on the set of all real numbers such that

$P = \{(a, b) : \sec^2 a - \tan^2 b = 1\}$. Then P is:

[Online April 9, 2014]

Options:

A. reflexive and symmetric but not transitive.

B. reflexive and transitive but not symmetric.

C. symmetric and transitive but not reflexive.

D. an equivalence relation.

Answer: D

Solution:

$$P = \{(a, b) : \sec^2 a - \tan^2 b = 1\}$$

For reflexive :

$$\sec^2 a - \tan^2 a = 1 \quad (\text{true } \forall a)$$

For symmetric :

$$\sec^2 b - \tan^2 a = 1$$

L.H.S

$$1 + \tan^2 b - (\sec^2 a - 1) = 1 + \tan^2 b - \sec^2 a + 1$$

$$= -(\sec^2 a - \tan^2 b) + 2$$

$$= -1 + 2 = 1$$

So, Relation is symmetric For transitive :

$$\text{if } \sec^2 a - \tan^2 b = 1 \text{ and } \sec^2 b - \tan^2 c = 1$$

$$\sec^2 a - \tan^2 c = (1 + \tan^2 b) - (\sec^2 b - 1)$$

$$= -\sec^2 b + \tan^2 b + 2$$

$$= -1 + 2 = 1$$

So, Relation is transitive.

Hence, Relation P is an equivalence relation

Question181

Let $f(n) = \left[\frac{1}{3} + \frac{3n}{100} \right] n$, where $[n]$ denotes the greatest integer less than or equal to n . Then $\sum_{n=1}^{56} f(n)$ is equal to:
[Online April 19, 2014]

Options:

A. 56

B. 689

C. 1287

D. 1399

Answer: D

Solution:

$$\text{Let } f(n) = \left[\frac{1}{3} + \frac{3n}{100} \right] n$$

where $[n]$ is greatest integer function,

$$= \left[0.33 + \frac{3n}{100} \right] n$$

For $n = 1, 2, \dots, 22$, we get $f(n) = 0$ and for $n = 23, 24, \dots, 55$, we get $f(n) = 1 \times n$ For $n = 56$, $f(n) = 2 \times n$

$$\text{So, } \sum_{n=1}^{56} f(n) = 1(23) + 1(24) + \dots + 1(55) + 2(56)$$

$$= (23 + 24 + \dots + 55) + 112$$

$$= \frac{33}{2}[46 + 32] + 112$$

$$= \frac{33}{2}(78) + 112 = 1399$$

Question182

Let f be an odd function defined on the set of real numbers such that for $x \geq 0$, $f(x) = 3 \sin x + 4 \cos x$ Then $f(x)$ at $x = -\frac{11\pi}{6}$ is equal to:

[Online April 11, 2014]

Options:

A. $\frac{3}{2} + 2\sqrt{3}$

B. $-\frac{3}{2} + 2\sqrt{3}$

C. $\frac{3}{2} - 2\sqrt{3}$

D. $-\frac{3}{2} - 2\sqrt{3}$

Answer: C

Solution:

Given f be an odd function

$$f(x) = 3 \sin x + 4 \cos x$$

$$\text{Now, } f\left(-\frac{11\pi}{6}\right) = 3 \sin\left(-\frac{11\pi}{6}\right) + 4 \cos\left(-\frac{11\pi}{6}\right)$$

$$f\left(-\frac{11\pi}{6}\right) = 3 \sin\left(-2\pi + \frac{\pi}{6}\right) + 4 \cos\left(-2\pi + \frac{\pi}{6}\right)$$

$$f\left(-\frac{11\pi}{6}\right) = 3 \sin\left\{-\left(2\pi - \frac{\pi}{6}\right)\right\} + 4 \cos\left\{-\left(2\pi - \frac{\pi}{6}\right)\right\}$$

$$\left\{ \begin{array}{l} \text{For odd functions} \\ \sin(-\theta) = -\sin \theta \\ \text{and } \cos(-\theta) = \cos \theta \end{array} \right\}$$

$$\therefore f\left(-\frac{11\pi}{6}\right) = -3 \sin\left(2\pi - \frac{\pi}{6}\right) - 4 \cos\left(2\pi - \frac{\pi}{6}\right)$$

$$\Rightarrow f\left(-\frac{11\pi}{6}\right) = +3 \sin\left(\frac{\pi}{6}\right) - 4 \cos \frac{\pi}{6}$$

$$\Rightarrow f\left(\frac{-11\pi}{6}\right) = 3 \times \frac{1}{2} - 4 \times \frac{\sqrt{3}}{2}$$

$$\text{or } f\left(\frac{-11\pi}{6}\right) = \frac{3}{2} - 2\sqrt{3}$$

Question 183

If g is the inverse of a function f and $f'(x) = \frac{1}{1+x^5}$, then $g'(x)$ is equal to:
[2014]

Options:

A. $\frac{1}{1 + \{g(x)\}^5}$

B. $1 + \{g(x)\}^5$

C. $1 + x^5$

D. $5x^4$

Answer: B

Solution:

Since $f(x)$ and $g(x)$ are inverse of each other

$$\therefore g'(f(x)) = \frac{1}{f'(x)}$$

$$\Rightarrow g'(f(x)) = 1 + x^5 \left(\because f'(x) = \frac{1}{1+x^5} \right)$$

Here $x = g(y)$

$$\therefore g'(y) = 1 + [g(y)]^5$$

$$\Rightarrow g'(x) = 1 + (g(x))^5$$

Question 184

Let $R = \{(x, y) : x, y \in \mathbb{N} \text{ and } x^2 - 4xy + 3y^2 = 0\}$, where $\mathbb{N}$ is the set of all natural numbers. Then the relation R is:
[Online April 23, 2013]

Options:

- A. reflexive but neither symmetric nor transitive.
- B. symmetric and transitive.
- C. reflexive and symmetric,
- D. reflexive and transitive.

Answer: D

Solution:

$$R = \{ (x, y) : x, y \in \mathbb{N} \text{ and } x^2 - 4xy + 3y^2 = 0 \}$$

$$\text{Now, } x^2 - 4xy + 3y^2 = 0$$

$$\Rightarrow (x - y)(x - 3y) = 0$$

$$\therefore x = y \text{ or } x = 3y$$

$$\therefore R = \{ (1, 1), (3, 1), (2, 2), (6, 2), (3, 3), (9, 3), \dots \}$$

Since (1, 1), (2, 2), (3, 3), are present in the relation, therefore R is reflexive.

Since (3, 1) is an element of R but (1, 3) is not the element of R, therefore R is not symmetric

Here (3, 1) ∈ R and (1, 1) ∈ R ⇒ (3, 1) ∈ R (6, 2) ∈ R and (2, 2) ∈ R ⇒ (6, 2) ∈ R

For all such (a, b) ∈ R and (b, c) ∈ R

$$\Rightarrow (a, c) \in R$$

Hence R is transitive.

Question 185

Let $R = \{ (3, 3), (5, 5), (9, 9), (12, 12), (5, 12), (3, 9), (3, 12), (3, 5) \}$ be a relation on the set $A = \{3, 5, 9, 12\}$. Then, R is :
[Online April 22, 2013]

Options:

- A. reflexive, symmetric but not transitive.
- B. symmetric, transitive but not reflexive.
- C. an equivalence relation.
- D. reflexive, transitive but not symmetric.

Answer: D

Solution:

Let $R = \{(3, 3), (5, 5), (9, 9), (12, 12), (5, 12), (3, 9), (3, 12), (3, 5)\}$ be a relation on set $A = \{3, 5, 9, 12\}$

Clearly, every element of A is related to itself.

Therefore, it is a reflexive.

Now, R is not symmetry because 3 is related to 5 but 5 is not related to 3.

Also R is transitive relation because it satisfies the property that if aRb and bRc then aRc .

Question186

Let $A = \{1, 2, 3, 4\}$ and $R : A \rightarrow A$ be the relation defined by $R = \{(1, 1), (2, 3), (3, 4), (4, 2)\}$. The correct statement is:
[Online April 9, 2013]

Options:

A. R does not have an inverse.

B. R is not a one to one function.

C. R is an onto function.

D. R is not a function.

Answer: C

Solution:

Domain = $\{1, 2, 3, 4\}$

Range = $\{1, 2, 3, 4\}$

$\therefore$ Domain = Range

Hence the relation R is onto function.

Question187

If $P(S)$ denotes the set of all subsets of a given set S , then the number of one-to-one functions from the set $S = \{1, 2, 3\}$ to the set $P(S)$ is
[Online May 19, 2012]

Options:

A. 24

B. 8

C. 336

D. 320

Answer: C

Solution:

Let $S = \{1, 2, 3\} \rightarrow n(S) = 3$

Now, $P(S)$ = set of all subsets of S

total no. of subsets = $2^3 = 8$

$\therefore n[P(S)] = 8$

Now, number of one-to-one functions from $S \rightarrow P(S)$ is ${}^8P_3 = \frac{8!}{5!} = 8 \times 7 \times 6 = 336$

Question188

If $A = \{x \in \mathbb{Z}^+ : x < 10 \text{ and } x \text{ is a multiple of 3 or 4}\}$, where $\mathbb{Z}^+$ is the set of positive integers, then the total number of symmetric relations on A is.

[Online May 12, 2012]

Options:

A. 2^5

B. 2^{15}

C. 2^{10}

D. 2^{20}

Answer: B

Solution:

A relation on a set A is said to be symmetric iff $(a, b) \in A \Rightarrow (b, a) \in A, \forall a, b \in A$

Here $A = \{3, 4, 6, 8, 9\}$

Number of order pairs of $A \times A = 5 \times 5 = 25$

Divide 25 order pairs of A times A in 3 parts as follows:

Part – A : $(3, 3), (4, 4), (6, 6), (8, 8), (9, 9)$

Part – B : $(3, 4), (3, 6), (3, 8), (3, 9), (4, 6), (4, 8), (4, 9), (6, 8), (6, 9), (8, 9)$

Part – C : $(4, 3), (6, 3), (8, 3), (9, 3), (6, 4), (8, 4), (9, 4), (8, 6), (9, 6), (9, 8)$

In part – A, both components of each order pair are same.

In part – B, both components are different but not two such order pairs are present in which first component of one order pair is the second component of another order pair and vice-versa.
 In part–C, only reverse of the order pairs of part –B are present i.e., if (a, b) is present in part – B, then (b, a) will be present in part –C
 For example (3, 4) is present in part – B and (4, 3) present in part –C.
 Number of order pair in A, B and C are 5, 10 and 10 respectively.
 In any symmetric relation on set A, if any order pair of part –B is present then its reverse order pair of part –C will must be also present.
 Hence number of symmetric relation on set A is equal to the number of all relations on a set D, which contains all the order pairs of part –A and part– B.
 Now $n(D) = n(A) + n(B) = 5 + 10 = 15$
 Hence number of all relations on set D = $(2)^{15}$
 $\Rightarrow$ Number of symmetric relations on set D = $(2)^{15}$

Question189

The range of the function $f(x) = \frac{x}{1+|x|}$, $x \in \mathbb{R}$, is
 [Online May 7, 2012]

Options:

- A. $\mathbb{R}$
- B. $(-1,1)$
- C. $\mathbb{R} - \{0\}$
- D. $[-1,1]$

Answer: B

Solution:

$$f(x) = \frac{x}{1+|x|}, x \in \mathbb{R}$$

$$\text{If } x > 0, |x| = x \Rightarrow f(x) = \frac{x}{1+x}$$

which is not defined for $x = -1$

$$\text{If } x < 0, |x| = -x \Rightarrow f(x) = \frac{x}{1-x} \text{ which is not defined for } x = 1$$

Thus $f(x)$ defined for all values of $\mathbb{R}$ except 1 and -1

Hence, range = $(-1, 1)$

Question190

Let A and B be non empty sets in R and $f : A \rightarrow B$ is a bijective function.

Statement 1: f is an onto function.

Statement 2: There exists a function $g : B \rightarrow A$ such that $fog = I_B$

[Online May 26, 2012]

Options:

A. Statement 1 is true, Statement 2 is false.

B. Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.

C. Statement 1 is false, Statement 2 is true.

D. Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation for Statement 1.

Answer: D

Solution:

Let A and B be non-empty sets in R.

Let $f : A \rightarrow B$ is bijective function.

Clearly statement - 1 is true which says that f is an onto function.

Statement -2 is also true statement but it is not the correct explanation for statement-1

Question191

Let R be the set of real numbers.

Statement-1: $A = \{ (x, y) \in R \times R : y - x \text{ is an integer} \}$ is an equivalence relation on R.

Statement- 2: $B = \{ (x, y) \in R \times R : x = \alpha y \text{ for some rational number } \alpha \}$ is an equivalence relation on R.

[2011]

Options:

A. Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

B. Statement-1 is true, Statement-2 is false.

C. Statement-1 is false, Statement-2 is true.

D. Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

Answer: A

Solution:

$\because x - x = 0 \in I$ ($\therefore R$ is reflexive)

Let $(x, y) \in R$ as $x - y$ and $y - x \in I$ ($\therefore R$ is symmetric)

Now $x - y \in I$ and $y - z \in I \Rightarrow x - z \in I$

So, R is transitive.

Hence R is equivalence.

Similarly as $x = \alpha y$ for $\alpha = 1$. B is reflexive symmetric and transitive. Hence B is equivalence.

Both relations are equivalence but not the correct explanation.

Question192

The domain of the function $f(x) = \frac{1}{\sqrt{|x|} - x}$ is

[2011]

Options:

A. $(0, \infty)$

B. $(-\infty, 0)$

C. $(-\infty, \infty) - \{0\}$

D. $(-\infty, \infty)$

Answer: B

Solution:

$f(x) = \frac{1}{\sqrt{|x|} - x}$, $f(x)$ is define if $|x| - x > 0$

$\Rightarrow |x| > x, \Rightarrow x < 0$

Hence domain of $f(x)$ is $(-\infty, 0)$

Question193

Let f be a function defined by

$$f(x) = (x - 1)^2 + 1, (x \geq 1)$$

Statement -1: The set $\{x : f(x) = f^{-1}(x)\} = \{1, 2\}$

Statement -2: f is a bijection and $f^{-1}(x) = 1 + \sqrt{x - 1}, x \geq 1$

[2011 RS]

Options:

- A. Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
- B. Statement-1 is true, Statement-2 is true; Statement-2 is NOT a correct explanation for Statement-1.
- C. Statement-1 is true, Statement-2 is false.
- D. Statement-1 is false, Statement-2 is true.

Answer: A

Solution:

Given f is a bijective function

$$\therefore f : [1, \infty) \rightarrow [1, \infty)$$

$$f(x) = (x - 1)^2 + 1, x \geq 1$$

$$\text{Let } y = (x - 1)^2 + 1 \Rightarrow (x - 1)^2 = y - 1$$

$$\Rightarrow x = 1 \pm \sqrt{y - 1} \Rightarrow f^{-1}(y) = 1 \pm \sqrt{y - 1}$$

$$\Rightarrow f^{-1}(x) = 1 + \sqrt{x - 1} \{ \because x \geq 1 \}$$

Hence statement- 2 is correct

$$\text{Now } f(x) = f^{-1}(x)$$

$$\Rightarrow f(x) = x \Rightarrow (x - 1)^2 + 1 = x$$

$$\Rightarrow x^2 - 3x + 2 = 0 \Rightarrow x = 1, 2$$

Hence statement- 1 is correct

Question194

Consider the following relations:

$R = \{ (x, y) \mid x, y \text{ are real numbers and } x = wy \text{ for some rational}$

number $w \}$; $S = \left\{ \left(\frac{m}{n}, \frac{p}{q} \right) \mid m, n, p \text{ and } q \text{ are integers such that } n, q \neq 0 \text{ and } qm = pn \right\}$

. Then
[2010]

Options:

- A. Neither R nor S is an equivalence relation
- B. S is an equivalence relation but R is not an equivalence relation
- C. R and S both are equivalence relations
- D. R is an equivalence relation but S is not an equivalence relation

Answer: B

Solution:

Let xRy .

$$\Rightarrow x = wy \Rightarrow y = \frac{x}{w}$$

$$\Rightarrow (y, x) \notin R$$

R is not symmetric

$$\text{Let } S : \frac{m}{n} S \frac{p}{q}$$

$$\Rightarrow qm = pn \Rightarrow \frac{p}{q} = \frac{m}{n}$$

$$\therefore \frac{m}{n} = \frac{m}{n} \therefore \text{reflexive}$$

$$\frac{m}{n} = \frac{p}{q} \Rightarrow \frac{p}{q} = \frac{m}{n} \therefore \text{symmetric}$$

$$\text{Let } \frac{m}{n} S \frac{p}{q}, \frac{p}{q} S \frac{r}{s}$$

$$\Rightarrow qm = pn, ps = rq$$

$$\Rightarrow \frac{p}{q} = \frac{m}{n} = \frac{r}{s}$$

$$\Rightarrow ms = rn \text{ transitive}$$

.S is an equivalence relation.

Question195

$$\text{Let } f(x) = (x+1)^2 - 1, x \geq -1$$

Statement -1: The set $\{x : f(x) = f^{-1}(x) = \{0, -1\}\}$.

Statement- 2: f is a bijection.

[2009]

Options:

- A. Statement-1 is true, Statement-2 is true. Statement-2 is not a correct explanation for Statement-1.
- B. Statement-1 is true, Statement-2 is false.
- C. Statement-1 is false, Statement-2 is true.
- D. Statement-1 is true, Statement-2 is true. Statement-2 is a correct explanation for Statement-1.

Answer: D

Solution:

Given that $f(x) = (x+1)^2 - 1$, $x \geq -1$

Clearly $D_f = [-1, \infty)$ but co-domain is not given. Therefore $f(x)$ is onto.

Let $f(x_1) = f(x_2)$

$$\Rightarrow (x_1 + 1)^2 - 1 = (x_2 + 1)^2 - 1$$

$$\Rightarrow x_1 = x_2$$

$\therefore f(x)$ is one-one, hence $f(x)$ is bijection

$\because (x+1)$ being something +ve, $\forall x > -1$

$\therefore f^{-1}(x)$ will exist. Let $(x+1)^2 - 1 = y$

$$\Rightarrow x+1 = \sqrt{y+1} \text{ (+ve square root as } x+1 \geq 0 \text{)}$$

$$\Rightarrow x = -1 + \sqrt{y+1}$$

$$\Rightarrow f^{-1}(x) = \sqrt{x+1} - 1$$

Then $f(x) = f^{-1}(x)$

$$\Rightarrow (x+1)^2 - 1 = \sqrt{x+1} - 1$$

$$\Rightarrow (x+1)^2 = \sqrt{x+1} \Rightarrow (x+1)^4 = (x+1)$$

$$\Rightarrow (x+1)[(x+1)^3 - 1] = 0 \Rightarrow x = -1, 0$$

$\therefore$ The statement- 1 and statement- 2 both are true.

Question196

Let R be the real line. Consider the following subsets of the plane $R \times R$:

$$S = \{ (x, y) : y = x + 1 \text{ and } 0 < x < 2 \}$$

$$T = \{ (x, y) : x - y \text{ is an integer} \}$$

Which one of the following is true?

[2008]

Options:

- A. Neither S nor T is an equivalence relation on R

- B. Both S and T are equivalence relation on R
- C. S is an equivalence relation on R but T is not
- D. T is an equivalence relation on R but S is not

Answer: D

Solution:

Given that

$$S = \{ (x, y) : y = x + 1 \text{ and } 0 < x < 2 \}$$

$$\therefore x \neq x + 1 \text{ for any } x \in (0, 2)$$

$$\Rightarrow (x, x) \notin S$$

So, S is not reflexive.

Hence, S is not an equivalence relation.

$$\text{Given } T = \{ (x, y) : x - y \text{ is an integer} \}$$

$$\therefore x - x = 0 \text{ is an integer, } \forall x \in \mathbb{R}$$

So, T is reflexive.

$$\text{Let } (x, y) \in T \Rightarrow x - y \text{ is an integer then } y - x \text{ is also an integer} \Rightarrow (y, x) \in T$$

$$\therefore T \text{ is symmetric}$$

$$\text{If } x - y \text{ is an integer and } y - z \text{ is an integer then } (x - y) + (y - z) = x - z \text{ is also an integer.}$$

$$\therefore T \text{ is transitive}$$

Hence T is an equivalence relation.

Question 197

Let $f : \mathbb{N} \rightarrow Y$ be a function defined as $f(x) = 4x + 3$ where

$$Y = \{ y \in \mathbb{N} : y = 4x + 3 \text{ for some } x \in \mathbb{N} \}.$$

Show that f is invertible and its inverse is

[2008]

Options:

A. $g(y) = \frac{3y+4}{3}$

B. $g(y) = 4 + \frac{y+3}{4}$

C. $g(y) = \frac{y+3}{4}$

D. $g(y) = \frac{y-3}{4}$

Answer: D

Solution:

Clearly $f(x) = 4x + 3$ is one one and onto, so it is invertible.

Let $f(x) = 4x + 3 = y$

$$\Rightarrow x = \frac{y-3}{4}$$

$$\therefore g(y) = \frac{y-3}{4}$$

Question198

Let W denote the words in the English dictionary. Define the relation R by $R = \{(x, y) \in W \times W \mid \text{the words } x \text{ and } y \text{ have at least one letter in common.}\}$ Then R is
[2006]

Options:

- A. not reflexive, symmetric and transitive
- B. reflexive, symmetric and not transitive
- C. reflexive, symmetric and transitive
- D. reflexive, not symmetric and transitive

Answer: B

Solution:

Clearly $(x, x) \in R, \forall x \in W$

$\therefore$ All letter are common in some word. So R is reflexive.

Let $(x, y) \in R$, then $(y, x) \in R$ as x and y have at least one letter in common. So, R is symmetric.

But R is not transitive for example

Let $x = \text{BOY}$, $y = \text{TOY}$ and $z = \text{THREE}$

then $(x, y) \in R$ (O, Y are common) and $(y, z) \in R$ (T is common) but $(x, z) \notin R$. (as no letter is common)

Question199

A real valued function $f(x)$ satisfies the functional equation
 $f(x - y) = f(x)f(y) - f(a - x)f(a + y)$ where a is a given constant and

$f(0) = 0$, $f(2a - x)$ is equal to
[2005]

Options:

A. $-f(x)$

B. $f(x)$

C. $f(a) + f(a - x)$

D. $f(-x)$

Answer: A

Solution:

Given that $f(0) = 0$ and put

$$x = 0, y = 0$$

$$f(0) = f^2(0) - f^2(a)$$

$$\Rightarrow f^2(a) = 0 \Rightarrow f(a) = 0$$

$$f(2a - x) = f(a - (x - a))$$

$$= f(a)f(x - a) - f(0)f(x)$$

$$= f(a)f(x - a) - f(x) = -f(x)$$

$$\Rightarrow f(2a - x) = -f(x)$$

Question200

Let $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)\}$ be a relation on the set

$A = \{3, 6, 9, 12\}$. The relation is
[2005]

Options:

A. reflexive and transitive only

B. reflexive only

C. an equivalence relation

D. reflexive and symmetric only

Answer: A

Solution:

R is reflexive and transitive only.

Here $(3, 3), (6, 6), (9, 9), (12, 12) \in R$ [So, reflexive]

$(3, 6), (6, 12), (3, 12) \in R$ [So, transitive]

$\therefore (3, 6) \in R$ but $(6, 3) \notin R$ [So, non-symmetric]

Question 201

Let $f : (-1, 1) \rightarrow B$, be a function defined by $f(x) = \tan^{-1} \frac{2x}{1-x^2}$, then f is both one – one and onto when B is the interval [2005]

Options:

A. $\left(0, \frac{\pi}{2}\right)$

B. $\left[0, \frac{\pi}{2}\right)$

C. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

D. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Answer: D

Solution:

$$\text{Given } f(x) = \tan^{-1} \left(\frac{2x}{1-x^2} \right) = 2\tan^{-1} x$$

for $x \in (-1, 1)$

$$\text{If } x \in (-1, 1) \Rightarrow \tan^{-1} x \in \left(-\frac{\pi}{4}, \frac{\pi}{4} \right)$$

$$\Rightarrow 2\tan^{-1} x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$$

$$\text{Clearly, range of } f(x) = \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$$

For f to be onto, codomain = range

$$\therefore \text{Co-domain of function} = B = \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$$

Question202

The graph of the function $y = f(x)$ is symmetrical about the line $x = 2$, then

[2004]

Options:

A. $f(x) = -f(-x)$

B. $f(2+x) = f(2-x)$

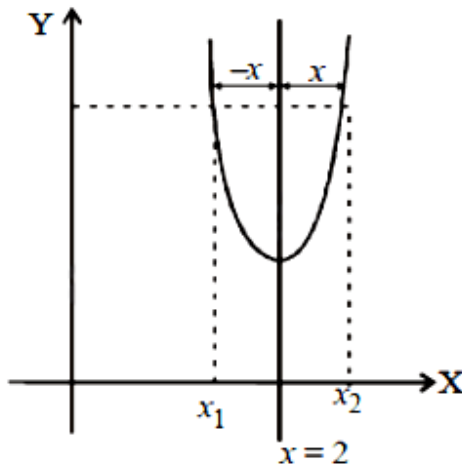
C. $f(x) = f(-x)$

D. $f(x+2) = f(x-2)$

Answer: B

Solution:

(b) Given that a graph symmetrical. with respect to line $x = 2$ as shown in the figure.



From the figure

$$f(x_1) = f(x_2), \text{ where } x_1 = 2 - x \text{ and } x_2 = 2 + x$$

$$\therefore f(2 - x) = f(2 + x)$$

Question203

Let $R = \{(1,3), (4, 2), (2, 4), (2,3), (3,1)\}$ be a relation on the set $A = \{1, 2, 3, 4\}$.. The relation R is

[2004]

Options:

- A. reflexive
- B. transitive
- C. not symmetric
- D. a function

Answer: C

Solution:

$\because (1, 1) \notin R \Rightarrow R$ is not reflexive
 $\because (2, 3) \in R$ but $(3, 2) \notin R$
 $\therefore R$ is not symmetric
 $\because (4, 2)$ and $(2, 4) \in R$ but $(4, 4) \notin R$
 $\Rightarrow R$ is not transitive

Question 204

If $f : \mathbb{R} \rightarrow S$, defined by $f(x) = \sin x - \sqrt{3} \cos x + 1$, is onto, then the interval of S is
[2004]

Options:

- A. $[-1, 3]$
- B. $[-1, 1]$
- C. $[0, 1]$
- D. $[0, 3]$

Answer: A

Solution:

Given that $f(x)$ is onto
 $\therefore \text{range of } f(x) = \text{codomain} = S$
Now, $f(x) = \sin x - \sqrt{3} \cos x + 1$

$$= 2 \sin\left(x - \frac{\pi}{3}\right) + 1$$

we know that $-1 \leq \sin\left(x - \frac{\pi}{3}\right) \leq 1$

$$-1 \leq 2 \sin\left(x - \frac{\pi}{3}\right) + 1 \leq 3 \therefore f(x) \in [-1, 3] = S$$

Question205

Domain of definition of the function $f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$, is [2003]

Options:

A. $(-1, 0) \cup (1, 2) \cup (2, \infty)$

B. $(a, 2)$

C. $(-1, 0) \cup (a, 2)$

D. $(1, 2) \cup (2, \infty)$

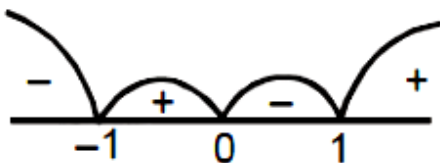
Answer: A

Solution:

$$f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$$

$$4 - x^2 \neq 0; x^3 - x > 0$$

$$x \neq \pm\sqrt{4} \text{ and } -1 < x < 0 \text{ or } 1 < x < \infty$$



$$\therefore D = (-1, 0) \cup (1, \infty) - \{\sqrt{4}\}$$

$$D = (-1, 0) \cup (1, 2) \cup (2, \infty)$$

Question206

If $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x + y) = f(x) + f(y)$, for all $x, y \in \mathbb{R}$ and $f(1) = 7$, then $\sum_{r=1}^n f(r)$ is

[2003]

Options:

A. $\frac{7n(n+1)}{2}$

B. $\frac{7n}{2}$

C. $\frac{7(n+1)}{2}$

D. $7n + (n + 1)$

Answer: A

Solution:

$$f(x+y) = f(x) + f(y)$$

$$\therefore f(1) = 7$$

$$f(2) = f(1+1) = f(1) + f(1) = 14$$

$$f(3) = f(1+2) = f(1) + f(2) = 21$$

$$\therefore \sum_{r=1}^n f(r) = 7(1+2+3+\dots+n)$$

$$= \frac{7n(n+1)}{2}$$

Question207

A function f from the set of natural numbers to integers defined by

$$f(n) = \begin{cases} \frac{n-1}{2}, & \text{when } n \text{ is odd} \\ -\frac{n}{2}, & \text{when } n \text{ is even} \end{cases} \text{ is}$$

[2003]

Options:

- A. neither one -one nor onto
- B. one-one but not onto
- C. onto but not one-one
- D. one-one and onto both.

Answer: D

Solution:

We have $f : \mathbb{N} \rightarrow \mathbb{I}$

Let x and y are two even natural numbers, and $f(x) = f(y) \Rightarrow \frac{-x}{2} = \frac{-y}{2} \Rightarrow x = y$

$\therefore f(n)$ is one-one for even natural number.

Let x and y are two odd natural numbers and $f(x) = f(y) \Rightarrow \frac{x-1}{2} = \frac{y-1}{2} \Rightarrow x = y$

$\therefore f(n)$ is one-one for odd natural number.

Hence f is one-one.

Let $y = \frac{n-1}{2} \Rightarrow 2y + 1 = n$

This shows that n is always odd number for $y \in \mathbb{I}$ (i)

and $y = \frac{-n}{2} \Rightarrow -2y = n$

This shows that n is always even number for $y \in \mathbb{I}$ (ii)

From (i) and (ii)

Range of $f = \mathbb{I} = \text{codomain}$

$\therefore f$ is onto.

Hence f is one one and onto both.
